Multi-vessel sea state estimation using artificial neural networks and vessel response spectral data

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MULTI-VESSEL SEA STATE ESTIMATION USING ARTIFICIAL NEURAL NETWORKS AND VESSEL RESPONSE SPECTRAL DATA

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Thesis Title

Multi-vessel Sea State Estimation using Artificial Neural Networks and Vessel Response Spectral Data

Thesis Abstract

The use of ships as wave buoy analogies (SAWB) provides a novel means to estimate sea states, namely wave heights, periods, and directions. While single vessels have shown reasonable results, a previous study has shown improved estimations when using multiple vessels. However, no investigation into the effects of vessel trajectory on multi-vessel sea state estimation (SSE) performance has previously been undertaken. It was hypothesised that the guidance of multiple vessels acting as wave buoys for SSE can improve upon single vessel estimation. Further, it was hypothesised that variation of vessel trajectories will reduce SSE error as it changes the filtering types of the vessels, when compared with common trajectories.

While several SAWB methodologies have been developed, the use of machine learning (ML) for SAWB has shown great promise in terms of improved estimation accuracy. Moreover, ML approaches to SAWB-based SSE are model-free, establishing relationships between causal wave properties and vessel motion responses using the input and output data alone. Therefore, neural networks have been designed to intake spectral vessel response data for heave, pitch, and roll, and output causal wave properties, with a high degree of accuracy when compared with similar approaches.

As spectra can represent time-series information statistically, it was hypothesised that the proposed SSE methodology could be used to update SSEs online. However, although the spectra had similar shapes over time, the density of spectral ordinates was lower, leading to poor real-time estimation accuracy.

Two uncertainty metrics were adopted to test their effectiveness as assessing SSEs, a time-based metric aimed at evaluating random errors, and a spectral agreement metric to assess bias errors. The results showed a combination of the metrics were able to associate uncertainty with SSE error distribution.

To investigate the difference in SSE performance when multiple vessels move along different trajectories, three guidance strategies were implemented, and the resulting estimation errors were compared. Further, each multi-vessel strategy was compared with single vessel estimation. The multi-vessel guidance approaches were all found to reduce errors in the estimations over the single vessel SSE, with the strategies involving varied trajectories decreasing estimation error more than the common heading approach.

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Publications and Awards

Publications

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Awards

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Extended Abstract

A novel approach to estimating sea states has seen recent advancements in terms of methodology and implementation. Alongside the rise of autonomous surface vessels, the use of autonomous ships as wave buoy (SAWB) analogies is becoming increasingly viable as a means to estimate local wave properties, specifically, wave heights, periods, and directions. While a single vessel has been shown to provide a reasonable measurement, a study using multiple vessels simultaneously indicated that estimations were improved, with each vessel essentially acting as a separate 'filter'; however, no study was conducted on the effects of vessel trajectories on sea state estimation (SSE) performance.

It was hypothesised that, by guiding the vessel trajectories, multi-vessel SAWB SSE error can be reduced when compared with a single vessel or multiple vessels travelling with a common heading. This hypothesis was based on the fact that vessels respond differently to wave conditions based on their wave encounter frequency, which is a function of their wave heading and forward speed. Therefore, vessels travelling with varied trajectories respond differently to the same sea state, acting as different filter types. As such, it was hypothesised that bias errors related to SSE using motion response data for a single vessel trajectory can be reduced by combining the SSEs from multiple filter types, as well as increase estimation accuracy when compared to multiple vessels with a common trajectory. While this could be achieved using a single vessel varying its trajectory, it would generally require more time to achieve equivalent accuracies.

A number of SAWB SSE methods have been created by researchers in recent years, which can be categorised in a number of ways. One major distinction is between those methods which rely on defining inverse transfer functions between wave motion and vessel response motion, which are based on the physical properties of each vessel, and the recent use of machine learning (ML) to define the same relationships using input and output data alone. The model-free ML approach shows great potential on numerous fronts. Firstly, no information about a vessel's physical parameters is required to train an effective SSE model, only experimental data. Secondly, the (often applied) linear assumption of vessel transfer functions can be removed from the assessment, allowing for a greater range of sea states to be estimated. Thirdly, assuming that the SSE model has been integrated into a ship's operational guidance, navigation, and con-

trol system, it can increase in accuracy over time as the model is trained on a greater variety of scenarios.

The rapid expansion of artificial intelligence has permeated the disciplines of engineering and computer science, leading to novel solutions for complex problems. ML has circumnavigated certain processes of human learning by automating pattern recognition and bypassing understanding of relationships between different variables. One type of ML that is now widespread is the use of artificial neural networks (NNs), which mimic the biological processes of brains, and are capable of mapping complex relationships using input and output data alone.

Therefore, it was decided to develop NNs which take vessel response spectral information as input, and output sea state information. It was hypothesised that the use of vessel response power spectral densities would provide a statistical representation of vessel responses, which can be used for SSE. Specifically, spectral data from the heave, pitch, and roll responses were given as input to the networks, and significant wave height, mean wave period and wave heading were each output from a different NN.

A Modified Pierson-Moskowitz spectrum was first used to simulate uni-directional wave elevation time-series to interact with the vessels. Vessel motion response time-series were then generated using a set of semi-analytical expressions, which provided a reasonable representation of wave frequency to vessel motion response frequency transfer functions. The vessel response time-series were then transformed into the frequency domain in the form of power spectral densities, and an analysis of the power in the spectra was conducted. The resulting information was combined with the vessel's forward speed and used as input to the NN models.

In order to understand what information can be extracted from each degree of freedom (heave, pitch, and roll), as well as the combination of degrees of freedom (DOF), an initial study was conducted into the SSE performance of the NNs trained on response data from one and two DOFs. A secondary study was also undertaken to investigate the effect of wave heading on the estimation performance of each DOF component. The performance of the 3-DOF models was finally analysed and compared with similar studies, with the NNs achieving good result in terms of estimation error and model fitness.

The SSE task was then expanded from a single vessel to multiple vessels. The aggregation of multiple estimates for vessels following different trajectories over the same sea state required the transformation of relative wave heading of each vessel to a global wave direction. A comparison between single vessel and three multi-vessel guidance approaches for SSE was presented. The first approach involved three vessels moving with a common heading, the second forced the vessels on trajectories with the greatest possible variation in headings, and the third was sent the vessels on randomised headings.

Results showed that all three of the multi-vessel guidance approaches resulted in SSE error reduction when compared with a single vessel. When comparing the different guidance approaches, the results from the varied heading and random heading approaches saw a greater reduction in estimation error than the common heading approach, with wave direction and mean wave period estimates showing a greater improvement than significant wave height estimation. A two sample t-test showed that there was a significant difference between the performance of the three different approaches, except for the difference between the significant wave height estimations of the common heading and varied heading strategies.

Two uncertainty metrics were introduced in order to assess confidence in the SSE results. The first was a time-based metric which was designed to ensure that the estimation results were statistically significant, reducing random errors by ensuring that a certain number of wave components were encountered. The second uncertainty metric was related to the difference in estimated spectra, where it was assumed that the convergence of spectral estimates would indicate that bias errors were reduced, given that multiple vessels converged on the same result. An offline analysis of the uncertainty metrics showed that the time-based metric could be applied to significant wave height and mean wave period estimation, while the spectral uncertainty metric could be applied to wave direction estimation.

Given the statistical nature of spectral data, it was hypothesised that the designed NNs would be able to estimate wave properties online. However, the NNs failed to effectively estimate wave conditions in real time. Although the shapes of the vessel response spectra were quite similar over time, the density of high energy spectral ordinates was lower at earlier times, meaning that the networks failed to map them accurately to the estimated wave properties due to the variation in spectral ordinate distribution. As such, the uncertainty metrics also failed to effectively assess estimation performance online, so their applicability to real time multi-vessel SSE could not be determined.

The results from the multi-vessel SSE task were generally found to support the primary hypothesis, with the varied heading and random heading SSE guidance strategies showing the greatest reduction in SSE errors when compared with a single vessel or the common heading approach. Further, the error distribution of the different approaches were found to be significantly different, indicating that the results were not random.

The guidance approaches explored in this study can be used as a baseline for future studies on multi-vessel strategies for SAWB-based SSE. The NNs designed using spectral response data show promise for SSE, and could potentially be modified to better estimate wave conditions in real time.

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Nomenclature

$\%_{RMSE}$ RMSE ratio for single versus multi-vessel estimations

α_{Φ} Vessel vertical transfer function variable

α_e Encounter conversion parameter

$\bar{T}_w$ Mean wave period, s

$\bar{z}_w$ Average wave elevation, m

$\Delta\hat{S}(\omega)$ Sea state estimation spectral variation

$\Delta\omega_{1/2}$ Half-power bandwidth (MPM), rad/s

$\Delta\omega_{1/2_e}$ Half-power encounter bandwidth (MPM), rad/s

Δ_{π} Vessel displacement, kg

δ_{π} Vessel section ratio (for roll)

ϵ_w Wave phase shift, rad

η_{Φ} Vessel vertical transfer function variable

γ_{π} Vessel section ratio (for roll)

κ_{Φ} Smith's correction factor

λ_w Wavelength, m

μ_{π} Vessel heading (NED), $^\circ$

μ_h Wave heading, $^\circ$

μ_w Wave direction, $^\circ$

ω Incidental wave frequency, rad/s

ω_e Wave encounter frequency, rad/s

ω_p Peak spectral frequency, rad/s

ω_{p_e} Peak spectral encounter frequency (MPM), rad/s

ϕ Roll angle about x-axis (NED), rad

Φ_ϕ Roll transfer function

Φ_z Heave transfer function

Φ_θ Pitch transfer function

Π Group of vessels

π_n Individual vessel

ψ Yaw angle about z-axis (NED), rad

ρ_w Density of water, kg/m^3

θ_π Pitch angle about y-axis (NED), rad

ξ_S Spectral uncertainty

ξ_T Standard deviation random error (time-based uncertainty)

ξ_{niel} Nielsen spectral uncertainty

ζ Wave elevation, m

$A1_\pi$ Area of first vessel section (for roll), m^2

$A2_\pi$ Area of second vessel section (for roll), m^2

a_p Perceptron activation function

A_w Wave amplitude, m

A_{PM} Modified Pierson-Moskowitz spectrum variable

$b1$ $b_{\pi_{44}}$ constant

$B1_{\pi}$ Breadth of first vessel section (for roll), m

$b2$ $b_{\pi_{44}}$ constant

$B2_{\pi}$ Breadth of second vessel section (for roll), m

$b3$ $b_{\pi_{44}}$ constant

B_{π} Vessel breadth, m

b_p Perceptron bias

b_s Spring damping coefficient

$B_{\pi_{44}}$ Hydrodynamic damping coefficient

$b_{\pi_{44}}$ Vessel sectional inviscid hydrodynamic coefficient

B_{PM} Modified Pierson-Moskowitz spectrum variable

c_{π} Type of wave heading

c_s Spring stiffness, N/m

$C_{\pi_{44}}$ Restoring moment coefficient

$C_{b_{\pi}}$ Vessel block coefficient

$C_{WP_{\pi}}$ Vessel water plane coefficient

d_w Ocean depth, m

e_R Residual

f_{Φ} Vessel vertical transfer function variable

F_w Wave excitation force, N

F_{0_w} Wave excitation force amplitude, N

F_{0_z} Heave forcing function

Fn Froude number

G_{0_θ} Pitch forcing function

GM_{T_π} Transverse metacentric height, m

H_s Significant wave height, m

H_w Wave height, m

k_w Wavenumber, rad/m

k_{cv} Number of k-folds in cross-validation

L_π Vessel length, m

M_ϕ Roll moment, kgm^2

m_{ζ_0} Wave spectral total power, m^2

m_{z_0} Heave PSD total power, m^2

N_π Number of vessels

N_w Number of wave components

p_{sig} T-test p-value

Q_π Vessel sectional hydrodynamic damping coefficient

R^2 Coefficient of determination

s_μ^2 Wave heading angle variance, deg^2

S_{ϕ_π} Roll response PSD, $rad^2/(rad/s)$

S_{θ_π} Pitch response PSD, $rad^2/(rad/s)$

S_ζ Wave spectra, $m^2/(rad/s)$

S_{z_π} Heave response PSD, $m^2/(rad/s)$

T_π Vessel draft, m

T_w Wave period, s

T_{N_ϕ} Natural period of roll, s

t_{sig} T-test t-value

T_{sse} Sea state estimation trial run length, s

U_π Linear velocity (BF), m/s

w_p Perceptron connection weight

x_π Position along x-axis (NED), m

x_p Perceptron input

x_s Spring displacement, m

x_w Position along wave x-axis (WR), m

y_π Position along y-axis (NED), m

y_w Position along wave y-axis (WR), m

z_π Position along z-axis (NED), m

z_w Wave elevation (WR), m

Acronyms

2D Two-dimensional

3D Two-dimensional

AI Artificial intelligence

BF Body-fixed reference frame

CNN Convolutional neural network

DOF Degree-of-freedom

EOM Equation of motion

FRF Frequency response function

IMU Inertial measurement unit

ML Machine learning

MLP Multi-layer perceptron

NED North-east-down reference frame

NN Neural network

NOMAD Navy oceanographic meteorological automatic device

RAO Response amplitude operator

RMSE Root mean square error

SAWB ship as a wave buoy

SSE Sea state estimation

USV Uninhabited surface vessel

WG Wave glider

WMO World Meteorological Organisation

WR Wave-relative reference frame

Chapter 1: Introduction

Humans have been a seafaring species for thousands of years [1]. In the pursuit of marine travel, the necessity to estimate the state of the sea, i.e. the heights, frequencies, and directions of the surface wave trains, naturally emerged due to the direct correlation between ship motion and sea surface wave properties [2]. Given that approximately 90% of world trade is delivered by sea [3], as well as numerous other major maritime-based industries, such as fishing and defence, the ability to estimate wave conditions in-situ and in real time remains of great importance today.

Sea states are often described statistically by their significant wave height and mean wave period, as well as by the direction in which they propagate [4]. The advantages of sea state estimation (SSE) are numerous. According to a study by Zhang and Li [3], 30% of maritime accidents are caused by poor weather, which directly affects sea surface conditions. Therefore, the ability to estimate local sea states could both provide an indication of regions to avoid due to rough seas, or help to plan safe navigation if avoidance is not possible. This same idea can be extended to reducing impact loads on ships to improve passenger and crew comfort, as well as protect vessel payloads. Further, by incorporating sea state information into future autonomous navigation systems, fuel consumption could be minimised and transit times could be reduced.

Several SSE techniques have been developed over the years, shifting from human to machine-based sensing. The longest continually used method of SSE is via direct visual observation by trained mariners [5]. Even to this day, the World Meteorological Organisation (WMO) continues to run its Voluntary Observing Ships' scheme, which compiles observed weather conditions from participating ships into a publicly available data set, including visually observed wave conditions [6]. However, errors and uncertainty in SSE via visual observation remain high due to the inherent inaccuracies of human visual interpretation of events [5]. One unique alternative to the estimation of wave properties using human sight was to use the sense of touch. Some Polynesian cultures were reported as being able to determine swell properties by submerging their testicles in the water and feeling the subtle differences in their motion [7], [8].

Technological advances led to the automation of SSE by applying the same underlying principle used by the Polynesians to a machine, the subtle variation in motion of a body in water could be used to calculate the causal wave properties. The Navy oceanographic meteorological automatic device (NOMAD) was one of the first automatic wave sensing systems, deployed from the 1950s [9]. NOMAD was a moored wave buoy with a hull-shaped form developed by the U.S. Navy, and heralded their widespread use today.

While moored wave buoy design is highly matured and their results reliable [10], the fact that they are in fixed locations means they are only practical for littoral maritime operations. Drifting wave buoys were introduced in the 1960s [11], which are free floating buoys deployed in the ocean and acted upon by the wind, waves, and currents. Drifting wave buoys permitted SSE in deep water, inaccessible to moored buoys; however, they have no propulsion system, so are only able to take wave measurements in whichever locations the ocean guides them to.

In order to overcome this, ship-based and satellite-based SSE methodologies have been developed, allowing for in-situ SSE for global maritime operations. Satellite altimetry and GNSS geodesy can provide SSEs anywhere on Earth; however, satellite altimetry is temporally coarse [4], while the motion of vessels mounted with a GNSS receiver itself introduces a significant source of noise when undertaking GNSS geodesy [12]. Ship-based radar allows for in-situ wave measurements to be taken directly from onboard a seagoing vessel, though the radar systems are expensive to install and require frequent calibration [13].

One novel approach to SSE, which can provide in-situ, real time estimations, has shown rapid advancement in recent times. The use of a ship as a wave buoy (SAWB) analogy applies the established methodologies of a wave buoy to a seagoing vessel [14]. By analysing the motion responses of a vessel to known wave inputs, a relationship between wave properties and vessel responses can be specified. Defining this relationship has traditionally been achieved by developing physics-based transfer functions between wave frequency components and vessel response frequency components in different degrees of freedom (DOFs), which often rely on the assumption that a linear relationship exists between the wave frequency and response frequency [14]. While this assumption tends to hold for moderate waves, a nonlinear relationship can form under rougher wave conditions.

In the fields of engineering and computer science (amongst others), researchers often look to nature when problem solving or looking for ways to augment existing processes. Artificial intelligence (AI), in particular, is concerned with mimicking biological systems in order to ascribe what humanity interprets as intelligence to artificial machines. Machine learning (ML) is a branch of AI which involves training computer programs to learn patterns and relationships, automatically updating (and hopefully improving) its modelling when exposed to new information, then using that information to make decisions or predictions [15].

With brains being perceived as the source of biological intelligence, the quest to artificially mimic the brain naturally emerged. Artificial neural networks (NNs) are simplified representations of the neural connections in the brain, which can be used to map complex relationships between input data and output variables [16].

As such, the use of ML and NNs for SSE using SAWB has recently been explored, with promising results for a number of reasons. Firstly, the linearity between the waves and vessel motion can be removed as ML models will simply map motion response inputs to wave property outputs, without having to define any physics-based models. Secondly, the same ML model can be easily adapted for multiple vessel types if input data can be provided. Thirdly, initial results are showing high accuracy SSEs [17]–[21]. NNs are the main form of ML which have been used to date for SAWB SSE [18]–[21].

Most of the current ML-based SAWB techniques have interpreted vessel response information directly in the time-domain [18]–[20]. One of the studies used a combination of time and frequency response information [21]; however, no studies have investigated the use of vessel response spectral data alone. It is hypothesised that NNs can be trained on vessel response spectral data to accurately estimate sea states. Further, as spectral data gives a statistical representation of a time-series, it is hypothesised that NNs trained on spectral data will be able to estimate sea states in real time. Three DOFs were selected as input to the NN, heave, pitch, and roll, being the three most commonly used DOFs for SAWB SSE (eg. [22]–[25]).

Although a single vessel acting as a wave buoy may provide sufficient SSEs for a range of circumstances, there are inherent uncertainties which arise from relying on the results from a single trial. Bias errors associated with a specific vessel's state (such as speed, heading, imperfect geometry, or weight distribution) may have an effect on estimations. Further, random statistical error from a single estimation cannot be accounted for if there is only a single result. Therefore, the use of multiple vessels acting as wave buoys presents a solution. Nielsen, Brodtkorb, and Sørensen [25] conducted one study on the use of multiple ships as wave buoys, which showed improved estimation results when using multiple vessels. However, the authors considered the vessels as already in transit, such that their headings and speed were random with respect to one another, so did not investigate the effects of vessel trajectories on the SSE performance.

It is hypothesised that SSE results can be improved by varying the trajectories of multiple vessels acting as wave buoys, when compared with a single vessel or multiple vessels moving with a common heading. This is due to the different response types of a vessel to different wave encounter frequencies, effectively acting like different filter types. As such, any biases associated with a single vessel state, which affects the SSE model, should be reduced when aggregated with SSEs from vessel with different states (with the state here primarily referring to wave heading).

In order to test this hypothesis, three different multi-vessel SSE guidance approaches were investigated, each using three vessels. The first approach is for the vessels to move with a common heading. The second forces the vessels to move at headings of maximum possible variation when considering the range of wave headings. The third is for the vessels to simply move along random trajectories.

Of course, guiding multiple crewed vessels in order to estimate sea states would be time and resource intensive. Therefore, uninhabited surface vessels (USVs) offer a cheaper, simple solution, while enabling the system to be completely autonomous. Wave gliders (WGs) are a type of USV which use an underwater glider to propel themselves, extracting power from wave motion (as well as using solar energy from two solar panels on the float) [26]. The renewable powered propulsion system grants the USVs extended range and endurance when compared with alternative power sources. Therefore, WGs could be used for SSE for long periods of time and over large distances. As such, a WG has been selected as the vessel model for this study.

Previous research on ML-based SAWB SSE has considered offline estimation by processing vessel response data for a fixed time period [17]–[20]. While this approach may suffice for hindcast SSEs, it is hypothesised that the time required for SSE can be optimised by utilising metrics which are able to determine the quality of updating SSEs in real time, so that the vessel continues until a level of confidence in the estimations is reached. Therefore, two uncertainty metrics were selected to analyse the SSEs, a time-based metric aimed at reducing estimation errors, and a spectral comparison metric aimed at reducing bias errors related to certain vessel configurations.

The thesis follows with an in-depth analysis of current SSE techniques in Chapter 2, the accumulation of which led to the proposed hypotheses and research questions. Chapters 3 and 4 then present the experimental simulation methodologies and results.

The wave model, ship model, vessel dynamics, and designed NN architectures for SSE are given in Chapter 3. Separate NNs were trained on the response data from each DOF (heave, pitch, and roll) to investigate how much information could be extracted for each wave condition estimated (significant wave height, mean wave period, and wave heading) using a single vessel. NNs were then trained using the combinations of two DOFs for each wave condition, and finally the NNs were trained using all three DOFs. In order to test the effects of wave encounter frequency on estimation performance (as a function of wave heading), results for each one of the NN models are analysed for a fixed forward speed and varied wave heading. The results from the final 3-DOF models are then compared with other SAWB SSE literature.

Chapter 4 introduces multi-vessel SAWB SSE, as well as the three simple guidance strategies using three vessels. While significant wave height and mean wave period can be directly aggregated for each of the guidance approaches, a global wave direction is introduced in order to aggregate the individual wave heading estimates for the varied heading and random heading guidance approaches. The performance of the guidance strategies for multi-vessel SSE are compared with that of a single vessel, as well as to one another. The reduction of errors is the primary measure of SSE performance, with the significance of the difference in estimation results for the different guidance strategies also tested. Finally, time and spectral uncertainty metrics are presented and evaluated for offline and online SSE.

It should be noted that the terms ship and vessel are equivalent and interchanged throughout the thesis to improve readability.

Chapter 2: Literature Review

2.1 Introduction

The ability to estimate sea states in-situ and in real time would provide mariners with information which could improve the safety and comfort of passengers and payloads, as well as limit impact loads on seagoing vessels. Current sea state estimation techniques have limitations which must be addressed in order to effectively estimate in-situ wave conditions. Therefore, the advantages and disadvantages of different SSE methodologies are investigated in this chapter, with respect to their applicability to solving this problem.

With the use of a ship as a wave buoy analogy emerging as a solution, a detailed analysis of the current literature on SAWB then proceeds. SAWB-based SSE methodologies have been divided into three categories: parametric versus non-parametric sea state representation, time versus frequency based analysis of the sea, and model-based versus model-free SSE methodologies. Two branches of SAWB are then further explored: multi-vessel SSE using SAWB, and ML-based approaches to SAWB analysis for SSE.

The culmination of this literature review resulted in the proposed research questions and hypotheses.

2.2 Sea State Estimation

Sea state estimation is the estimation of wave properties, such as wave heights, frequencies, and directions [14]. Numerous SSE methods are already being employed globally to improve maritime operations, aid in weather forecasting, for the study of oceanography, and to help understand coastal degradation [27].

Popular SSE techniques include the use of wave buoys [27], ship-based radar [13], satellite-based remote sensing [28], and an emerging method of using a ship as a wave buoy analogy [14]. Table 2.1 gives a summary of sea state estimation techniques and their corresponding advantages and disadvantages. The most common method of taking sea state measurements is via wave buoys - floating sensor-mounted platforms which are primarily located along continental margins [29]. Although wave buoys are useful for monitoring global sea conditions, they are few and far apart (see Appendix 5.1), and contain no means of locomotion, thus making them impractical for dynamic operability for in-situ applications. Ship based radar systems solve this problem by allowing measurements to be taken during ocean transit; however, the radar requires frequent calibration, is expensive to install, and interpretation of the radar data is computationally demanding [30].

Table 2.1: Sea state estimation techniques.

Method	Advantages	Disadvantages
Classic wave buoy	Reliability [10]	Fixed, limited placement (in coastal regions) [31]
Ship-based radar	In-situ measurements	Expensive installation, frequent calibration, high computational power [13]
Ship as a wave buoy	In-situ measurements, dynamically operable	Relatively high uncertainty [14]
Satellite altimetry	Global coverage	Temporally coarse [4]
GNSS geodesy	Global coverage, high accuracy [32], [33]	Little operational heritage for application, complex implementation

Therefore, two main solutions to estimating sea states in-situ during maritime operations exist. Firstly, the use of a ship as a wave buoy (SAWB) analogy, whereby relationships between ship motions and causal wave properties are defined in order to then estimate sea states. Secondly, SSE via remote sensing by satellites.

There are two main branches of remotely detecting sea surface properties using satellites: satellite altimetry and GNSS geodesy. Satellite altimetry encompasses the procedures which involve measuring the time taken for a radar pulse, transmitted from a satellite, to reflect from the sea surface and travel back to the satellite or a separate receiver. The spatial resolution of satellite altimetry, however, is generally quite crude (on the order of 10-100km), as well as having coarse temporal resolutions [34]. GNSS geodesy is the triangulation between a GNSS receiver at the ocean surface and global navigation satellites. GNSS geodesy is capable of measuring displacements on the scale of decametres to centimetres.

Foster et al. [12] conducted a study into the use of a ship-based geodetic GPS receiver, in conjunction with a radar water level gauge, to take sea state surface height measurements. The inclusion of the water level gauge was for the purpose of removing the noise introduced by the variation in ship draft depending on loads, velocities, and sea water densities. Though the focus of the study was on measuring the geoid, the authors tested the use of the GPS-water level gauge combination for accurate wave system properties, using the radar gauge to remove the vertical displacements of the GPS receiver due to rotation of the ship, with promising results when compared to altimetry-based data. However, a reference station was required to fine-tune the GPS localisation, limiting the study to maximum distances of $\sim 100km$ from the reference point. A later study by Foster et al. [35] concentrated specifically on the estimation of sea states using a geodetic GPS and a radar gauge. The wave height estimates had errors of less than 11% with respect to the measurements obtained by a wave buoy, while also showing similarities with hindcast data obtained from WAVEWATCH III. As with [12], the GPS positioning data had to be calibrated using GPS reference stations in post-processing, although the authors discuss the use of precise point positioning with ambiguity resolution to obtain a real time solution for the same problem.

A 13-day experiment in the North Sea, conducted by Penna et al. [33], involved a GNSS wave glider (Liquid Robotics Wave Glider SV2) to estimate both the long wavelength sea surface changes (geoid and dynamic ocean topography), and short, local sea states (wind waves and swell). The wave glider was capable of connecting to both GPS and GLONASS signals. GNSS data was interpreted using kinematic precise point positioning such that the location of the wave glider could be calculated anywhere without the need of a separate reference station. The measurements showed that waves from $< 1m$ to $3 - 4m$ in height were encountered. The authors compared hourly significant wave heights and the dominant hourly wave periods with both the daily wave height prediction data from WAVEWATCH III [36], and wave property data collected by the Cefas Tyne/Tees WaveNet buoy (located south of the wave glider's path). The results showed differences of approximately $20cm$ between the hourly wave height measurements from the wave glider and both WAVEWATCH and the WaveNet buoy, while wave period estimations were within $2s$ difference. It was noted that the measurements taken aboard the wave glider would include errors due to the pitching and rolling caused by the surface waves to the wave glider's float, and the effects of the subsurface glider dynamics, which cause small amplitude signal distortions on the antenna elevation.

While GNSS geodesy offers a dynamic solution to the SSE problem, its overall complexity, where previous studies have used knowledge of the geoid [33], [35], and additional noise source from a ship's responses to wave excitation [12] make it less appealing for simple, in-situ SSE while at sea when compared to the alternative SAWB method. SAWB further shows promise due to its adaptability to autonomous systems and flexibility in application.

2.2.1 Ship as a wave buoy

Marine vessels are often equipped with sensors which are capable of recording their translational and rotational displacements about their centre of gravity. Therefore, relationships between a vessel's responses in each degree of freedom and the causal wave properties can be defined via experimentation and simulation, where the wave inputs are varied and corresponding vessel outputs are recorded. Using this information, the process can then be reversed to solve the inverse problem, where the responses of a vessel in a wave system can be used to estimate the properties of the waves, acting as an analogy to the methodology used in wave buoys. This process can be seen in the simple schematic given in Figure 2.1. SAWB is a novel method for SSE which shows promise due to its potential for relatively simple, in-situ, near real time estimations.

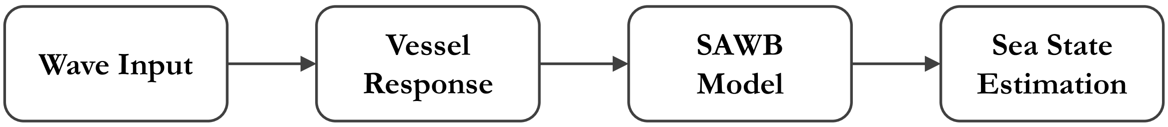

Figure 2.1: Schematic outlining the sea state estimation process when using a ship as a wave buoy.

While all of the SAWB methods interpret the vessel responses to wave excitation, there are a few key defining features that split the methodologies. Figure 2.2 provides a taxonomy of the different types of SAWB techniques, where it can be seen that there are three main defining features. The first divide is between the analysis of the sea state in the time domain or the frequency domain, the second is the separation between wave spectra that are parameterised versus nonparameterised, and the third is the division between a vessel model-based or model-free dependency. Each of these three divisions (domain/parameterisation/model dependency) have been combined with each other in various ways, but are generally applied independently [1], as shown in Figure 2.2.

[1]Nielsen at al.'s step-wise procedure [37], [38] is an example of SAWB which uses a combined model-based and model-free approach

In model-based SAWB approaches, transfer functions, also known as response amplitude operators (RAOs), describe how the wave motion is transferred to vessel responses. RAOs can be calculated using three-dimensional (3D) panel codes based on potential wave theory, using strip theory, and/or with the use of computational fluid dynamics programs [14]. RAOs are vessel-dependent, taking into account ship characteristics, such as geometry, dimensions, and mass. One of the main assumptions when developing RAOs is that moderate wave properties form a linear relationship with the six degree-of-freedom (DOF) motion response, and corresponding structural loads. For example, in regular waves, the amplitude response of a vessel is considered proportional to the encounter wave amplitude. When encountering an irregular wave system, a vessel's motion response can be calculated by invoking the principle of superposition.

As RAOs rely on the linearity assumption, they can provide unreliable results in severe wave conditions, where the linear relationship between wave and response no longer holds. Further, the resulting SSEs are subject to a certain level of uncertainty due to the fidelity of the transfer functions used, and the associated complexity involved with the influence of ship geometry on wave motion response [25]. Model-free approaches, however, are designed independently to the specific vessel, instead defining a relationship between input and output data. In other words, a connection between known vessel responses to known wave input is created without passing through a physics-based model. Therefore, model-free approaches rely only on data interpretation, usually via machine learning or some type of filter, circumventing the creation of complex transfer functions.

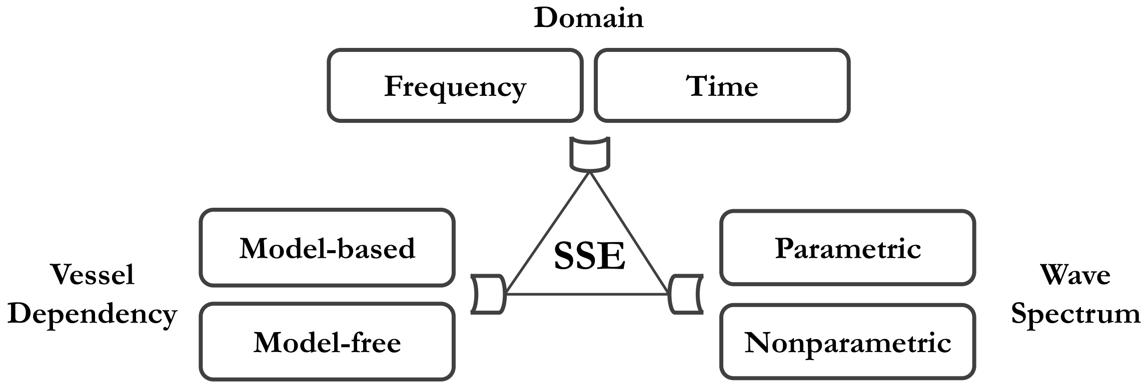

Figure 2.2: Taxonomy of SAWB SSE methods.

Wave properties can be considered either in the time domain or the frequency domain. The frequency domain is associated with waves given in terms of their energy density spectrum, which involves taking a vessel motion response spectrum, then transforming it into the incident wave energy spectrum [14]. SSE using SAWB is most commonly done in the frequency domain; however, some studies have focused on estimating sea states directly from the time domain response of ships [23], [37], [38]. Analyses of wave properties in the frequency domain give a much more general overview of the sea state than in the time domain, but require a minimum of 10-15 minutes of measured vessel response data in order to analyse the response spectra, under statistically stationary conditions [14]. Thus, a ship's direction and speed must be maintained as constant with respect to the waves over the measurement period. The required time of SSE in the frequency domain, therefore, means that SSE cannot really be considered as in "real time", but can be considered as near real time, and is more information rich.

SSE in the time domain aims to provide a continuous estimation of the real time sea state using real time vessel responses; however, time domain analysis currently only provides a portion of the sea state information [23]. Wave data can be transferred between the time and frequency domains, but the inclusion of phase shift means that transforming from the frequency into the time domain can have an unlimited number of results. In contrast, the transformation from the time domain to the frequency domain has only a single solution.

When considering how to represent ocean waves, two main approaches have been taken: parametric and non-parametric. The parametric representation defines an idealised spectral shape which has been derived analytically to describe different ocean conditions, but need to be parameterised [24]. Examples include the Pierson-Moskowitz spectrum which describes fully developed seas, and the JONSWAP spectrum which describes seas with limited fetch [2].

For a non-parametric description, there are no prior assumptions about the spectral shape of the sea [22]. A minimisation task is required to determine the components of the discretised frequency-directional domain which comprise the spectrum, where the only constraint on the spectral amplitudes is that they are non-negative (also known as a hyper-parametric representation). A parametric approach may be better suited to describe a sea state when the amount of information available to analyse it is limited, though it may be biased by the spectrum chosen, while a non-parametric approach can in theory provide a more detailed description of the sea state, but without adequate input, would not necessarily provide a better estimation.

While six DoF responses can be used as input for estimating wave properties (as used by [17]–[19]), it is common to only use three. Either roll or sway are required as port/starboard asymmetry is necessary to distinguish between port and starboard wave encounters, thus, to capture wave direction. Nielsen [22] showed similar estimation results when using either roll or sway, alongside heave and pitch, but chose to focus on roll. This is likely due to the fact that a ship's main mode of maneuverability is controlled by its rudder, which directly affects horizontal rotation and translation, making sway more susceptible to noise from the rudder or sway motion being directly input from the vessel controllers. Similarly, [23]–[25] chose to use heave, pitch, and roll as input to their SSE models, while [39], [40] opted for heave, pitch, and sway.

Takekuma and Takahashi [41] were two of the first researchers to discuss the use of a ship as a wave buoy analogy in 1972 by calibrating response amplitude operators for the response behaviour of a vessel in waves of a known sea spectra, focusing only on the pitching motion. The authors then tested their RAOs on data from different ships in operation, and found that the significant wave heights and mean wave periods could be calculated with reasonable accuracy when compared with estimations from an observer on-board, assuming a Bretshneider spectrum. They noted that their technique only worked in head seas with an encounter angle of less than 60°.

Iseki and Ohtsu [42] published one of the first studies on SSE using SAWB for a vessel with forward speed in 2000. Their methodology used a Bayesian modelling procedure to estimate directional wave spectra, where the relationship between the cross-spectra of three vessel responses were related to the directional wave spectra using a multi-variable linear regression model, and the Bayesian model was used to evaluate its coefficients. Testing their method using a towing tank and operational vessel experiments, they concluded that SAWB was possible, even when the vessel had forward speed, and in both head seas and following seas.

Using sequential quadratic programming, Pascoal, Soares, and Sørensen [39] estimated wave spectra by minimising the difference between how the estimated spectrum generates a vessel response and the actual measured vessel response using cost functions. This was done for parametric and non-parametric sea spectra. A genetic algorithm was used for minimisation in the parametric case, and was found to provide good initial estimates for the sequential quadratic programming algorithm. A frequency by frequency minimisation, with initial values given by a non-negative least-square formulation, was done for the non-parametric approach, and was shown to be more flexible and robust overall. However, their initial study did not handle multi-modal wave spectra very well, resulting in a modified minimisation procedure being subsequently developed by Pascoal and Soares [40] which was able to handle multi-peaked spectra. Hinostroza and Soares [24] further extended the parametric model from [39] to include the influence of a vessel's length, with better results achieved for small vessels.

In 2006, Nielsen [22] developed a SAWB SSE method based on the comparison of spectral energy distributions for both parametric and non-parametric wave modelling. This method essentially parameterises complex transfer functions that describe the relationship between vessel response spectra (pitch, roll, and heave) and the directional wave spectrum via a minimisation process which equates their energy densities. Nielsen termed the non-parametric SSE technique the Bayesian method, which aimed to estimate a discretised wave field comprised of 18 wave headings and 30 wave frequency components. Bayesian modelling requires prior knowledge assumptions about the wave spectra in order to predict, so Nielsen introduced the following prior assumptions: the wave spectrum smoothly transitions in both frequency and direction, and both the very high and very low wave frequencies are negligible. The minimisation problem is then converted into a maximisation of posterior probability of the distribution of the wave spectrum, and solved using a QR factorisation algorithm. When considering a parametric model, a bimodal spectrum with frequency-dependent spreading was chosen, where the parameters were optimised via a gradient search algorithm. Results from simulations showed the Bayesian method outperforming the parametric method given four different wave conditions, with the Bayesian method showing good agreement between the estimated and actual wave spectrum, while the parametric method under-performed for two of the cases. When the methods were compared on actual full-scale response data, it was difficult to conclude which method performed better as ground truth knowledge of the actual sea state was not known; however, the estimations from the two methods agreed quite well with each other.

Nielsen et al. [37], [38] developed a step-wise procedure for SSE in the time domain which utilises a signal-based wave frequency estimator, as well as a nonlinear least squares fitting operation to calculate wave amplitude and phase. The step-wise SSE procedure is unique in that it is a combined model-based and model-free approach, where the frequency estimator is model-free, using a Aranovskiy filter to interpret measured vessel responses, while the nonlinear least squares fitting operation is dependent on RAOs. After testing the step-wise procedure using numerical simulations on vessels with zero forward speed in [37], it was expanded to model-scale experiments in [38] with positive results. However, both the numerical simulations and experiments were only able to analyse sea states comprised of regular sinusoidal wave trains.

Multi-vessel Estimation

To date, only one sea state estimation study utilising multiple vessels as wave buoys has been undertaken.

Nielsen, Brodtkorb, and Sørensen [25] describe the use of a network of sea state measurement platforms in order to improve sea state estimation performance. While the use of platforms using different measurement techniques is mentioned, the authors focus on the use of multiple ships as wave buoys for their study. It was assumed that the ships maintain a large enough distance from each other so that ship-to-ship interactions were negligible, that they were operating in deep water, and that the ships were not distributed over an excessively large area. As such, the wave energy density spectrum is presumed constant across space and time. The authors chose to aggregate individual ship SSEs by first applying an estimate weighting to each ship's transfer functions. As the SSE task was assumed to be performed not by a specialised task-force of ships, but rather ships already at sea completing different operations, the size and shape of the ships varied. Therefore, the weightings were applied as a function of the ship's filtering characteristics, dependent on the dimensions and geometry of each individual vessel.

The authors performed a number of simulations involving three different types of ship encountering four different wave systems, each derived from a Bretschneider spectrum with the inclusion of a spreading function, which remained the same across the entire simulated environment (with an example shown in Figure 2.3). Two of the sea states include both wind and swell waves, though propagating in the same direction. Each of the four sea states were tested for two different operational setups (varying vessel speeds and headings), while the mean wave direction remained constant for all cases. Further, the effects of three different weighting values were tested, one based on heave response, one based on pitch response, and one equally weighted. Using these conditions, the motion responses of each vessel (heave, pitch, and roll) were simulated for 30-minute time sequences, then used as input into the SSE model. The brute force SSE method used [43] estimates the full wave spectrum, not just statistical representations.

The non-weighted single vessel estimates failed to capture the location and number of spectral peaks, with the two larger ships performing worse than the smallest ship. Once weighted, however, the aggregated estimates proved to match the frequency of the true peaks quite accurately (refer to figures in [25]).

The weighting of the vessel estimates, as a function of their size, was shown to improve results by reducing the low-pass filtering effects of the larger vessels (as the authors'expected). The weighted aggregate estimation of both the mean wave periods and mean wave direction were quite accurate, while the estimations of significant wave height were shown to be worse at higher frequencies. Data from the 20 simulation runs of each of the eight scenarios presented the standard deviation of the aggregated estimation errors for each weighting type. For mean wave period the estimation errors for the weighting based on heave response had a standard deviation of 4.38%, the standard deviation of the pitch weighting was 4.35%, and the equal weighting was 8.25%. For the wave direction, the heave weighting gave an error standard deviation of 3.75%, pitch was 3.63%, and equal weighting was 5.38%. As such, it can be seen that both the heave and pitch-based weightings improved upon the equally weighted aggregated estimations for mean wave period and wave direction, with little difference found between them. However, for significant wave height, the heave response weighting had a standard deviation of error of 8.50%, the pitch response was 6.50%, and the equal weighting was 5.50%. The reasoning for this was that significant wave height estimations were described as being more closely related to the total energy of the estimated spectrum, which did not vary much between estimations, while the wave period and direction estimations were more closely related to the distribution of energy in the estimated spectra.

The standard deviations of the estimation errors for the mean wave period and significant wave height were low, but were significantly larger for the mean wave direction.

Overall the weighted aggregate estimates performed better than the non-weighted or single vessel estimates. This study, thus, supports the notion that multiple vessels can be used to improve upon in-situ SSE while at sea. Nielsen, Brodtkorb, and Sørensen [25] mention that further studies should look into the use of direct measurement, rather than transfer function estimation, to establish if there is a more appropriate real time SSE technique to apply to multiple vessels. The authors did not investigate the effect of vessel trajectories on SSE performance.

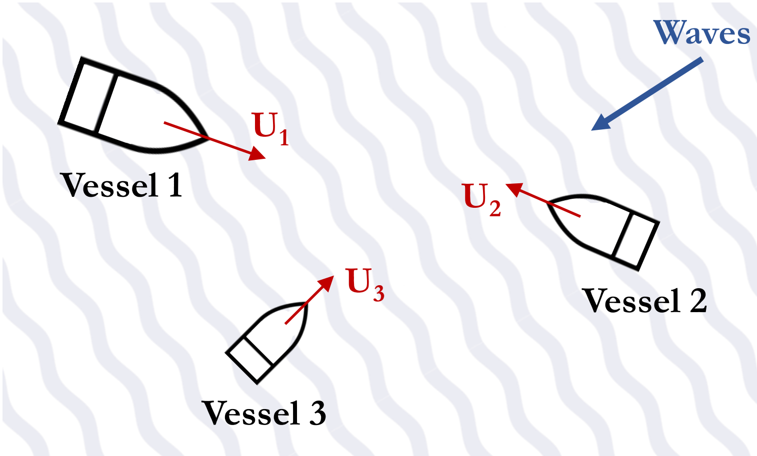

Figure 2.3: Multi-vessel SSE operational setup, modified from [25].

A thesis by Fangbemi [44] is currently the only other literature which considers multiple vessels taking SSEs. However, Fangbemi only considered the sea state elevation as a point of measurement at two locations, using this information as input to a Kalman filter in order to estimate a rectangular 3D sea surface (although the author discusses the potential use of IMU measurements onboard WG platforms). Fangbemi's thesis was primarily concerned with establishing local inter-platform communications, where knowledge of the sea surface elevation in real time was viewed as useful for improving line of sight communications, aligning antennas when both vessels were at wave crests.

ML-augmented Estimation

The augmentation of SSE, using SAWB, via the application of ML techniques, has recently gained attention. The techniques used include quadratic discriminant analysis and partial least square regression [17], as well as deep learning [18]–[21]. This literature forms a new branch of the model-free SAWB methodologies.

Arneson, Brodtkorb, and Sørensen [17] separated the SSE problem into two components, wave heading estimation and wave height and frequency estimation, applying two different ML models to solve them respectively. The first method was the use of quadratic discriminant analysis (QDA) to extract wave heading information, and the second was partial least squares regression (PLSR) for the estimation of peak wave height and peak wave period. Wave headings were discretised for the QDA, representing a classification problem, then PLSR was undertaken for each heading classification to estimate wave heights and periods. Results showed that wave heading estimation was quite accurate with an average error of 4°, while the results for the significant wave height showed an average deviation of $0.7m$ (from the range of $2.0m$ to $12.9m$ tested), while results for the peak wave period had an average deviation of $1.5s$ (from the range of $8.3s$ to $14.2s$ tested). While the authors'described their approach as underperforming when compared with model-based approaches, its ability to estimate from low and high sea states (ranging from 1 to 8> in the WMO sea state code [45]) highlights one of the major benefits of model-free SAWB, the ability to handle rougher sea states where the linear assumption breaks down. However, fluctuations in the results indicate that further modifications are necessary to improve the robustness of the techniques.

Wave direction is cited as being the most difficult parameter to accurately estimate when using traditional wave estimation methods [18]. Therefore, Mak and Düz [18] focused on improving directional wave estimates by training three different types of convolutional neural networks (CNNs) to estimate wave headings using ship response data: a CNN for multivariate regression (CNN-REG), a long short-term memory recurrent neural network (MLSTM-RNN) CNN, and a Sliding Puzzle CNN. The authors took 6-DOF time-series responses of a ship, recorded over six years, and compared them to simultaneously taken wave measurements using an on-board wave scanning radar. Instead of transforming the time-series into the frequency domain, Mak and Düz [18] use the time-series responses as input in order to maintain the phase differences for individual events. As data was taken over the course of regular ship operation, the resulting spread of vessel headings was not evenly distributed, meaning that the training data was much denser for certain headings. After 5-fold cross validation was performed on the data, the CNNs were compared when data was input in chronological order and when the data was shuffled. Results showed lower error for the shuffled data, and the MLSTM-RNN CNN and Sliding Puzzle CNN generally outperformed the CNN-REG.

Mak and Düz [19] then tested the performance of their NNs on simulated data using a 6-DOF seakeeping and manoeuvring simulation tool called FREDYN. As opposed to their experiments using real ship motion data, the simulated data was evenly distributed in terms of sea states (significant wave height and mean wave period), and tested at 6° heading intervals (for a complete 360°) at a speed of 15 knots for 1800 seconds. The simulations were conducted using three ship types grouped by size, where 7140 simulations per ship type were conducted with the same combinations of wave input (JONSWAP spectra as a function of significant wave height and mean wave period). The Sliding Puzzle and MLSTM-RNN CNNs were able to estimate the headings with an approximate 1° error for 95% of results, while the CNN-REG was able to estimate the heading with an approximate 4° error, for all ship types, indicating high performance for a NN-based SAWB approaches. The authors approach can estimate in real time; however, is only concerned with wave heading.

Another study by Cheng et al. [20] experimented with using NNs for SSE, developing a densely connected CNN (dubbed SSENET) to estimate wave heights and directions using ship response data for a vessel with forward speed. Cheng et al.'s method is unique, in that they introduced a zigzag shaped path for the vessel to follow when collecting response data, stating that "zigzag is an element of the basic maneuverability of modern ships", so "the use of motion data of zigzag is more common and the trained model on this kind of motion data can be applied to more ships". The authors grouped the sea states into five wave height categories (based on the WMO sea state code [45]), and separated the wave direction into eight categories, resulting in 40 different sea state combinations. The authors also discuss the requirement for a NN to be able to extract fine-grained features due to the slight differences in vessel responses when headings are similar. 9-DOF time-series vessel response measurements were used to train the SSENET, taken from 12 public data sets, as well as two simulated data sets based on the Norwegian University of Science and Technology's (NTNU's) R/V Gunnerus vessel. After fine tuning SSENET, it was able to achieve high accuracy on most of the public data sets, reaching a maximum of 99.61% accuracy when classifying the Daily Sport data, and an average accuracy of 82.99% for the 12 public data sets. SSENET, further, achieved an 89.81% and 93.46% accuracy for the two NTNU data sets. Cheng et al. [20] compared their results with a number of other NNs, finding the SSENET outperformed each of the others.

Cheng et al. [21] also developed another SSE technique using CNNs, which used vessel response spectrograms as input instead of raw time-series data (SpectralSeaNet). Spectrograms describe a combination of both time and frequency information as two dimensional (2D) images via Fourier transformation using a shifting window of fixed length. Simulated data was generated using the Offshore Simulator Centre AS. As in [20], the authors classified the sea states as one of five based on their significant wave height, and used nine DOFs as input. The spectrograms from each DOF were combined into single spectrogram images to use as inputs into the CNNs, then the 9-DOF combined images were compared with single DOF images, 2-DOF images, and 4-DOF images. The 9-DOF combined spectrograms used as input to the CNN were found to outperform the others. The spectrogram approach was also compared to a CNN and LSTM trained on raw time-series data, where the CNN trained on spectrogram images had an average accuracy of 94%, while the raw time-series had 91% and the LSTM had 79%. This study indicates that use of spectral data for NNs designed for SSE shows promise.

The benefit of developing ML models for sea state estimation using vessel response data and known wave parameters, is that no ship specific information is required, such as ship transfer functions in model-based approaches, though any change in vessel or vessel parameters will require new models to be trained. The ML models are able to learn the relationship between response information and the wave conditions using only the system inputs and outputs themselves. However, conversely, the ML approach is dependent on the quality of the training data used. Interestingly, each of the ML-based approaches chose to use six or more DOF responses as input for their learning models, when compared with the common three DOFs of the model-based SAWB approaches assessed [22]–[25], [39], [40]. Perhaps it was assumed that more features could be extracted from the response data information when using more data; however, some DOFs may be more susceptible to noise from control surface motion than others. It should be noted that the benefit of bypassing inverse transfer functions, and thus in-depth knowledge of a ship's properties, may be useful for some applications but not others. For example, a simple response input and sea state output ML approach would not capture *why* a vessel responds to wave input as it does.

Most of the ML-based SSE methods using SAWB have thus far used time-series response data as input (for example, [18]–[20]), though the use of response data in the frequency domain was tested in Arneson, Brodtkorb, and Sørensen's study [17], as well as combined with time-series data in Cheng et al.'s work [21]. However, no NN has so far been designed for frequency domain input alone.

As response spectra offer a simple, yet information dense, statistical representation of a vessel's motion, it is hypothesised to be able to be used to train effective SSE models to estimate wave properties with a high degree of accuracy. Due to the high performance shown when using NNs for time-domain SSE analysis, their applicability to frequency-domain analysis should be investigated.

2.3 Conclusion

The use of SAWB for SSE permits an SSE system which is dynamically operable in-situ and in real time. However, the coordination of multiple vessels as wave buoys could create an even more dynamic system capable of SSE, with improved results by reducing uncertainties associated with a single vessel alone. Further, model-free approaches can be more easily adapted to new vessel platforms, requiring input and output data alone, with ML-based techniques showing promising results in terms of estimation accuracy.

With the majority of ML-based SSE focusing on NNs, current literature has primarily used times-series response data as input [18]–[20], with one study using a combination of time and frequency response components [21]. There are current no studies which use frequency response data alone as input to NNs in order to estimate sea states. Given that frequency data provides a more statistical analysis of a vessel response, it is hypothesised that it could also prove to provide better information for an SSE technique which updates online.

Therefore, this literature review has identified that an effective solution for SSE using the coordination of multiple vessels has not yet been shown. Further, a simple and effective ML model for SSE, which can be updated in real time and estimate the wave properties significant wave height, mean wave period and wave direction, has yet to be thoroughly investigated.

Chapter 3: Machine Learnt Sea State Estimation Using Artificial Neural Networks and Vessel Response Spectral Data

3.1 Introduction

A wide range of SSE techniques using a ship as a wave buoy already exist in literature; however, each has its own limitations, and further research is required to mature the methodologies before they can be widely implemented.

The use of ML to model the relationships between vessel response data and causal wave properties has shown great promise. ML offers improvement upon other techniques in terms of estimation precision, the removal of assumptions-based physical models, and easy adaptability to new conditions.

Currently, the few existing studies on ML for SAWB-based SSE have either used time-series response data as input to artificial NNs [18], [19], or have used limited spectral information for discriminant analysis and regression [17]. It is hypothesised that by appropriately selecting features from vessel response spectra, NNs can be trained to be able to efficiently estimate sea states. As the spectral representation of a vessel's responses is more information dense than the raw time-series data, being a statistical representation, the features extracted from the response spectra are assumed to provide more information than time-series data when used as input to NNs for SSE.

In order to test this hypothesis, it was decided to simulate the motion responses of a vessel to wave inputs with varying conditions. Uni-directional wave elevation time-series were generated from the parametric Modified Pierson-Moskowitz spectrum for a range of significant wave heights and mean wave periods. The wave heading angle and vessel speed were also modified to ensure that the networks could respond to the different filtering characteristics of the vessels to different wave encounter frequencies. Further, it was decided to only analyse the heave, pitch, and roll responses as they have commonly been used for other SAWB SSE studies [23]–[25], and are assumed to provide sufficient information to determine statistical wave properties.

A methodology was also designed to transform the vessel motion time-series data into their response power spectral density (PSD) components, and then to extract important features from the PSDs. This information was then used as input for the NN models.

Three properties of the sea have been selected as the most important for analysis: the significant wave height, mean wave period, and wave heading, where three separate NNs have been designed to estimate each wave property independently. Significant wave height, mean wave period, and mean wave heading were selected as the wave conditions to estimate as they are three of the most commonly used statistical representations of a wave system for SSE using SAWB [14] (though the mean wave heading is simply the wave heading in a unidirectional wave train).

While it was assumed that using the response data from all three DOFs would provide the best estimation performance when using NNs, it was decided to explore what information could be extracted from each DOF alone. Networks were then trained using the various combinations of 2-DOF responses to assess their combined performance. Ultimately, the response data from all three DOFs was used to train the final NNs for each wave property. Each of these networks were calibrated to improve the estimations, with 5-fold cross validation performed on them to gauge how well the models fit the data. The performance of final networks is then compared to previous SAWB techniques.

3.2 Methodology

3.2.1 Sea States

The highly dynamic motion of the ocean can be decomposed into a number of different wave categories as a function of their period [46] (given in Table 3.1), ranging from capillary waves to trans-tidal waves. While the entire spectrum of wave types contribute to the motion of sea surface waves at any given time, the waves of main concern to maritime operations are generated by the interaction between the wind and the water surface, ordinary gravity waves. Ordinary gravity waves can be dissected further into local wind waves and swell. Swells are waves previously generated by wind, but that no longer interact with the local wind and generally have longer periods, whereas local wind waves are produced by in-situ wind interactions and have shorter periods. Wind waves develop via two physical processes: the tangential drag force of the wind on the surface of the water, and the additional pressure imparted on the windward side of waves (as opposed to the leeward side). These effects have a similar magnitude and combine to create the waves most commonly encountered by sea-going vessels. Wave amplitudes and lengths are dependent on the duration of time that wind blows over a given surface, its velocity, and the fetch, where fetch is the area of an ocean where wind blows in a constant direction resulting in wave generation. A *fully developed sea* is assumed to have an unlimited fetch.

Table 3.1: Ocean wave classification [46].

Wave type	Wave period
Capillary waves	$< 0.1 sec$
Ultra-gravity waves	$0.1 sec$ to $1 sec$
Ordinary gravity waves	$1 sec$ to $30 sec$
Infra-gravity waves	$30 sec$ to $5 min$
Long-period waves	$5 min$ to $12 hour$
Ordinary tides	$12 hour$ to $24 hour$
Trans-tidal waves	$24 hour >$

Another form of wave classification was developed by the World Meteorological Organization (WMO) [45], known as the sea state code (basically reflecting the Douglas Sea Scale), given in Table 3.2. The WMO sea state code is based on wave amplitude rather than period, and describes the sea from being calm and glassy, to moderate, to rough, through to phenomenal.

Table 3.2: WMO sea state code [45].

WMO Sea State Code	Significant wave height (m)	Sea type
0	0.00	Calm (glassy)
1	0.00 to 0.10	Calm (rippled)
2	0.10 to 0.50	Smooth (wavelets)
3	0.50 to 1.25	Slight
4	1.25 to 2.50	Moderate
5	2.50 to 4.00	Rough
6	4.00 to 6.00	Very rough
7	6.00 to 9.00	High
8	9.00 to 14.00	Very high
9	14.00 >	Phenomenal

Wave Modelling

Assuming a wave is sinusoidal (a regular wave), as shown in Figure 3.1, its amplitude (A_w) describes its maximum elevation displacement (at the crest or trough) from the water level when flat and calm, i.e. the mean water level ($\bar{z}_w$). The wavelength (λ_w) gives the distance from wave crest to crest (or from any point on the wave until its cycle repeats itself), and the angular wave frequency (ω) represents the rate of angular displacement of the wave, or number of radians (within a wave cycle) per second. Given the wave period (T_w), representing the time required for a wave to complete a full cycle, the wave's angular frequency can be calculated as $\omega = \frac{2\pi}{T_w}$. Furthermore, the wave height (H_w) is simply given as twice its amplitude ($H_w = 2A_w$).

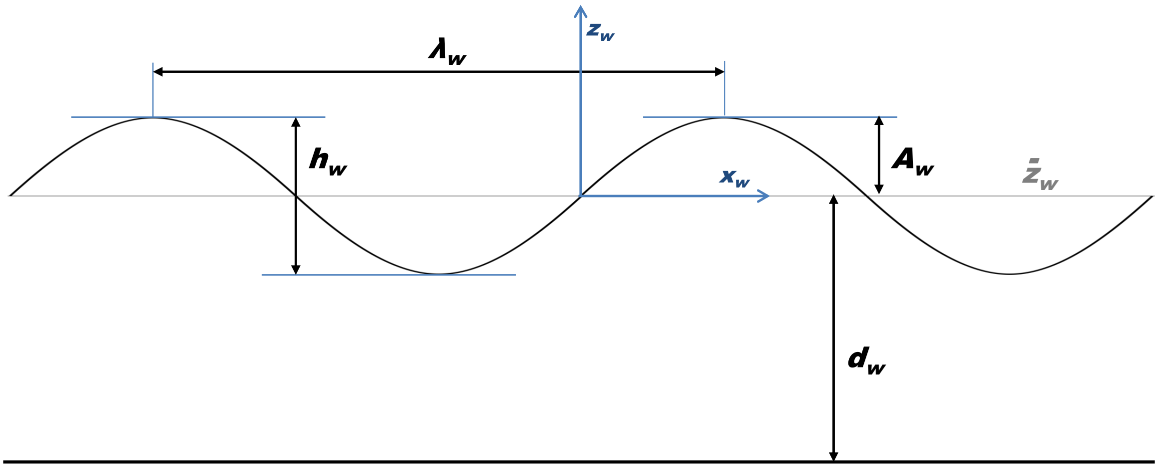

Figure 3.1: Wave properties.

The wave elevation (ζ) at a given point in time, assuming constant wave properties, is calculated as a function of time and distance using Equation 3.1[1], where x_w is the longitudinal distance along a wave, k_w is the wavenumber (radians displaced per unit distance), and t represents time in seconds.

$$\zeta(x_w, t) = A_w \sin(k_w x_w - \omega t) \tag{3.1}$$

A further relationship can be described between wave frequency and wave number, known as the *dispersion relation*. If deep water is assumed, where deep water is defined as being when water particles in motion due to waves do not interact with the sea floor, then the *deep water dispersion relation* can be derived as Equation 3.2. Gravitational acceleration (g) is assumed to be $9.81 m/s^2$ at sea level.

[1]Equations 3.1 to 3.3 were taken from [23].

$$k_w = \frac{\omega^2}{g} \tag{3.2}$$

The state of the sea can seldom be modelled as a single sinusoidal wave. However, the stochastic nature of the ocean surface [47] can be approximated by modelling it as a composition of many sinusoidal waves with varying characteristics. By introducing a wave phase shift (ϵ_w), which describes the relative angular displacement of each wave, an irregular ocean surface can be found using Equation 3.3. The phase shift is considered random in this case, and N_w defines the number of wave components, with n representing the individual components in this equation.

$$\zeta(x_w, t) = \sum_{n=1}^{N_w} A_{w_n} \ \sin(k_{w_n} x_w - \omega_n t + \epsilon_{w_n}) \tag{3.3}$$

Sea states are often characterised in terms of statistical properties, where wave height and period are the most common variables used, specifically the significant wave height (H_s) and mean wave period (T_1) [48]. The significant wave height represents the average height of the tallest one-third of wave heights measured, though also commonly defined as four times the standard deviation of individual wave heights (approximately equivalent). The mean wave period is the average period of the waves measured.

Equation 3.3 presents a uni-directional wave elevation as a time-series within 2D space (expandable to 3D space by replication along the wave surface $y_w - axis$); however, the wave properties can also be represented in the frequency domain, generally in the form of a wave energy density spectrum.

Wave Spectra

In a similar manner to how an irregular wave train is created by combining multiple regular waves of varying frequency (Equation 3.3[2]), the energy in the wave spectrum can be found by breaking it down into its constituent frequency components. The frequency-wise distribution of the wave energy components is called the wave's *energy density spectrum*, or *power density spectrum* (PSD), and can be calculated using Fourier transformations of the wave elevation time-series.

By discretising the spectra into n frequency components, the relationship between the spectral ordinate of a frequency ($S_{\zeta_n}(\omega_n)$) and the wave component's amplitude can be calculated

[2]Equation 3.4 to 3.9 were obtained from [2].

using Equation 3.4, where $(d\omega)$ represents the frequency component.

$$S_{\zeta_n}(\omega_n) = \frac{{A_n}^2}{2\, d\omega} \tag{3.4}$$

Equation 3.4 can then be rearranged to solve for a wave amplitude for each discretised frequency component using Equation 3.5.

$$A_{w_n} = \sqrt{2 S_{\zeta_n}(\omega_n)\, d\omega} \tag{3.5}$$

In order to completely recreate a wave time-series, the phase shift of each frequency component must be known. If the phase shift of each component is selected randomly, the energy of the wave system will be the same, but a derived time-series will be different [49].

The total power of the wave system can also be calculated via integration of the area under a wave PSD, as shown in Equation 3.6.

$$m_{\zeta_0} = \int_0^\infty S_\zeta(\omega)\, d\omega \tag{3.6}$$

Wave spectra have been analytically formulated to represent certain types of sea states. The Modified Pierson-Moskowitz (MPM) represents fully developed seas, parameterised by the significant wave height and average zero-crossing wave period (T_z). Equation 3.7 defines the MPM spectrum, where the variables A_{PM} and B_{PM} are given in Equations 3.8 and 3.9 respectively [2].

$$S_{\zeta_{PM}}(\omega) = \frac{A_{PM}}{\omega^5}\ \exp\left(\frac{-B_{PM}}{\omega^4}\right) \tag{3.7}$$

$$A_{PM} = \frac{H_s^2}{4\pi}\left(\frac{2\pi}{T_z}\right)^4 \tag{3.8}$$

$$B_{PM} = \frac{1}{\pi}\left(\frac{2\pi}{T_z}\right)^4 \tag{3.9}$$

The average zero-crossing wave period is approximately equal to the average time between upward wave crossings of the mean sea level [50]. However, this study focuses on the average wave period (T_1), rather than average zero-crossing wave period, as it is more intuitive, and is easily transformed as input to the wave spectra using $T_z = 0.921 T_1$.

Figure 3.2 provides an example of how the MPM spectra vary with different H_s values, while Figure 3.3 shows how the MPM spectra vary with different T_1 values. Specifically, Figure 3.2 shows spectra parameterised with differing H_s and a fixed T_1 of 8.5s, and Figure 3.3 shows spectra with differing T_1 and a fixed H_s of 1.2m. As can be seen, the impact of an increase in H_s results in an increase in energy for each frequency component. An increase in T_1 also results in increased energy for the frequency components; however, the decrease in T_1 also widens the spectra and shifts it towards larger frequency values.

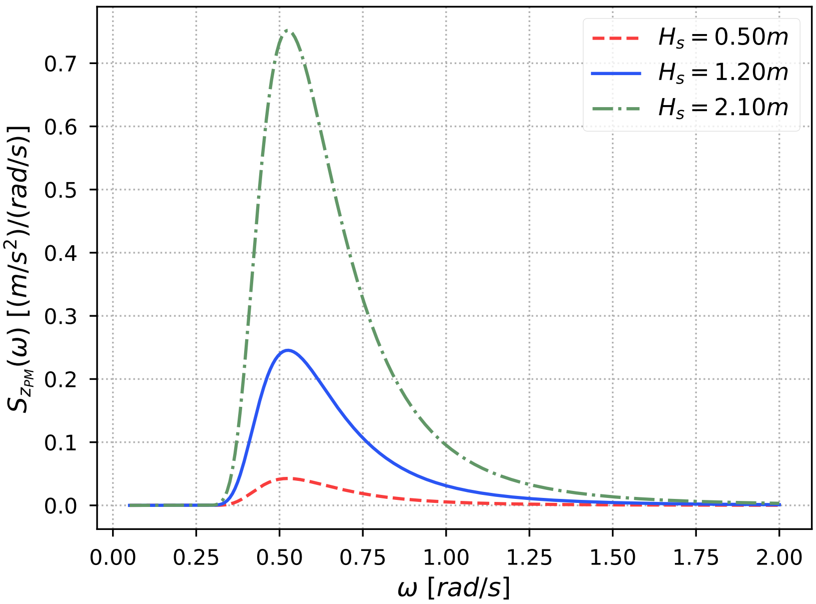

Figure 3.2: Comparing MPM wave spectra for varying H_s and fixed T_1.

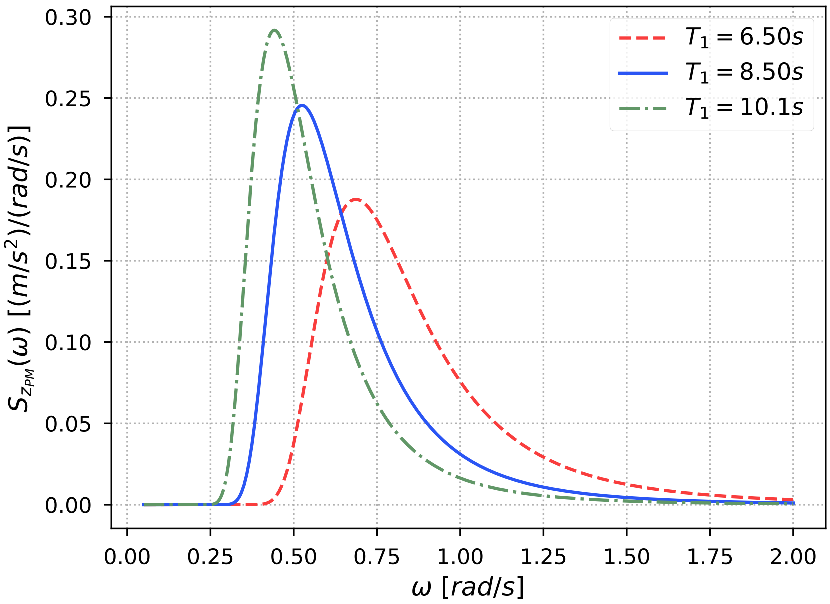

Figure 3.3: Comparing MPM wave spectra for varying T_1 and fixed H_s.

MPM spectra were used to simulate uni-directional wave systems by defining a range of wave frequency components with randomised phase shifts, then Equation 3.5 was used to convert the spectra to wave amplitudes, and summed using Equation 3.3 to develop wave time-series. Figure 3.4 gives an example of a randomly generated wave elevation time-series from MPM spectrum with an H_s of $1.2m$ and a T_1 of $8.5s$ for $1200s$ using this procedure.

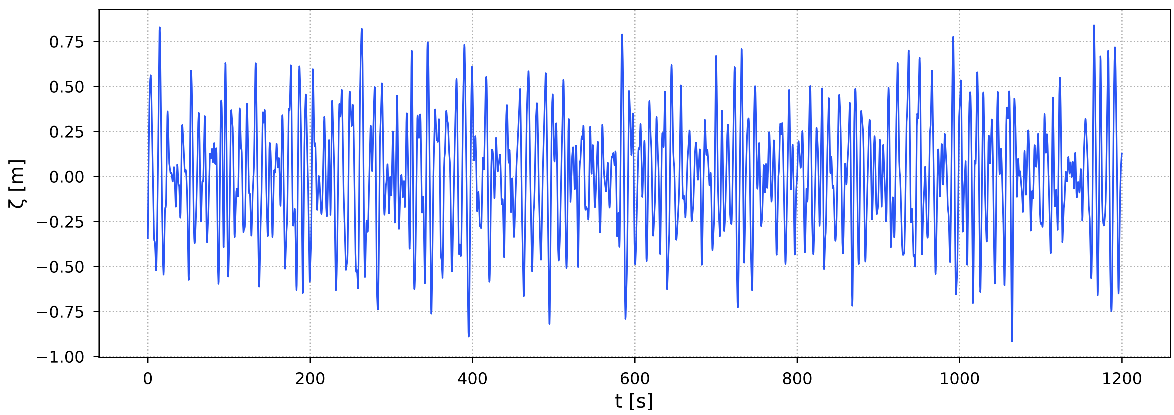

Figure 3.4: Wave elevation generated from an MPM spectrum with $H_s = 1.2m$ and $T_1 = 8.5s$.

3.2.2 Ship Model

A Wave Glider (WG) is a low-profile uninhabited surface vessel which uses an underwater glider, a simple frame with mounted wings and rudder, attached by an umbilical cable, to convert wave energy into propulsive power [26]. WGs use top-mounted solar arrays to power their onboard systems, and as a means of backup, battery stored energy. WGs are, therefore, powered entirely by renewable energy, giving them a very large range and long endurance. WGs are able to propel itself by taking advantage of the differential movement between their float and subsurface components. One limitation to the WG is the necessity of regular sun exposure to maintain power levels, as well as a minimum water depth requirement of $10m$. A WG was used for the GNSS geodetic SSE research done by Maqueda et al. [34] and Penna et al. [33].

The increased endurance and range of WGs provide an advantage for SSE in that they are able to survey a larger area, or survey the same area for a longer period of time, than traditional USVs. As such, it was decide to use a WG as the vehicle model for this study, using the dimensions given in [26]. Further, while larger ships act as low-pass filters, being less responsive to high-frequency waves [25], the small scale of the vessel means that it should be less affected by this. However, to reduce the complexity of the model, the WG profile was assumed to only include the float, ignoring the effects of the glider and tether, where the float is shaped much like a surfboard (as can be seen in the example of a WG given in Figure 3.5[3]).

The omission of the effects of the glider for simplicity was done as this study is a proof of concept for the sea state estimation methodology. It should be noted that the dynamics of the glider would introduce another source of motion affecting the float, and should be considered in higher fidelity studies when using the WG for SAWB. Further, the inclusion of the extra DOFs of the glider could actually be harnessed for improved SSE by training an SSE model which takes into account subsurface wave induced motion of the glider as well as surface wave induced motion of the float.

Table 3.3 outlines the dimension and mass parameters of the WG float used for this project.

[3]Creative Commons licence: creativecommons.org/licenses/by/2.0/deed.en

Table 3.3: Wave glider float parameters.

Parameter	Value
Mass (m_π)	$55kg$
Length (L_π)	$2.05m$
Breadth (B_π)	$0.61m$
Draught (T_π)	$0.16m$
Water plane coefficient (C_{wp_π})	0.877
Block coefficient (C_{b_π})	0.731
Displacement (Δ_π)	$150kg$
Transverse metacentric height (GM_{T_π})	$0.264m$

Additionally, it should be noted that WGs are generally slow moving, reaching maximum speeds between $1m/s$ and $1.5m/s$ [51]. However, this study aims to be a baseline for future studies on using USVs for SSE, some of which are capable of achieving speeds much greater than $1.5m/s$ (for example, the SeaFox USV can reach speeds of up to $20m/s$ [52]). Kessel [51] discussed ways in which the maximum speed of WGs could be increased, primarily by increasing the size of the float, while modifying their design could further increase the maximum speed of WGs. Therefore, it was decided to increase the maximum simulated speed of the WG to $5m/s$ so that the SSE results could more easily be extrapolated to other USVs, while it is assumed that such speeds could be attained by future WG designs.

Figure 3.5: Wave glider float image modified from [53].

3.2.3 Ship Dynamics and Transfer Functions

A vessel has six degrees of freedom, three translational and three rotational. Each of the degrees of freedom are aligned with an axis (x, y, z), which in this case are assumed to be in relation to the north-east-down (NED) reference frame. In translation, a ship may displace along the x-axis, known as surge (x_π), the y-axis, called sway (y_π), and the z-axis, termed heave (z_π). The Euler angles of roll, pitch, and yaw are used to describe the angular rotation about each axis. Roll (ϕ_π) gives rotation about the x-axis, pitch (θ_π) about the y-axis, and yaw (ψ_π) about the z-axis [54]. The subscript π in this section refers to the ship-based variables and parameters.

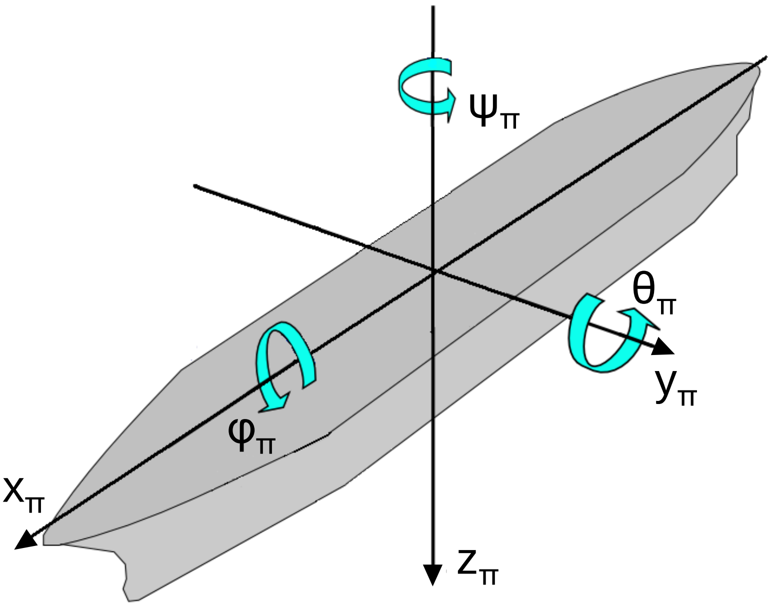

Figure 3.6: Vessel DOFs.

As discussed in Section 2.2, while the use of 6-DOF responses is possible for SSE, it is common to only use three. However, most of the previous literature on using ML for SAWB included six (or more) DOFs. It was chosen to reduce this to three DOFs in order to investigate how much information could be extracted from them when using a NN for analysis, while simultaneously reducing the memory and processing load of each vessel for the ultimate goal of guiding multiple vessels. Therefore, heave, pitch, and roll have been selected as the response DOFs used to interpret wave properties.

When a ship has zero velocity, the incidental wave frequency relates directly to the ship response frequency; however, when the ship is in motion, the excitation frequency becomes dependent on the wave *encounter frequency* (ω_e). The encounter frequency, calculated in Equation 3.10, is a function of the speed of the vessel (U_π) and relative wave heading angle (μ_h) between a ship's body-fixed axes and the direction of wave propagation [25]. Further, encounter frequency is a function of wave number, which is directly related to the speed of wave propagation. For further information on the derivation of wave encounter frequency, please refer to [55].

$$\omega_e = \omega - k_w U_\pi cos(\mu_h) \equiv \alpha_\Phi \omega \tag{3.10}$$

The relative wave heading angle describes what sort of impact waves have on ship motion. Figure 3.7 defines the terminology used for the various incoming wave heading angles in this chapter. If the ship is moving with a similar heading to the direction of the waves ($\mu_h \in [0°, 60°]$), then the ship experiences a *following sea*. If the ship has a heading which opposes the heading of the wave ($\mu_h \in (120°, 180°]$), then it is experiencing a *head sea*. If the ship is traveling in a direction approximately perpendicular to the waves ($\mu_h \in [60°, 120°]$), then it is in a *beam sea*. Wave heading classifications can further be broken up into bow sea (in between beam sea and head sea), and quartering sea (in between beam sea and following sea), but will not be addressed in this study.

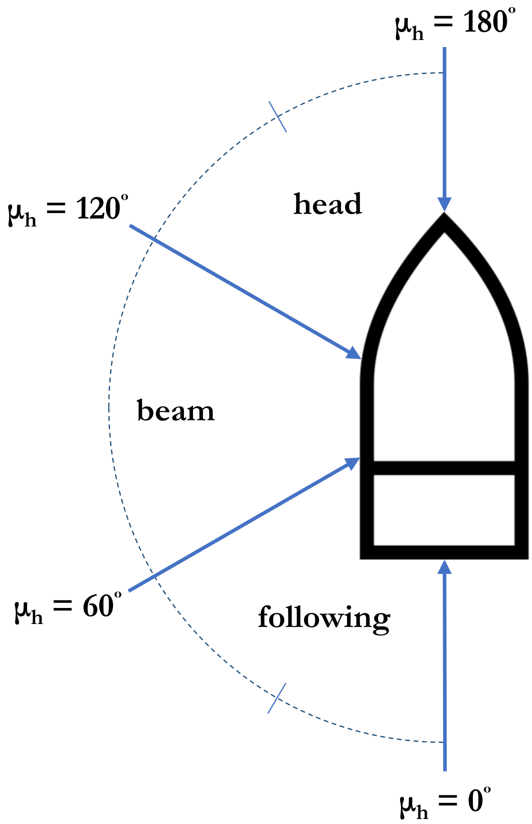

Figure 3.7: Relative wave heading angles.

Wave encounter frequency can be better understood by examining Figure 3.7. As the blue arrows indicated the direction of wave propagation, and the vessel is considered as moving only with forward speed, it can be seen that the wave components encountered will not equate those encountered when the vessel is stationary and merely responding to the motion of the waves in the direction of the wave propagation. A ship travelling in a head sea would have a higher encounter frequency than the incidental wave frequency, while a ship travelling in a following sea would have a lower encounter frequency than the incidental wave frequency, for example (unless it has a sufficiently high speed).

Each degree of freedom of a ship can be approximated as a mass-spring-damper system, where the 'spring' stiffness, or restoring force, is caused by the buoyancy force imparted on the ship by the water, and the system is hydrodynamically damped due to the viscosity of the water [49] (alongside other damping sources, such as wave radiation).

The equation of motion (EOM) of a simple damped point mass spring system is given in Equation 3.11, where c_s is the spring stiffness, b_s is the damping coefficient, x_s is the distance the spring is displaced from its neutral position, and its first and second derivatives are the velocity ($\dot{x}_s$) and acceleration ($\ddot{x}_s$) of the mass (including gravitational acceleration).

$$0 = m\ddot{x}_s + b_s\dot{x}_s + c_s x_s \tag{3.11}$$

However, ships are under almost constant excitation due to the external forces caused by wave motion. When considering only a regular wave, then the excitation force of a wave is given as Equation 3.12. F_{0_w} is known as the forcing amplitude and is related to the causal wave amplitude, while ω_e is the encountered wave frequency (discussed below).

$$F_w(t) = F_{0_w}\ sin(\omega_e t) \tag{3.12}$$

Therefore, the EOM of the mass-spring-ship system becomes Equation 3.13.

$$m\ddot{x}_s + b_s\dot{x}_s + c_s x_s = F_{0_w}\ sin(\omega_e t) \tag{3.13}$$

The terms of the EOM can be grouped and classified as the: inertial term ($m\ddot{x}_s$), damping term ($b_s\dot{x}_s$), stiffness term ($c_s x_s$), and excitation force term ($F_0\ sin(\omega_e t)$).

By harnessing the semi-analytical approach developed by Jensen, Mansour, and Olsen [56], this mass-spring-damper approximation can be expanded further to calculate ship-specific responses to wave excitation. The equations derived by [56] only require the following vessel-specific information: vessel length, breadth, draught, displacement, block coefficient, water plane area, and transverse metacentric height (given in Section 3.2.2). Previous SSE studies using SAWB which have simulated ship responses using the equations derived in [56] include [38], [43], [57], [58].

The closed-form expressions presented in [56] provide simplified representations of the complex interactions between monohull vessels and encountered waves. The derived transfer functions were designed for "the early stages of design", and the responses of the vessel should not be considered as highly accurate. However, as this thesis presents an early stage design of an SSE technique, as well as a vessel guidance approach, the equations have been deemed appropriate for the application. Further, should the NNs designed to estimate wave properties be able to extract sufficient information from the responses obtained from these simplified transfer functions, it is predicted that experimental response data, or responses simulated using more complex transfer functions, would provide even more information for SSE.

However, one major limitation is introduced by Jensen, Mansour, and Olsen's model. As the vessel response DOFs are assumed to be decoupled, information about DOF cross-spectra information cannot be extracted. Therefore, due vessel lateral-plane symmetry, differentiation between port and starboard wave headings cannot be made.

The transfer functions for heave and pitch are first derived together, then the transfer function for roll is derived separately as it uses a slightly different set of assumptions.

Heave and Pitch

By assuming that pitch and heave motion are uncoupled, that the vessel has a constant sectional added mass equal to the displaced water, and that the vessel is box-shaped and homogeneously loaded (which is a reasonable approximation for WG floats), the vertical EOMs for a vessel in regular waves are given in Equations 3.14 and 3.15 respectively.

$$\left(\frac{2k_w T_\pi}{\omega^2}\right)\ddot{z}_\pi + \left(\frac{Q_\pi^2}{k_w B_\pi \alpha_\Phi^3 \omega}\right)\dot{z}_\pi + z_\pi = A_w F_{0_z}\ cos(\omega_e t) \tag{3.14}$$

$$\left(\frac{2k_w T_\pi}{\omega^2}\right)\ddot{\theta}_\pi + \left(\frac{Q_\pi^2}{k_w B_\pi \alpha_\Phi^3 \omega}\right)\dot{\theta}_\pi + \theta_\pi = A_w G_{0_\theta}\ sin(\omega_e t) \tag{3.15}$$

The inertial terms of the vessel EOMs encompass not only the mass moment of inertia of the vessel itself, but also the added mass of the water displaced by the vessel in motion.

The parameter α_Φ is calculated using Equation 3.16, where the *Froude number* is calculated as $Fn = U_\pi/\sqrt{gL_\pi}$, and is related to the encounter frequency by $\omega_e = \alpha_\Phi \omega$.

$$\alpha_\Phi = 1 - Fn\sqrt{k_w L_\pi} cos(\mu_h) \tag{3.16}$$

The sectional hydrodynamic damping coefficient Q_π is modelled as a dimensionless ratio between incoming and diffracted wave amplitudes in Equation 3.17 (derived in [59]).

$$Q_\pi = 2\ \sin\left(\frac{\omega_e^2 B_\pi}{2g}\right)\ \exp\left(-\frac{\omega_e^2 T_\pi}{g}\right) = 2\ \sin\left(\frac{1}{2}k_w B_\pi \alpha_\Phi^2\right)\ \exp\left(-k_w T_\pi \alpha_\Phi^2\right) \tag{3.17}$$

The forcing functions F_{0_z} and G_{0_θ} are then given in Equations 3.18 and 3.19, where the encounter wave number is $k_e = |k_w \cos(\mu_h)|$.

$$F_{0_z} = \kappa_\Phi f_\Phi \frac{2}{k_e L_\pi}\ \sin\left(\frac{k_e L_\pi}{2}\right) \tag{3.18}$$

$$G_{0_\theta} = \kappa_\Phi f_\Phi \frac{24}{(k_e L_\pi)^2 L_\pi}\ \left[\sin\left(\frac{k_e L_\pi}{2}\right) - \frac{k_e L_\pi}{2}\ \cos\left(\frac{k_e L_\pi}{2}\right)\right] \tag{3.19}$$

The parameter f_Φ is defined by Equation 3.20, and the Smith correction factor is calculated as $\kappa_\Phi = \exp(-k_e T_\pi)$.

$$f_\Phi = \sqrt{(1 - k_w T_\pi)^2 + \left(\frac{Q_\pi^2}{k_w B_\pi \alpha_\Phi^3}\right)^2} \tag{3.20}$$

The solutions to Equations 3.14 and 3.15 result in the frequency response functions (FRFs), which describe the frequency-wise response of a vessel to wave excitation for heave ($\Phi_z = \eta_\Phi F_{0_z}$) and pitch ($\Phi_\theta = \eta_\Phi G_{0_\theta}$), where η_Φ is shown in Equation 3.21.

$$\eta_\Phi = \left(\sqrt{(1 - 2k_w T_\pi \alpha_\Phi^2)^2 + \left(\frac{Q_\pi^2}{k_w B_\pi \alpha_\Phi^2}\right)^2}\right)^{-1} \tag{3.21}$$

The FRFs vary with vessel speed and wave heading, where examples of heave and pitch FRFs for the WG are given in Figures 3.8 and 3.9, respectively. The left plots give an example of how the FRFs change with varying wave headings (and a forward speed of $2m/s$), while the right plots show how the FRFs change as the vessel speed is varied (for a fixed wave heading of 130°).

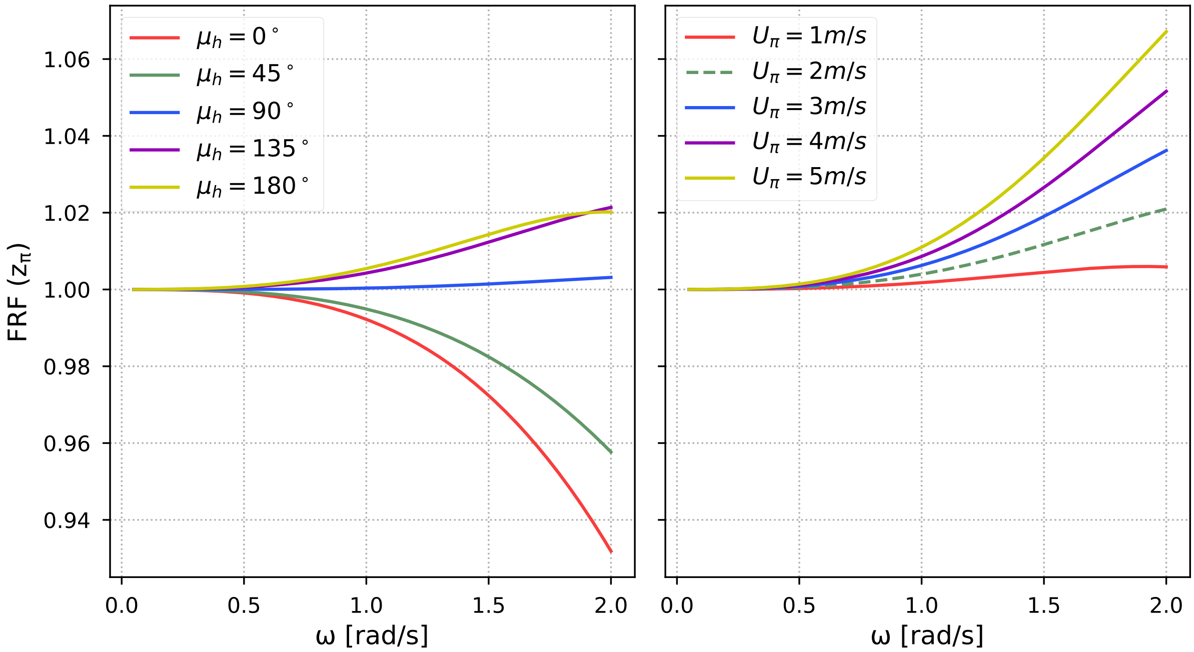

Figure 3.8: Heave FRF versus wave frequency for varying μ_h at $U_\pi = 2m/s$ (left) and for varying U_π at $\mu_h = 130°$ (right).

While it can be seen that heave responses vary a lot depending on the vessel's speed, the pitch response varies very little with a change in forward speed. As is expected, there is no pitch response for a wave heading of 90° as the vessel axis of rotation (y_π for pitch) is perpendicular to the heading of the waves.

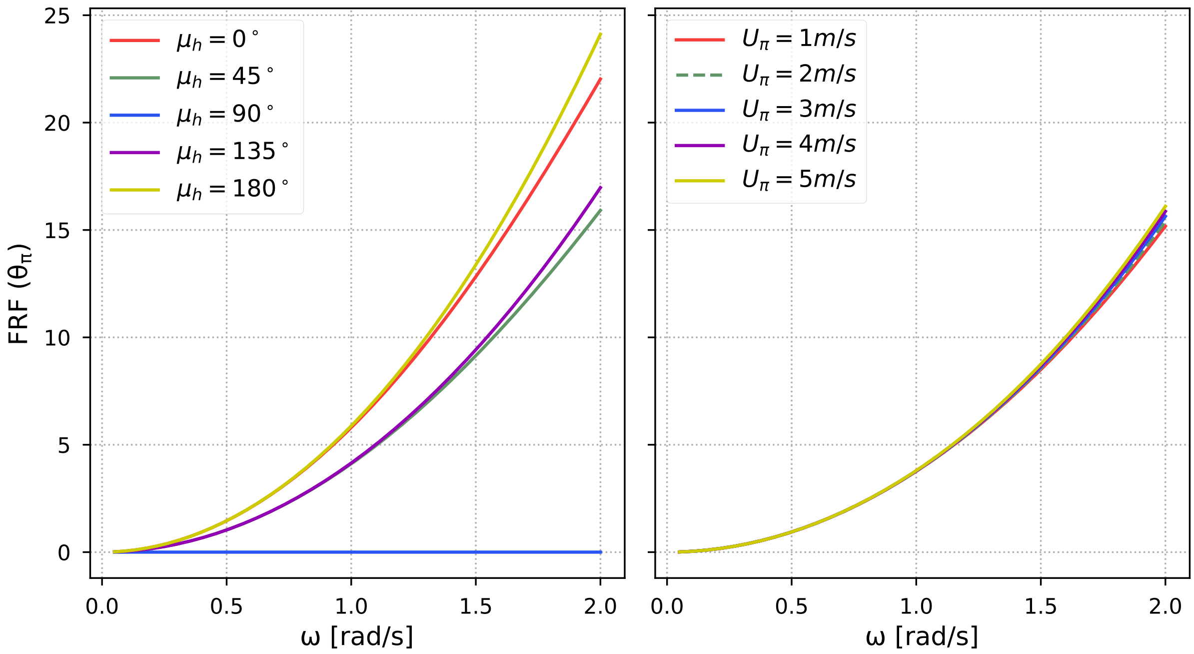

Figure 3.9: Pitch FRF versus wave frequency for varying μ_h at $U_\pi = 2m/s$ (left) and for varying U_π at $\mu_h = 130°$ (right).

The heave and pitch time-series responses for the WG moving at $2m/s$ with a wave heading of 130°, resulting from the FRFs in Figures 3.8 and 3.9 (shown as the dashed green line) using the same sea state as generated in Figure 3.4, are given in Figures 3.10 and 3.11.

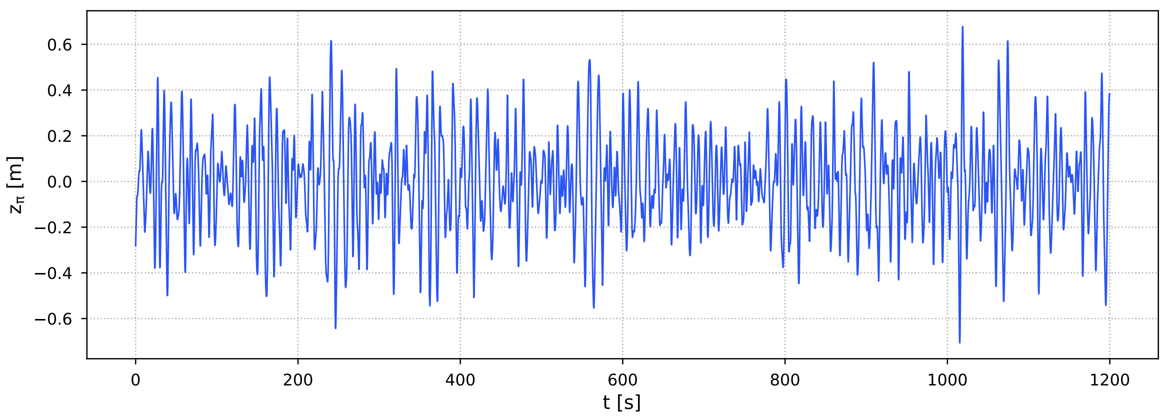

Figure 3.10: Heave time-series response to wave input with $H_s = 1.2m$ and $T_1 = 8.5s$.

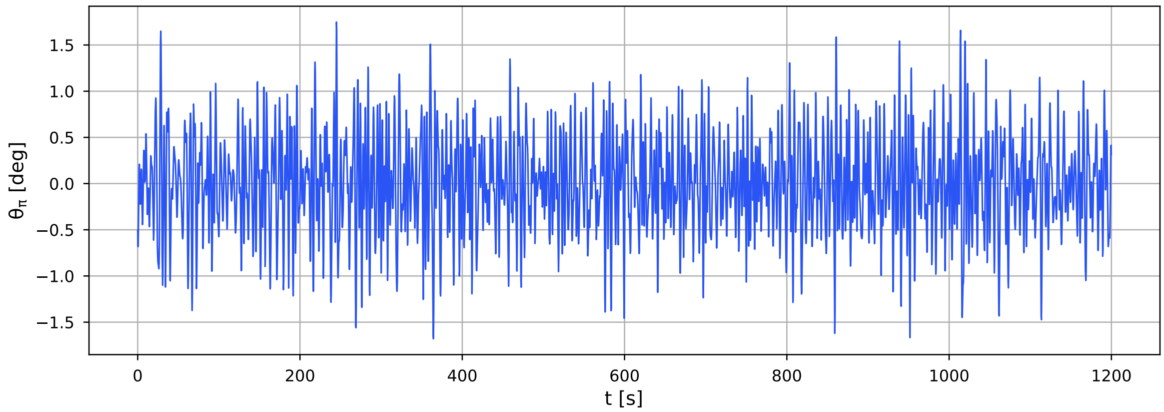

Figure 3.11: Pitch time-series response to wave input with $H_s = 1.2m$ and $T_1 = 8.5s$.

Roll

The EOM for roll, assuming roll is decoupled from other vessel responses, can be derived as Equation 3.22.

$$\left(\frac{T_{N_\phi}}{2\pi}\right)^2 C_{\pi_{44}}\ddot{\phi} + B_{\pi_{44}}\dot{\phi} + C_{\pi_{44}}\phi = M_\phi \tag{3.22}$$

Here, T_{N_ϕ} represents the natural period for roll, which was calculated using Equation 3.23 (from [60]). As the natural period replaces the mass moment of inertia and added mass terms, the variation in added mass with respect to wave frequency has been ignored.

$$T_{N_\phi} = \frac{\sqrt{gGM_{T_\pi}}}{2\sqrt{Ix_\pi/\Delta_\pi}}2\pi \tag{3.23}$$

The restoring moment coefficient is calculated as $C_{\pi_{44}} = g\Delta_\pi GM_{T_\pi}$, where Δ_π is the vessel displacement, and GM_{T_π} is the transverse metacentric height, given in Table 3.3.

In order to calculate the hydrodynamic damping coefficient of the vessel ($B_{\pi_{44}}$), [56] assumes that the vessel is comprised of two prismatic sections of equal draught, but with different breadths ($B1_\pi$ and $B2_\pi$) and corresponding cross-sectional areas $A1_\pi$ and $A2_\pi$, which are visualised in Figure 3.12 (replicated from [56]). As such, the vessel model is more complex for roll than heave and pitch, which is required to capture the desired roll characteristics.

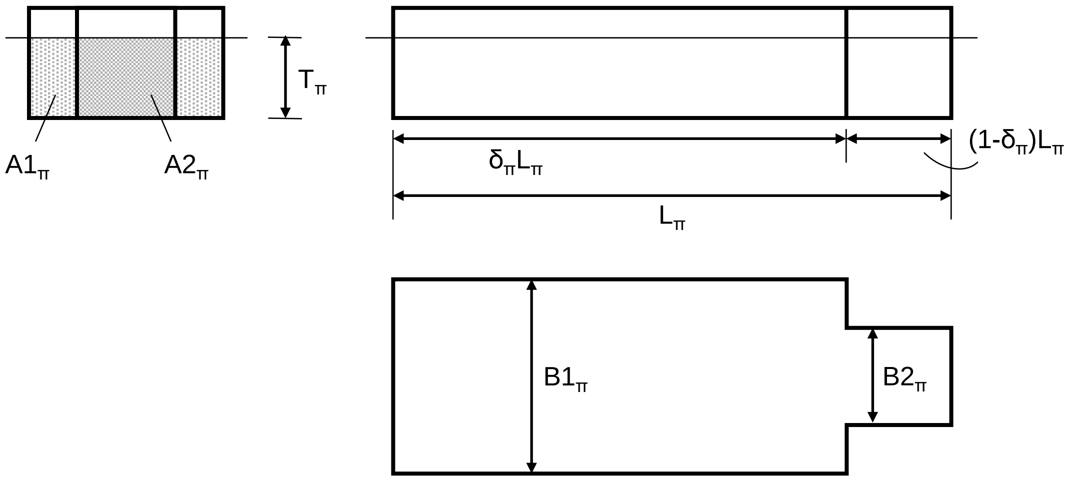

Figure 3.12: Simplified vessel model used for calculation of roll motion analysis (replicated from [56]). The vessel is modelled as two rectangular prisms with equal draft but varying breadths, where γ_π is the ratio of the breadths.

The ratio γ_π is found such that the water plane area coefficient is equal for the actual vessel and dual beam model (Equation 3.24), while δ_π can be seen in Figure 3.12. For this study, a δ_π value of 0.7 was assumed.

$$\gamma_\pi = \frac{B1_\pi}{B2_\pi} = \frac{C_{WP_\pi} - \delta_\pi}{1 - \delta_\pi} \tag{3.24}$$

Similarly, the area $A1_\pi$ is derived such that the block coefficient (C_{b_π}) is equal for the real vessel and dual beam model (Equation 3.25), and $A2_\pi = \gamma_\pi A1_\pi$.

$$A1_\pi = \frac{C_{b_\pi} B1_\pi T_\pi}{\delta_\pi + \gamma_\pi(1 - \delta_\pi)} \tag{3.25}$$

Jensen, Mansour, and Olsen derived an expression (Equation 3.26) for the sectional inviscid hydrodynamic coefficient ($b_{\pi_{44}}$) using the Frank close fit method, where $b_{\pi_{44}}$ is a function of the encounter frequency and vessel parameters: breadth, draught, and water plane area.

$$\frac{b_{\pi_{44}}}{\rho_w A_{WP_\pi} B_\pi^2}\sqrt{\frac{B_\pi}{2g}} = b1(B_\pi/T_\pi)\ \exp\left[b2(B_\pi/T_\pi)\omega_e^{-1.3}\right]\omega_e^{b3(B_\pi/T_\pi)} \tag{3.26}$$

The constants $b1$, $b2$, and $b3$ are calculated as a function of breadth and draught (Equations 3.27 to 3.29), assuming that $3 \leq (B_\pi/T_\pi) \leq 6$, which is valid for the WG.

$$b1 = 0.256(B_\pi/T_\pi) - 0.286 \tag{3.27}$$

$$b2 = -0.11(B_\pi/T_\pi) - 2.55 \tag{3.28}$$

$$b3 = 0.033(B_\pi/T_\pi) - 1.419 \tag{3.29}$$

The total hydrodynamic damping coefficient is then calculated in Equation 3.30 using the ratio of the two sectional damping coefficients (b_{44_1} and b_{44_2}).

$$B44 = L_\pi b_{44_1}\left[\delta_\pi + \left(\frac{b_{44_2}}{b_{44_1}}\right)^2 (1-\delta_\pi)\right] \tag{3.30}$$

The roll excitation moment (M_ϕ) can then be calculated using Equation 3.31, with a full derivation given in [56].

$$\begin{aligned} M_\phi = &\sin(\mu_h)\sqrt{\frac{\rho_w g^2}{\omega}}\frac{1}{k_w\cos(\mu_h)}\ \exp(i\omega t)\ \times\ \{\sqrt{b_{44_1}}\sin(\delta_\pi L_\pi k_w \cos(\mu_h)) \\ &+ 2\sqrt{b_{44_2}}\ \cos(0.5(1+\delta_\pi)L_\pi k_w\cos(\mu_h))\ \sin(0.5(1-\delta_\pi)L_\pi k_w\cos(\mu_h)) \\ &+ i[\sqrt{b_{44_1}}(\cos(\delta_\pi L_\pi k_w\cos(\mu_h)) - 1) + 2\sqrt{b_{44_2}}\sin(0.5(1+\delta_\pi)L_\pi k_w\cos(\mu_h)) \\ &\times \sin(0.5(1-\delta_\pi)L_\pi k_w\cos(\mu_h))]\} \end{aligned} \tag{3.31}$$

The FRF for roll motion is found in Equation 3.32.

$$\Phi_\phi = \frac{|M_\phi|}{\left(\left[\omega_e^2(T_{N_\pi}/2\pi)^2+1\right]^2 C_{\pi_{44}}^2 + \omega_e^2 B_{\pi_{44}}^2\right)^{1/2}} \tag{3.32}$$

The absolute roll moment must be found by taking the real part of the moment, as shown in Equation 3.33.

$$\begin{aligned} Re(M_\phi) = |M_\phi|\cos(\omega t+\epsilon) = |\sin(\mu_h)|\sqrt{\frac{\rho_w g^2}{\omega}}\frac{2}{k_e} \\ \times\sqrt{b_{44_1}}[\sin^2(0.5\delta_\pi L_\pi k_e) + \left(\frac{b_{44_2}}{b_{44_1}}\right)^2\sin^2(0.5(1-\delta_\pi)L_\pi k_e) \\ + 2\left(\frac{b_{44_2}}{b_{44_1}}\right)\sin(0.5\delta_\pi L_\pi k_e)\ \sin(0.5(1-\delta_\pi)L_\pi k_e)\ \cos(0.5L_\pi k_e)]^{1/2} \\ \cos(\omega t+\epsilon) \end{aligned} \tag{3.33}$$

Example roll FRFs for the WG with varying wave headings and a forward speed of $2m/s$ (left plot), and varying vessel speed for a fixed wave heading of 130° (right plot) is shown in Figure 3.32. Similarly to pitch, the FRFs for roll do not change greatly as speed is modified. When considering the relative wave heading, there is no roll response to the waves when $\mu_h = 0°$ or $\mu_h = 180°$ as the axis of rotation (x_π) is perpendicular to the wave heading at these angles.

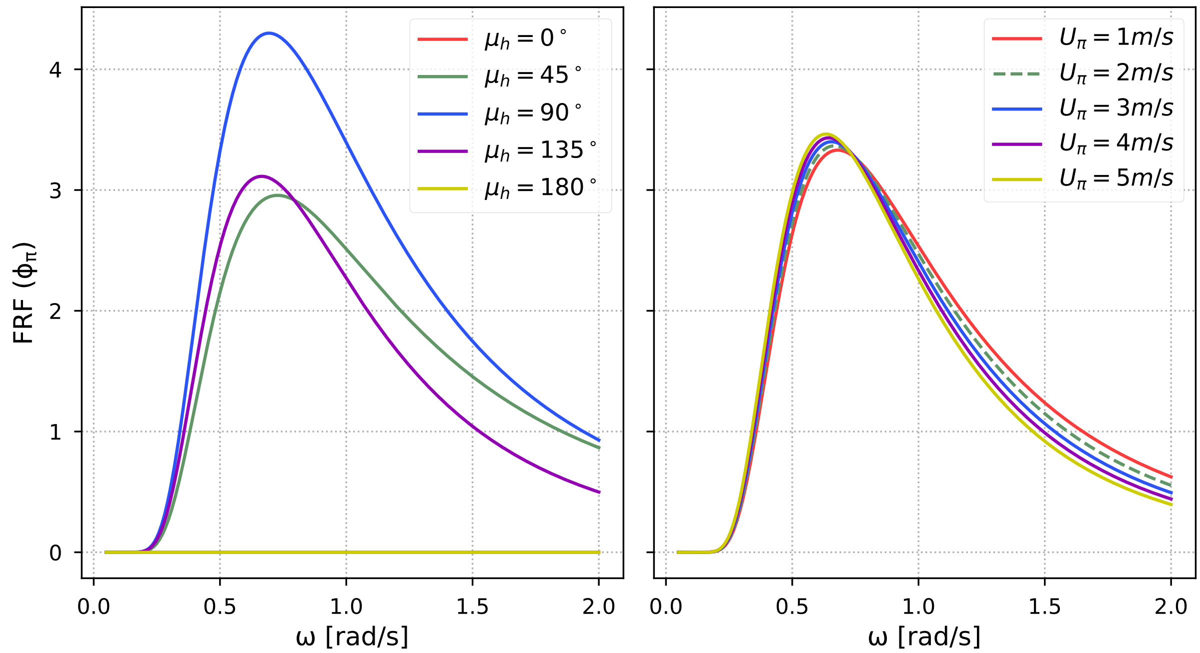

Figure 3.13: Roll FRF versus wave frequency for varying μ_h at $U_\pi = 2m/s$ (left) and for varying U_π at $\mu_h = 130°$ (right).

The example roll time-series response for the WG moving with a speed of $2m/s$ and at a wave heading of 130°, given in Figure 3.14, was generated using the FRF shown in dashed green in Figure 3.32 and the sea state in Figure 3.4 (with $H_s = 1.2m$ and $T_1 = 8.5s$).

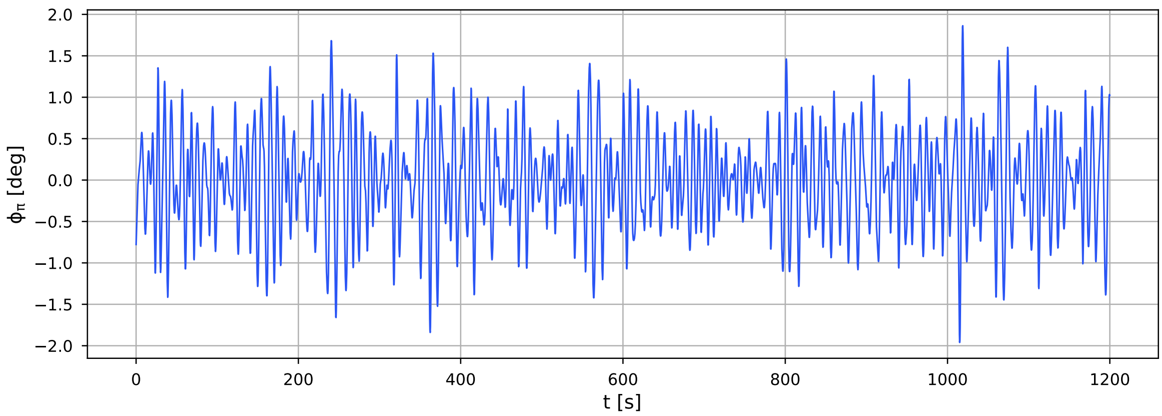

Figure 3.14: Roll time-series response to wave input with $H_s = 1.2m$ and $T_1 = 8.5s$.

Response Power Spectral Density

One way to statistically analyse a signal, in this case the time-series vessel responses, is to transform the time-series into its PSD representation [61]. The time-series are generated by using the FRFs to translate wave motion to vessel motion in the heave, pitch, and roll DOFs. As discussed in Section 3.2.1, a PSD describes how power in the time-series is distributed across frequency components. In order to estimate a PSD from a discrete time-series, a discrete Fourier transform (DFT) can be used, where the fast Fourier transform (FFT) is a one of the most popular and computationally efficient DFTs [62].

As discussed by Nielsen [22], some type of smoothing function is desirable when transforming a vessel time-series response to the frequency domain due to the volatility of a direct DFT. As such, it was decided to use Welch's method to estimate the PSD of the vessel response time-series. Welch's method is based on Bartlett's method, designed to reduce noise within a signal while also reducing the frequency resolution [63]. The input signal is divided into K_{wel} segments, each containing M_{wel} data points, where the segments overlap each other, shifted by S_{wel} data points. Thus, the percentage of overlap can be calculated as $100[(M_{wel} - S_{wel})/M_{wel}]$. Segments are then windowed, and an FFT is performed for each segment. The average value of each segment is then taken for the frequency and corresponding spectral ordinate. For a more in-depth description of Welch's method, refer to [63].

In order to extract important features of the vessel response PSDs, an iterative process was followed to configure SciPy's Welch function [64]. A number of different window types have been developed for Welch's method, where the most common are rectangular windows [65]. One of the most popular, the *Hann* window, was chosen for this analysis. Fifteen windows were used for heave and roll transformations, with only 13 used for the pitch transformation. Likewise, the same 50% overlap between windows was selected for heave and roll, while 80% was used for pitch. The Welch parameters were selected following experimentation with each of them to achieve optimal results.

After transforming the signal into a PSD via Welch's method, the spectral ordinates with the greatest values and their corresponding frequency components were extracted to be used as input for the SSE ML algorithms. Furthermore, in a similar manner to the study by Arneson, Brodtkorb, and Sørensen [17], the PSD was integrated to get the total power of the response spectra to be used as input to the NNs. Equation 3.34 gives an example of the calculation of the total power of a response PSD spectra for heave, with the same process followed for pitch and roll.

$$m_{z_0} = \int_0^{\infty} S_z(\omega)\, d\omega \tag{3.34}$$

Figures 3.15 to 3.17 give examples of the PSDs produced using the time-series signals from Figures 3.10, 3.11, and 3.14. The raw PSD is shown in green, the Welch approximation is given in blue, and the highest energy components selected as input for the NN are highlighted in red. The left plots in Figures 3.15 to 3.17 give an example of the 30 PSD components using the Welch method with the highest energy, while the right plots show the 80 components with the highest energy.

As can be seen, the raw PSD is highly volatile, making feature extraction more difficult, while the smoothed PSD derived by Welch's method provides a good overall approximation of the power distribution. It is hypothesised that the components with the greatest energy of the PSD constitute the most important response information, providing feature-extraction prior to SSE NN training, which reduces the overall complexity required for the NNs, as well as reducing the computational time to train and use the networks.

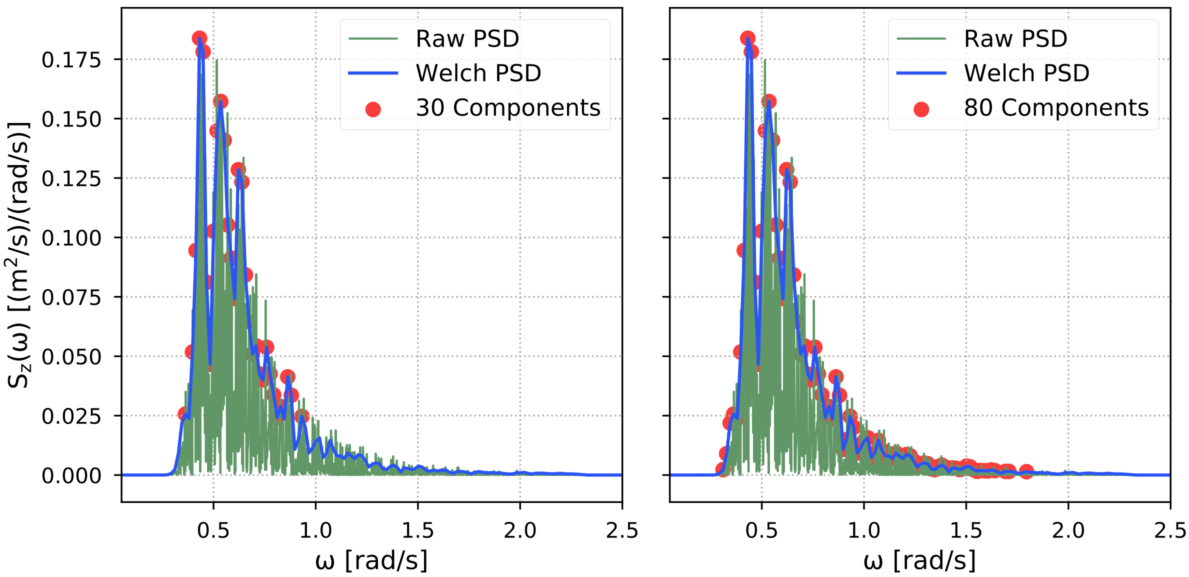

Figure 3.15: Heave response PSD ($U_\pi = 2m/s$, $\mu_h = 130°$, $H_s = 1.2m$, $T_1 = 8.5$).

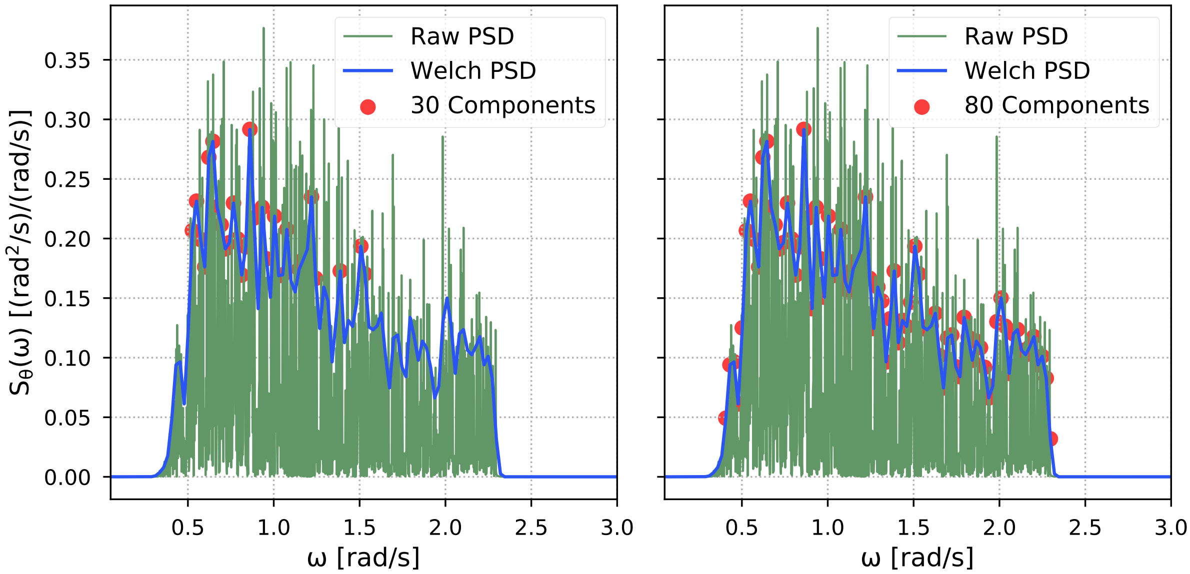

Figure 3.16: Pitch response PSD ($U_\pi = 2m/s$, $\mu_h = 130°$, $H_s = 1.2m$, $T_1 = 8.5$).

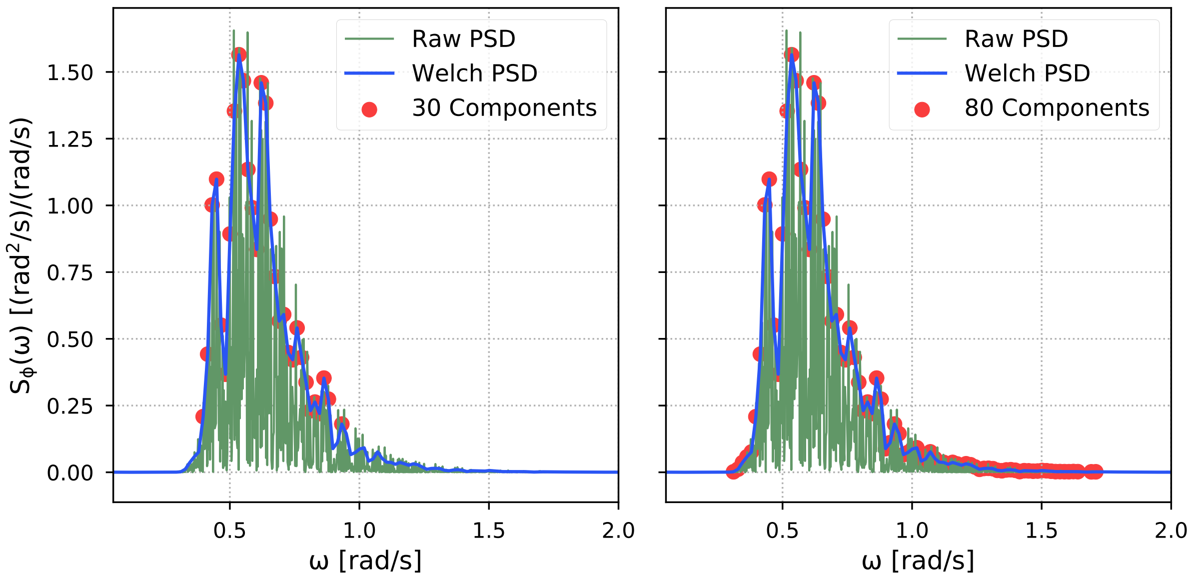

Figure 3.17: Roll response PSD ($U_\pi = 2m/s$, $\mu_h = 130°$, $H_s = 1.2m$, $T_1 = 8.5$).

3.2.4 Sea State Estimation

In order to estimate sea states, vessel motion responses were analysed for different sea and vessel states so that they could be used as input for artificial NNs, which were designed to output the significant wave height, mean wave period, and relative wave heading angles.

To summarise, the vessel motion formulations developed by Jensen, Mansour, and Olsen [56] were used to simulate the motion responses of the WG, which were then transformed into PSDs, and top spectral components and the total power of the spectra extracted from each response. Following experimentation, it was shown that 30 PSD components provided enough information to estimate significant wave height and mean wave period well, while 80 PSD components performed better when estimating the wave heading.

The response data was then combined with the vessel's speed (assumed to be known) and used as input for the different NN architectures, where each architecture only output one parameter: H_s, T_1, or μ_h.

While other NN-based SAWB SSE techniques used raw time-series data as input, it was hypothesised that the manual feature extraction process described could prove a better means to estimate wave properties continually in real time. This is hypothesised as frequency domain responses provide a more statistical representation of the data.

In order to build the NN for SSE, a significant amount of training data was required. Therefore, it was decided to run 48000 simulations, each lasting for 40 minutes (2400 seconds), broken into 8192 time steps (a factor of 2 for the Fourier transformation). Each of the simulations was run using uniformly sampled variables from Table 3.4.

For each simulation, an MPM sea spectrum was generated and transformed into a wave system in time and space, where the spectra were comprised of 500 wave frequencies ranging from $0.5rad/s$ to $2.0rad/s$ (where a previous study on SSE using SAWB also used 500 wave frequency components to generate their parametric wave spectra [57]), then combined with the FRFs from [56] to produce the vessel response time-series. The wave conditions simulated correspond to ordinary gravity waves based on the ocean wave classification scale from [46], and Sea State Codes 3 to 4 (slight to moderate) based on the WMO scale [45].

Table 3.4: Simulation variable ranges

Variable	Range
Significant wave height	$0.5m$ to $2.5m$
Mean wave period	$4s$ to $13s$
Relative wave heading	0° to 180°
Vessel speed	$0m/s$ to $5m/s$

However, in reality no system runs perfectly. While the response time-series for this study were simulated, it is intended as a benchmark for potential future implementation in an actual USV. As such, the time-series responses would normally be obtained using some type of sensor.

For this reason, it was decided to add noise into the resulting response signals in order to accommodate more realism. The level of noise present in the inertial measurement unit of a waverider buoy was selected as likely candidate for a motion sensor on a WG. Table 3.5 gives the standard deviation of the noise present in the inertial measurement unit from [66] for the vertical and rotational DOFs (assuming a Gaussian distribution of the noise). Random noise was sampled for each step of the response time-series from distributions generated suing the values in Table 3.5.

Table 3.5: Response signal noise (from [66]).

DOF	Noise Std.
Heave	$0.01m$
Pitch	$0.028°$
Roll	$0.011°$

Neural Network Architecture

A traditional NN has a number of layers, where each is comprised of a number of perceptrons (nodes), which are associated with a function that takes the weighted sum of the previous layer of the network as their input, and produces output for the next layer [16]. In a fully-connected NN, each perceptron in one layer is connected to every perceptron in the next layer. Weights applied to individual connections within the layers are then modified to converge on optimal outputs from the NN overall. Equation 3.35 defines the activation function of a single perceptron, where x_{p_j} represents the inputs, w_{p_j} are the connection weights, and b_p is the bias.

$$a_p = f\left(\sum_j w_{p_j} x_{p_j} + b_p\right) \tag{3.35}$$

Figure 3.18 gives a generic example of a simple multi-layer perceptron (MLP) model, taking response data and vessel speed as input and producing a wave property as output. The input data (x_{p_j}) is shown in blue, the weighted connections between the nodes (w_{p_j}), which map the input to output, can be clearly seen. Each node in the hidden layers has its own activation function (Equation 3.35), as well as the final output node. It should be noted, however, that NNs are often more complex than the simple architecture shown in Figure 3.18. MLPs are feedforward NNs, a form of supervised ML which can be used for both regression and classification problems. The models designed for the SAWB SSE task are regression MLPs.

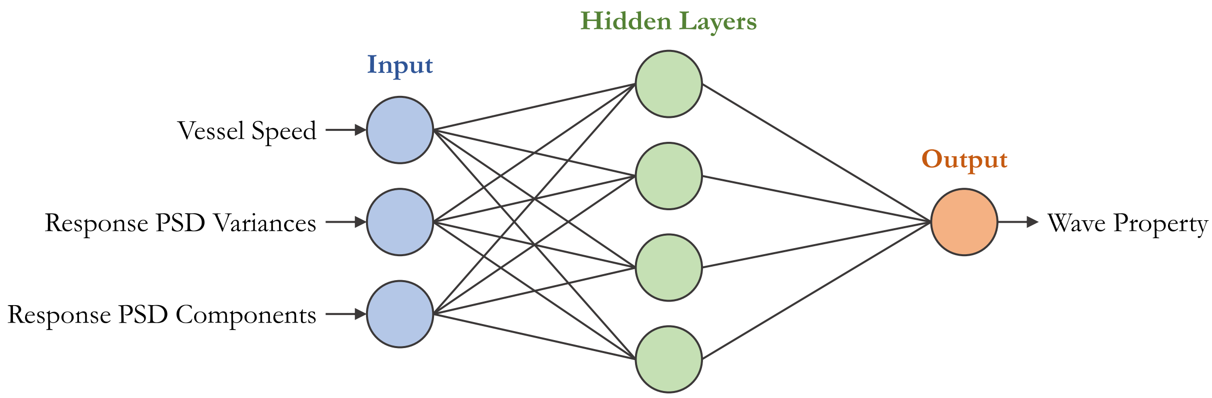

Figure 3.18: Simple MLP schematic.

In order to create the NN for SAWB-based SSE, the Keras open-source Application Programming Interface (API), built on top of TensorFlow 2.0, for Python was used as it is simple and intuitive, while permitting extensive customisation [67]. The first step in defining the network architecture involved selecting a model. In this case, the *Functional* API was selected as it offered greater flexibility for inputting multiple different data types (or outputting multiple variables) [68]. Fully connected layers can then be added to the model using the *Dense* class, and customised in terms the number of nodes within each layer, activation function etc.

Listing 1 in Appendix 5.1 gives an example of the functions developed to define and compile the NNs for SSE, specifically the function for 1-DOF responses, while Figure 3.19 displays its structure. Here, $input\_1$ and $input\_2$ were defined for the spectral ordinates and associated frequency components, while $input\_3$ contains the vessel speed and response PSD total power. The PSD ordinates and frequencies were then concatenated, forming the first *branch* of the NN, then passed to the first hidden layer. The output from this single hidden layer was then concatenated with $input\_3$, together comprising the NN *trunk*, which was then passed to another hidden layer, before outputting one of the sea state parameters in the final layer.

By first concatenating the PSD ordinates and frequencies then passing them through a hidden layer, the relationship between them is reduced to the size of the number of nodes of the hidden layer. As such, a greater weighting is placed on the $input\_3$ variables than the individual PSD ordinates or frequencies (initially comprising 60 components when combined) when the output from the branch hidden layers are concatenated with the $input\_3$ variables then passed through the rest of the network.

As an example, a 1-DOF model would take either heave, pitch, or roll response data as input, then output the significant wave height, mean wave period, or wave heading.

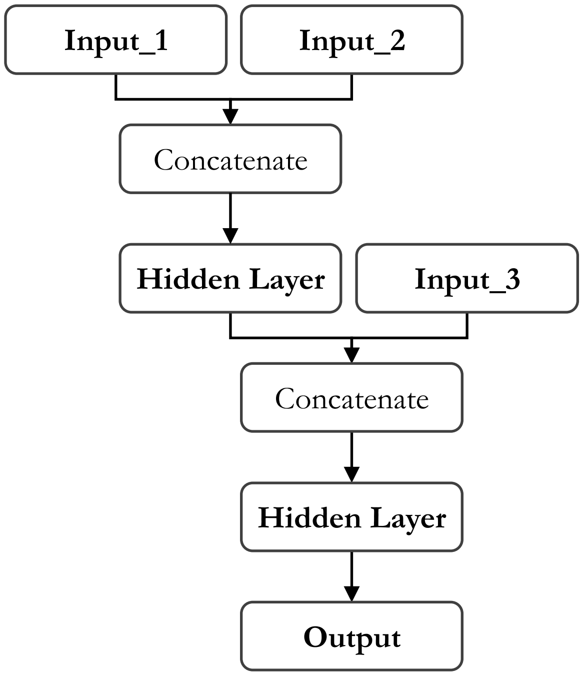

Figure 3.19: NN structure for 1-DOF data.

After the performance of single DOF responses to estimate wave properties were analysed, the network architecture was expanded to input 2-DOF response data in order to determine what sea state information could be extracted from the combinations of heave/pitch, heave/roll, and pitch/roll. As is shown in Figure 3.20, two branches exist in the 2-DOF models, one for each response. The same branch architecture for the 1-DOF model was replicated for the second DOF; however, the output of the hidden layer of each branch was concatenated with the response spectra total power and vessel speed information before being passed to the trunk hidden layers.

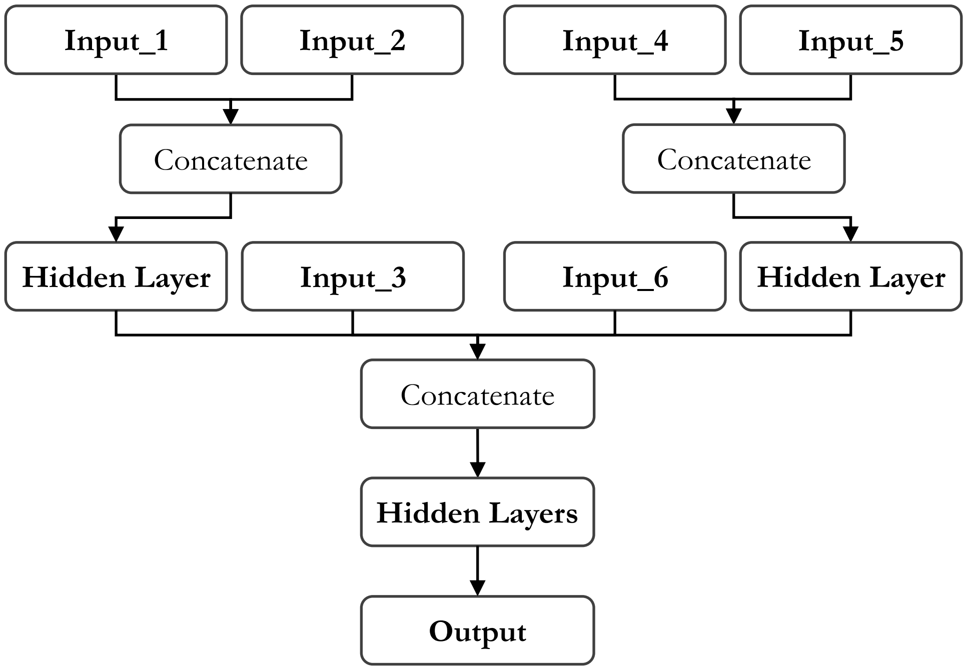

Figure 3.20: NN structure for 2-DOF data.

Finally, the model architecture for the complete 3-DOF variant was designed (Figure 3.21). Three network branches were established to accommodate the three responses, where each was treated in the same manner. The output from the hidden layers from each of the branches were then concatenated, alongside the PSD response total power and vessel speed, then passed to the hidden layers of the trunk to then output the estimations (just as for the 2-DOF models).

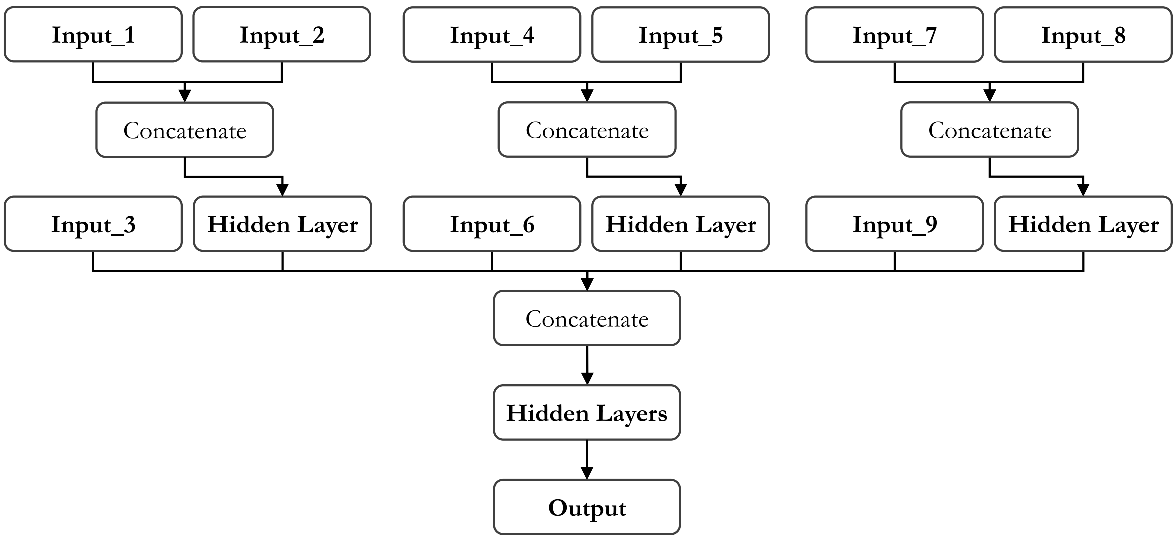

Figure 3.21: NN structure for 3-DOF data.

The *rectified linear unit* activation function (ReLu) was used for each layer as it is one of the most successful and widely-used activation functions [69]. Likewise, the *Adam* optimiser was used to find the best set of node connection weights to map the input data to the desired output, and a *mean square error* loss function was implemented to improve the model fitness.

The model was then trained using a set of training and validation data, and a given number of *epochs* and *batch* size. The number of epochs defines the number of times the algorithm will work through the entire data set, and the batch size defines the number of samples to pass through the network before the model parameters are updated in each epoch [67].

It was chosen to perform *k-fold cross validation* to ensure there was no (or minimal) overfitting or selection bias in the model [70]. K-fold cross validation splits the data into two sets, training data and validation data, where here a split of 80% training and 20% validation data was used. The model is then trained using the training data, and evaluated using the validation data. However, if there exists some bias in the 'randomised' data set, then the resulting model may not accurately predict future relations. Therefore, k-fold cross validation *folds* the data set k_{cv} times (for an 80/20 split, $k_{cv} = 5$), where the training-validation data split is repeated until every sample in the data has been used exactly once in the validation data set. The performance variance of the model can then be used as a metric to understand the variation in the model's fit, where a lower variance indicates a better fit. For example, if the model is overfitting a particular split, the variation will be higher.

The metrics used to assess the performance of the models were the root mean square error (RMSE) value and the coefficient of determination (R^2). RMSE represents the sample standard deviation of the difference between the predicted and actual values, where a higher weighting is placed on greater errors, and its units are the same as the predicted variable's units [71]. R^2 is a metric of how well the model fits the data, where a value of 0 represents a model which does not fit the data at all, and a value of 1 represents a perfect fit [71]. Equations 3.36 and 3.37 are examples of RMSE and R^2 for significant wave height, where $\hat{H}_s$ is the predicted height, H_s is the actual height, and $\bar{H}_s$ is the average height. The numerator in Equation 3.36 calculates the prediction residuals ($e_R = \hat{H}_{s_i} - H_{s_i}$), which gives the difference between the predicted value and true value, and can be used to assess how well the data fits the model [71].

$$RMSE = \sqrt{\frac{\sum(H_{s_i} - \hat{H}_{s_i})^2}{N}} \tag{3.36}$$

$$R^2 = \frac{\sum(\bar{H}_s - \hat{H}_{s_i})^2}{\sum(H_{s_i} - \bar{H}_s)^2} \tag{3.37}$$

In this case, the trained models were evaluated using a set of test data, 10% of the original data set (4,800 samples), split before the training and validation data sets were split (34,560 and 8,640 samples, respectively).

3.3 Results and Discussion

3.3.1 Response Component Analysis

While it was assumed that a NN model trained using all three DOF responses (heave, pitch, and roll) would perform better than one trained using fewer DOFs, it was decided to investigate the information which could be extracted when a model was trained using the DOFs separately. Firstly, separate NNs were trained using only 1-DOF, then NNs were trained using 2-DOF responses, resulting in six NN models for each wave condition to be estimated: significant wave height, mean wave period, and relative wave heading.

It was found that using the 30 spectral ordinates with the highest energy as input to the NNs gave the best estimation performance for H_s and T_1, while estimation of μ_h was improved when using the 80 spectral ordinates with the highest energy.

Different NN configurations were experimented with, in terms of model architectures and training parameters, in order to assess their effect on SSE performance, with the results presented in Appendices 1 to 7. After testing, the networks developed for H_s and T_1 showed similar performance, so the same design was used for them both for the 1-DOF and 2-DOF network models.

The final model configuration for the 1-DOF and 2-DOF models given in Tables 3.6 and 3.7 are derived from the results presented in Appendices 1 and 3. It was decided to use only a single hidden layer in the network branches to extract the relationship between PSD and frequency components, before combining the resulting weights with the total PSD power and vessel speed information. Two hidden layers were then used in the trunk of the 1-DOF architecture, while three improved estimation performance in the 2-DOF model. The same number of nodes were shared with the first two layers of the trunk in the 2-DOF model and layers in the 1-DOF model.

The only other difference was the use of a batch size of 32 in the 1-DOF model and 16 in the 2-DOF model. An Adams optimisation learning rate of 0.001 showed the best results The performance of each of the 1-DOF and 2-DOF models is further evaluated in the proceeding subsections.

Table 3.6: 1-DOF NN model configuration.

Element	Details
Branch structure	1 hidden layer (16 nodes)
Trunk structure	2 hidden layers ([16,8] nodes)
Epochs	100
Batch size	32
Loss rate	0.001

Table 3.7: 2-DOF NN model configuration.

Element	Details
Branch structure	1 hidden layer (16 nodes)
Trunk structure	3 hidden layers ([16,8,8] nodes)
Epochs	100
Batch size	16
Loss rate	0.001

Significant Wave Height

The performance of the network configurations presented in Tables 3.6 and 3.7 for the different component models for H_s are given in Table 3.8. The estimation performance of pitch and roll alone were shown to be quite similar; however, the performance of heave was almost an order of magnitude greater than both of them in terms of RMSE, and with a model fitness doubled (R^2). Heave alone had an R^2 value of 0.990, implying a very close fit of the model to the data, with an average RMSE of only 5.83mm. When heave was combined with pitch, the NN estimation error only slightly reduced, with heave and roll showing similar performance to heave alone. Pitch and roll together were capable of improved estimation performance; however, still with an RMSE nearly double that of heave's performance. It can, therefore, be concluded that heave is the DOF most closely related to wave height for a vessel of similar size to the WG. The models were also found to have little variation between cross-validations, implying little overfitting or selection bias, as the standard deviations of the RMSE values across the five k-folds were all very low when compared to the average RMSE values for each of the DOFs.

Table 3.8: SSE NN component model performance for $\mathrm{H_s}$.

Response(s)	RMSE Avg.	RMSE Std.	R-Squared Avg.
Heave	5.83mm	0.192mm	0.988
Pitch	38.5mm	0.464mm	0.485
Roll	38.4mm	0.743mm	0.486
Heave + Pitch	5.51mm	0.097mm	0.990
Heave + Roll	5.85mm	0.070mm	0.988
Pitch + Roll	10.1mm	0.505mm	0.966

Figure 3.22 portrays the residuals for each of the component models trained to estimate H_s, where the **estimated** H_s values are plotted against the residual errors (as all proceeding residual plots show the estimated wave condition versus residual errors). Aligned with the data given in Table 3.8, the residual plots show the least error in the models which include heave motion. For each of the models using heave, the errors were found to be slightly worse for greater values of $\hat{H}_s$, which was also present for the pitch and roll model (although with greater spread). This implies that the models tended to overestimate H_s.

The 1-DOF pitch and roll models both saw a steady increase in error as their H_s predictions increased, before reducing again, showing the greatest errors associated with H_s estimations between $1m$ and $2m$. This indicates that the models underestimated high seas and overestimated low seas as the largest errors were associated with estimates about the middle of the $\hat{H}_s$ range and distributed relatively evenly between positive and negative senses, thus, the real H_s values must have been closer to the limits of the range. The perceived straight edges of the distribution shape of the errors, particularly for the 1-DOF pitch and roll models, could be attributed to how the NN was trained. Given that the model was trained using a supervised learning technique, the relationship between the vessel response input and H_s output would take into account the limits of the range of H_s values simulated. As such, as the estimations tend towards greater values, the errors become less and less likely to have a large positive value as estimations are generally not going to exceed the H_s range limit of $2.5m$. The opposite can also be said for negative errors as the estimations tend towards lower values, where estimations will tend to not be less than the lower limit of the range of H_s values simulated of $0.5m$. The errors around the middle of the range of estimated values show the greatest magnitude both in the positive and negative direction (overestimating and underestimating), indicating that large real values of H_s were rarely estimated as low, and low real values of H_s were rarely estimated as high (though a few outliers do exist).

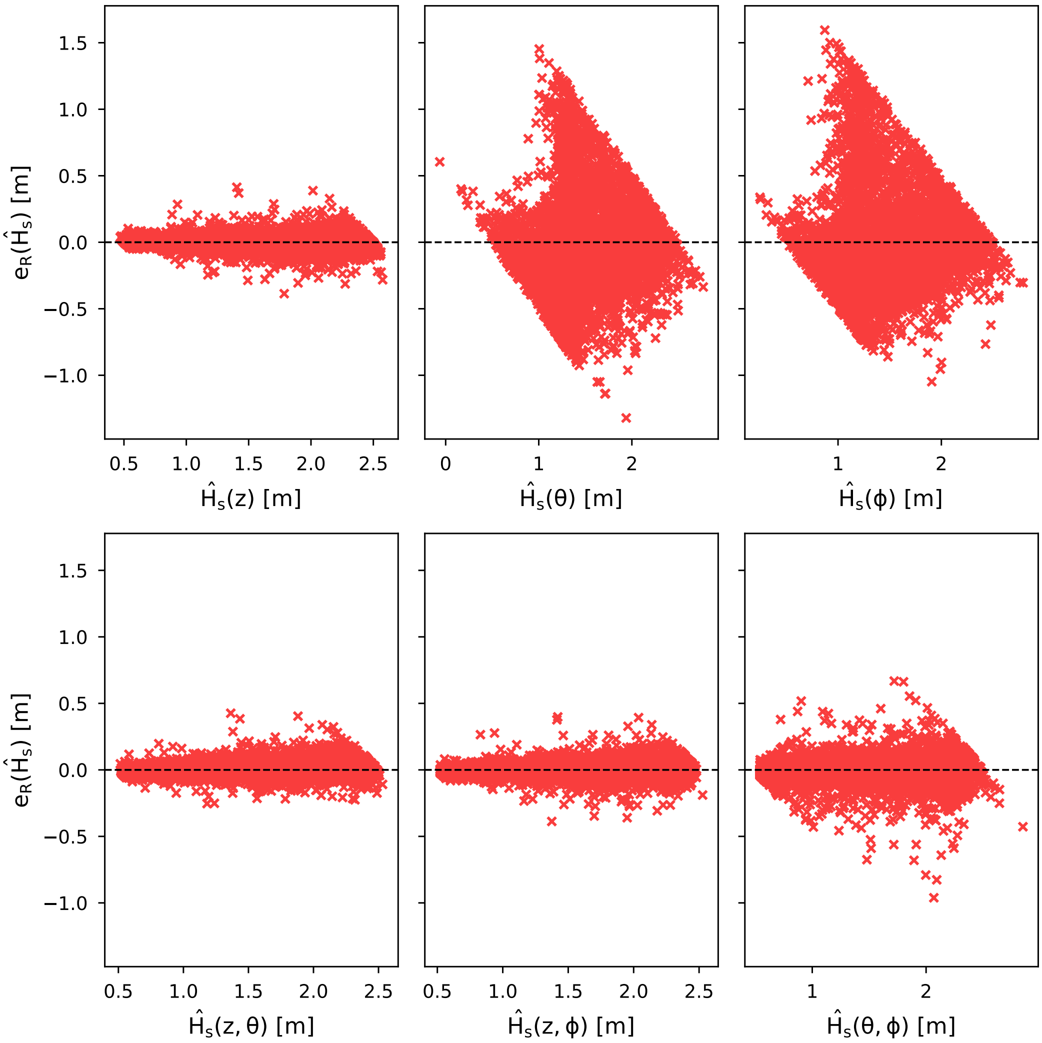

Figure 3.22: Residual plots for H_s component estimations.

Due to the influence of wave heading on vessel FRFs (presented in Figures 3.8, 3.9, and 3.13), it was decided to analyse its effects on the NN performance. Therefore, it was decided to run 4,800 test simulations with a fixed vessel speed of $2.5m/s$. The NN models were then re-trained using the 48,000 training cases and each of the six component models were tested on the new data.

Once the vessel speed was fixed, the absolute residuals for varying wave headings were plotted on half-polar plots for each of the component models in Figure 3.23. The higher H_s estimation performance of the models trained using the heave response can be immediately seen, with the only clear trend found for the models which included heave being that they tended to have greater errors in following seas. This was likely due to the vessel encountering fewer wave components when travelling in following seas, so statistically having less information. The errors present when only pitch motion was used as input to the network were larger when the vessel experienced beam seas. As pitch motion would approach zero as μ_h approaches 90°, it is likely that the reduced motion of the vessel would lead to less pronounced pitch PSD components, making it harder for the designed algorithm to extract significant information from the PSD. A similar trend was found for roll, where the reduced motion as μ_h approached either 0° or 180° resulted in greater errors.

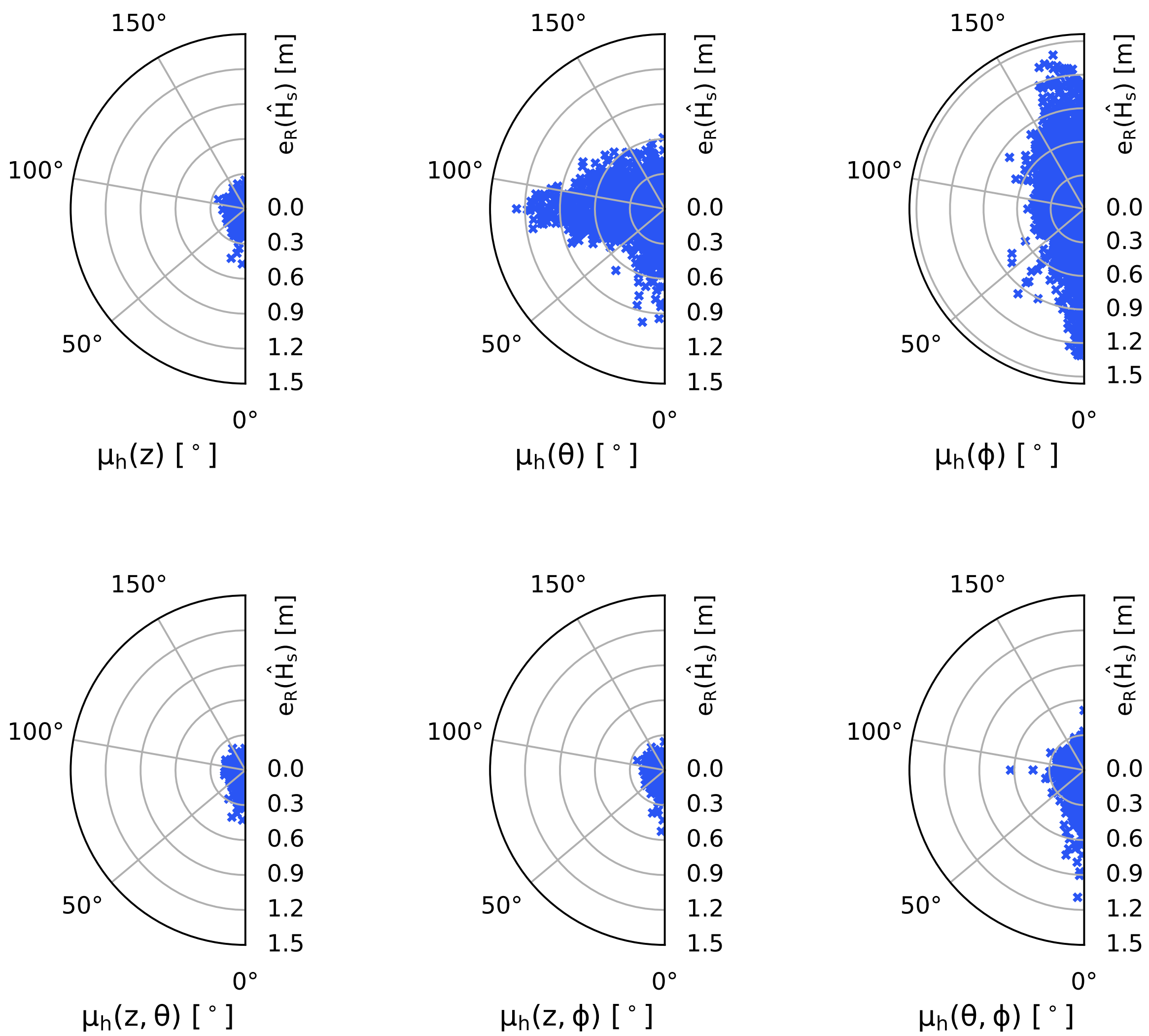

Figure 3.23: Residual polar plots for $\mathrm{H_s}$ with fixed U_π of $2.5m/s$ and varying μ_h.

Mean Wave Period

Table 3.9 presents the results for T_1 estimations using the network architectures given in Tables 3.6 and 3.7. The performance of all of the response component models was found to be quite good, with less variation found between the NNs than for H_s. Again, heave was found to be the single DOF which could be used to extract the most information to estimate T_1, with pitch being second. The combination of both heave and pitch had the lowest errors and best model fit, with an average RMSE of $0.228s$ and an R^2 value of 0.991. The models all had low RMSE standard deviations, again demonstrating that there was little overfitting or selection bias.

Table 3.9: SSE NN component model performance for T_1**.**

Response(s)	RMSE Avg.	RMSE Std.	R-Squared Avg.
Heave	$0.368s$	$0.018s$	0.944
Pitch	$0.573s$	$0.023s$	0.928
Roll	$0.708s$	$0.021s$	0.877
Heave + Pitch	$0.228s$	$0.022s$	0.991
Heave + Roll	$0.294s$	$0.042s$	0.985
Pitch + Roll	$0.401s$	$0.021s$	0.973

The residual plots for the different T_1 estimation models (Figure 3.24) were all quite similar, except those without heave had a greater spread. As expected, the model for heave and pitch showed the least spread with few outliers, and the model for heave and roll showed the second best performance. Greater errors were associated with estimates closer to the centre of the T_1 range than the limits of the range for the roll-only model, which indicates that the NNs underestimated large T_1 seas, and overestimated small T_1 seas, with the same trend as for $\hat{H}_s$ seen with respect to the simulated T_1 range limits not being exceeded by very much by the estimations (due to the NN model mapping).

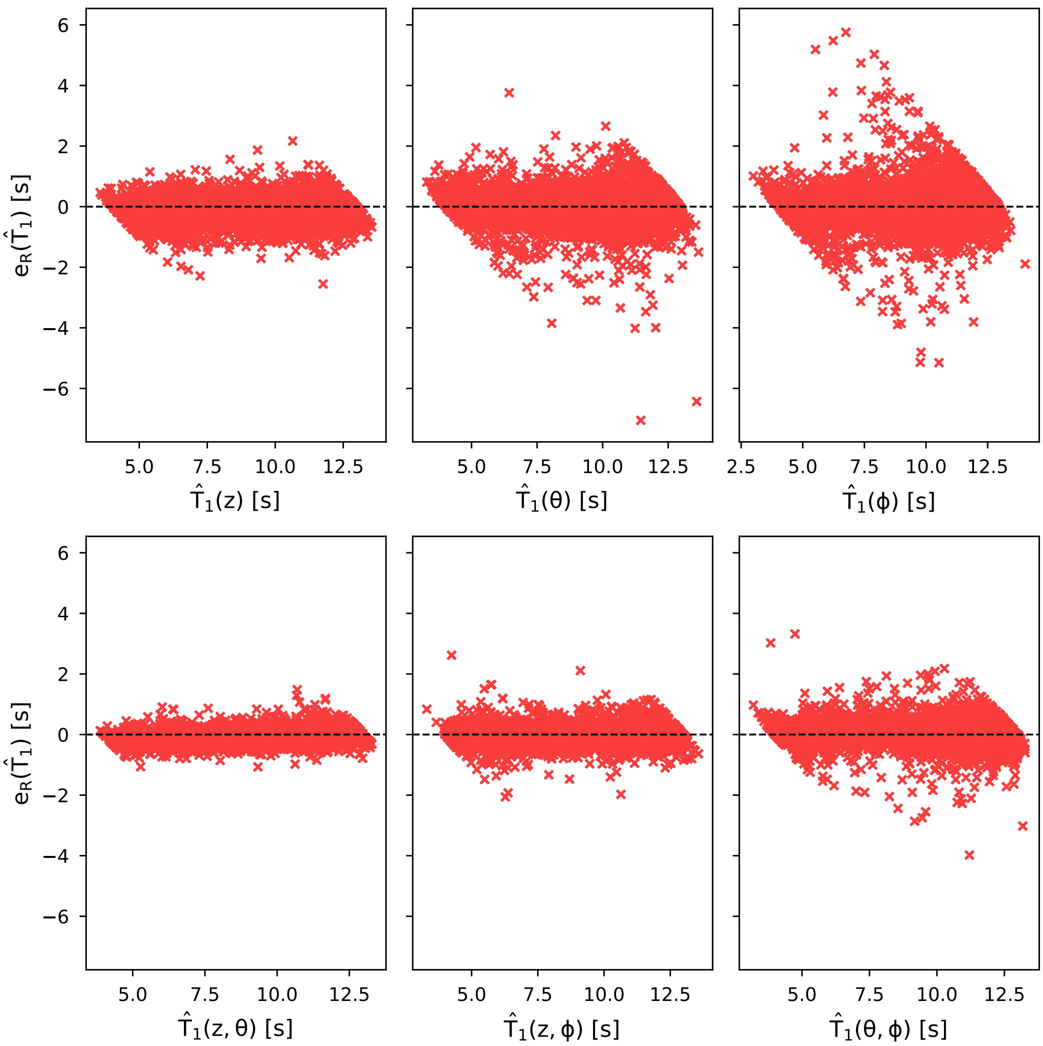

Figure 3.24: Residual plots for T_1 component estimations.

Fixing the vessel speed, effects of wave heading variation on estimation performance of the different component NNs was tested and plotted in Figure 3.25. A line of residual outliers can be seen for the pitch model at $\mu_h \approx 90°$, while a similar line can be seen in for the roll model at $\mu_h \approx 180°$, and a trend to greater errors at $\mu_h \approx 0°$. In line with the results from Figure 3.23, this is likely due to the PSD feature selection algorithm failing to extract relevant information when pitch or roll motion approach zero.

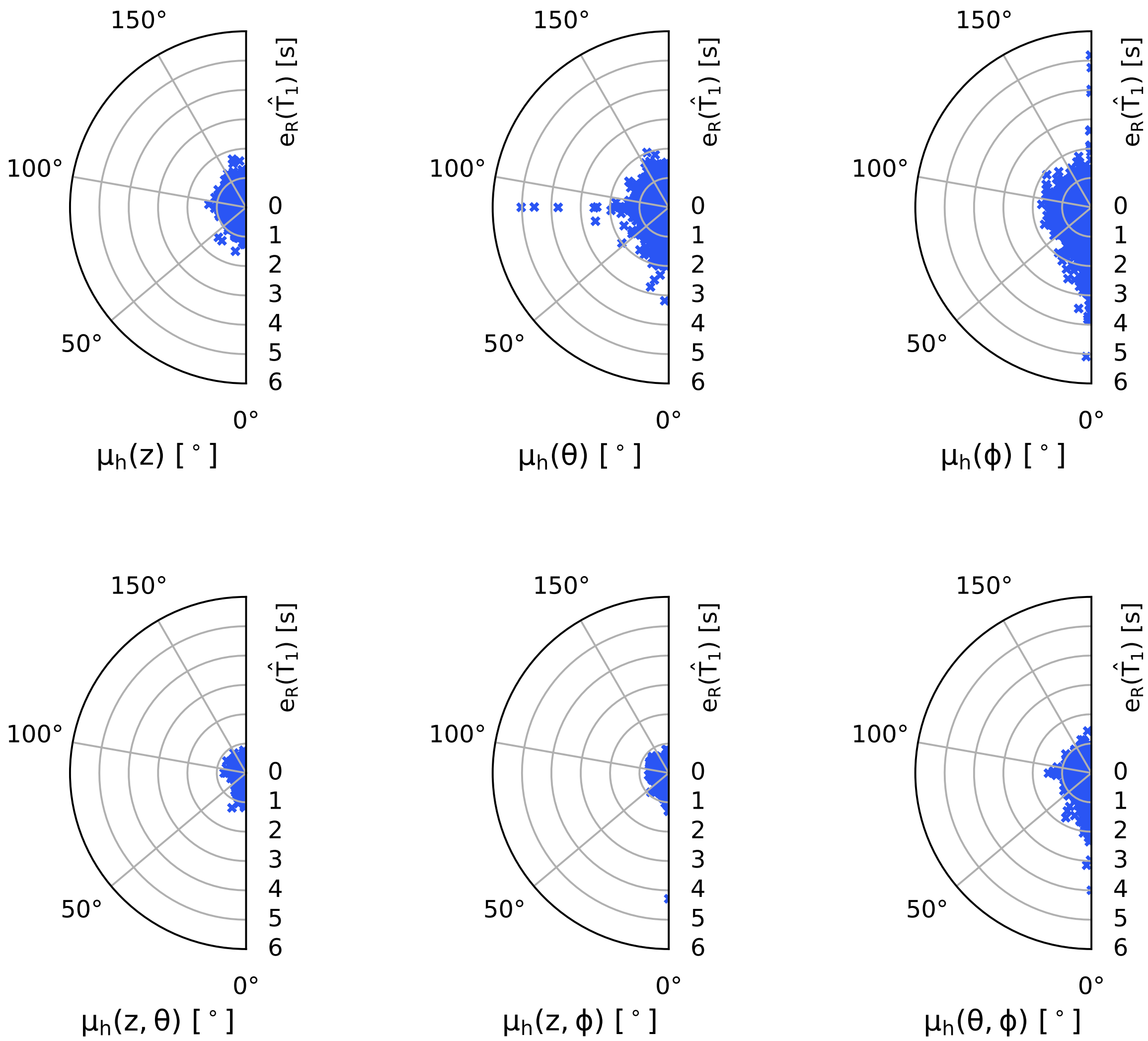

Figure 3.25: Residual polar plots for T_1 with fixed U_π of $2.5m/s$ and varying μ_h.

Wave Heading

Experimentation showed that the models for μ_h performed better when using the configurations given in Tables 3.10 and 3.11. Experimental results from varying the μ_h model architectures and training parameters using pitch are given in Appendix 5, and results using pitch and roll are given in Appendix 6.

Table 3.10: 1-DOF NN model configuration for μ_{h}.

Element	Details
Branch structure	1 hidden layer (16 nodes)
Trunk structure	2 hidden layers ([32,16] nodes)
Epochs	100
Batch size	16
Loss rate	0.001

Table 3.11: 2-DOF NN model configuration for μ_{h}.

Element	Details
Branch structure	1 hidden layer (16 nodes)
Trunk structure	3 hidden layers ([32,32,16] nodes)
Epochs	100
Batch size	32
Loss rate	0.001

The 1-DOF and 2-DOF models for μ_h had the same amount of hidden layers as the ones for H_s and T_1; however, each layer required more nodes. This could be due to the fact that 80 PSD components are used as input to the μ_h networks, as opposed to 30, so a greater number of connections could better capture the extra information. The 1-DOF was shown to perform better when trained with a batch size of 16, while the 2-DOF models performed better with a batch size of 32 (the inverse of the H_s and T_1 models).

The performance results for each of the μ_h component models are given in Table 3.12. The models were able to estimate wave headings with relatively high accuracy. Heave alone performed the worst of all models trained, achieving an average RMSE of 18.90° and R^2 of 0.852 (averaged across the 5 k-folds). Roll also performed similarly. However, when roll was paired with pitch, the best estimation results were achieved, with an average RMSE of only 8.40° and R^2 of 0.973. The standard deviations of the RMSE varied quite a bit, with heave alone showing significant variation, thus showing more selection bias and/or overfitting.

Table 3.12: SSE NN component model performance for μ_{h}.

Response(s)	RMSE Avg.	RMSE Std.	R-Squared Avg.
Heave	18.9°	1.13°	0.852
Pitch	10.5°	0.195°	0.956
Roll	18.3°	0.732°	0.866
Heave + Pitch	8.88°	0.473°	0.970
Heave + Roll	12.8°	0.157°	0.935
Pitch + Roll	8.40°	0.221°	0.973

By looking at the residuals for the μ_h estimations for each component model shown in Figure 3.26, a number of patterns can be seen. For heave alone, the greatest errors are generally estimated around beam sea. This is similar to what was seen for $\hat{H}_s$ and to a lesser extent $\hat{T}_1$, with the limits of the wave property range being underestimated and overestimated. However, the greatest outlying errors for most of the models increased in magnitude as they approach head and following sea estimations. This shows that at times the models would estimate head seas as following seas, and vice versa.

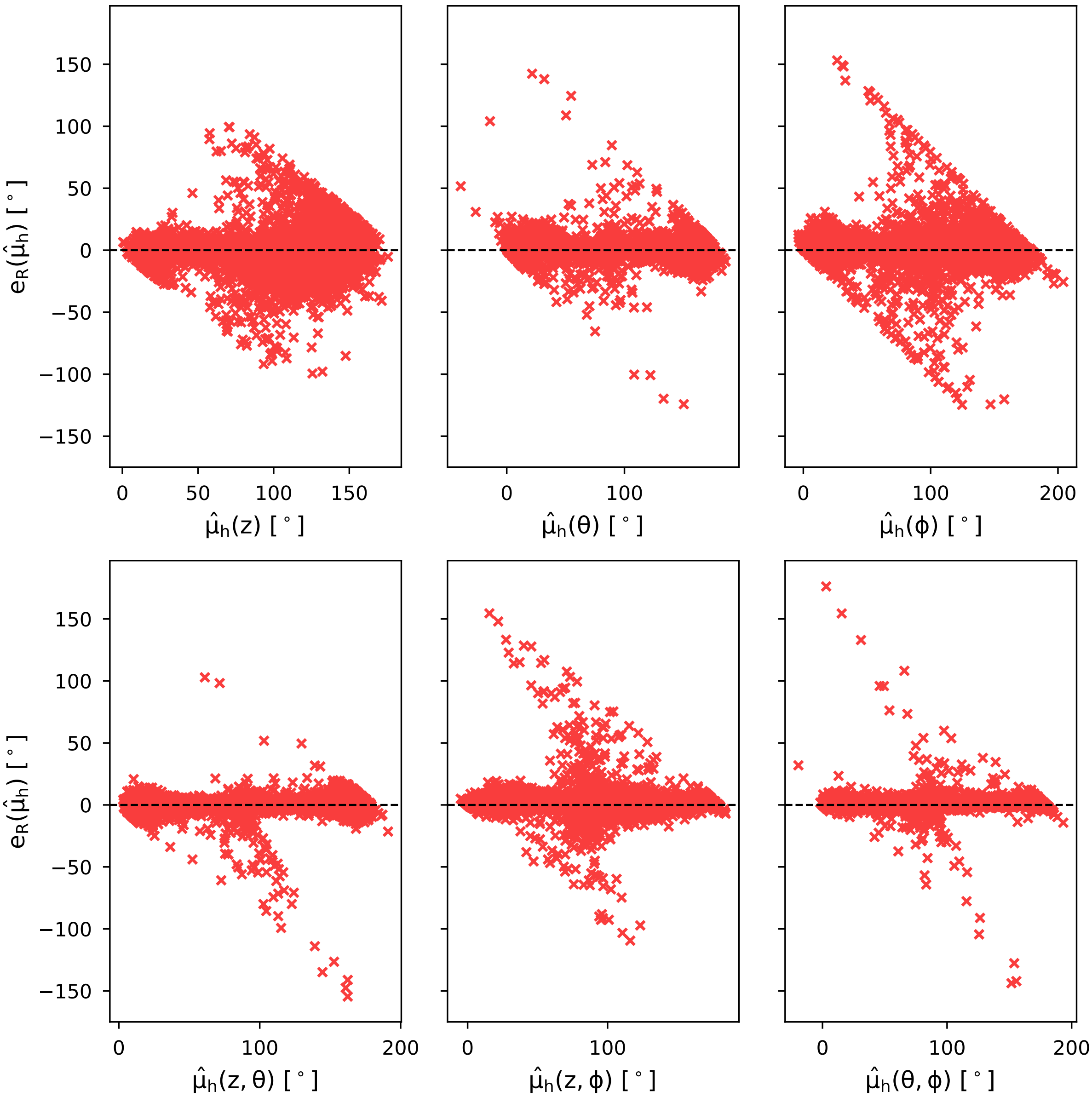

Figure 3.26: Residual plots for $\hat{\mu}_h$ component estimations.

The networks were then tested by simulating cases with a fixed vessel speed and varying μ_h, with the absolute residual errors presented for each component in Figure 3.27. The greatest estimation error was present when using only roll at wave headings at approximately head sea or following sea. As discussed, the PSD component analysis for roll motion would likely not be able to capture sufficient information to effectively estimate μ_h at these angles. The largest density of errors was seen around beam sea for the heave only model. Considering Figure 3.8, this makes sense as there is minimal heave response at beam sea.

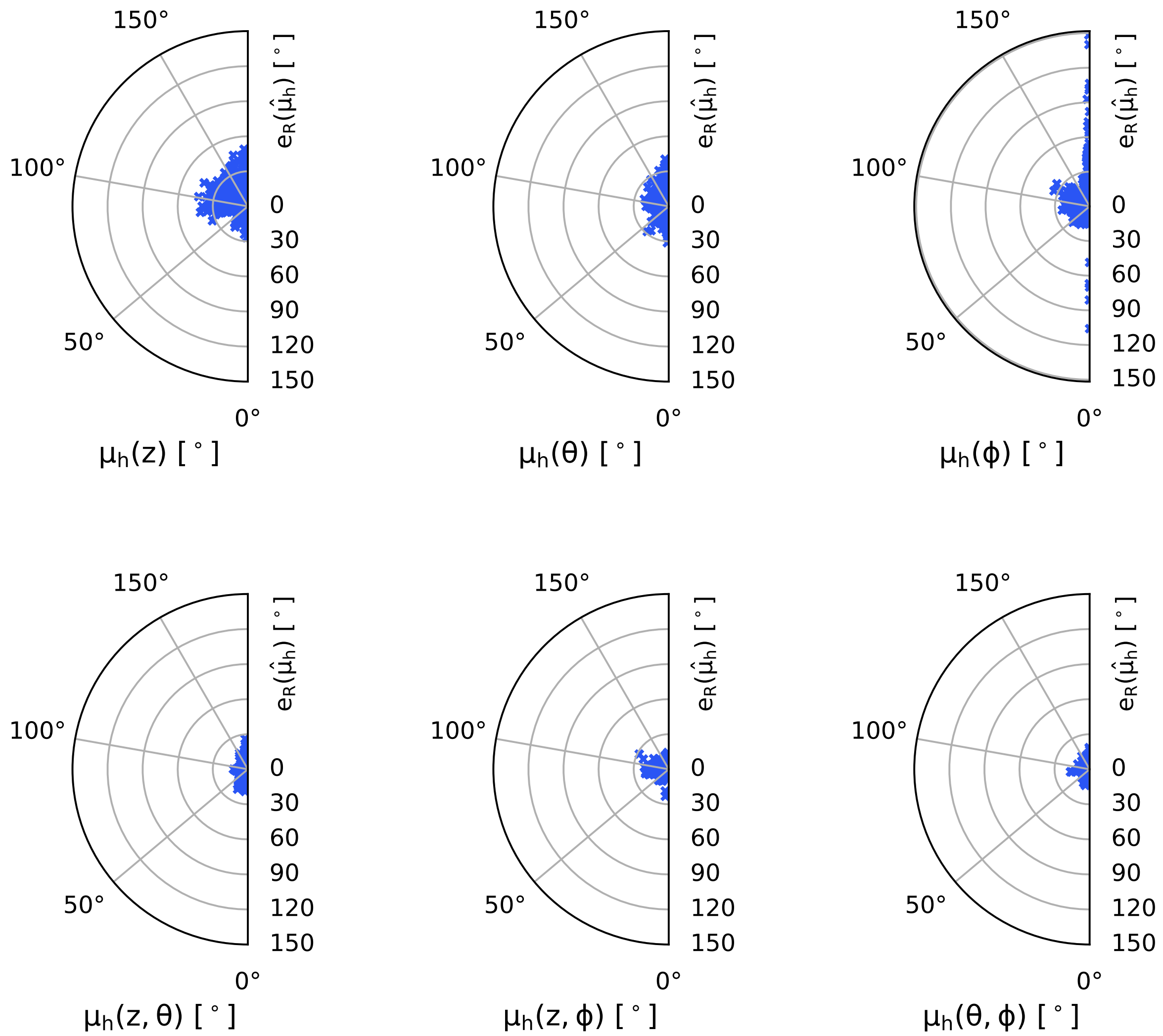

Figure 3.27: Residual polar plots for μ_h with fixed U_π of $2.5m/s$ and varying μ_h.

3.3.2 Final Network Models

Finally, the three vessel response DOFs, heave, pitch, and roll, were combined into a single NN to estimate the wave conditions: significant wave height, mean wave period, and wave direction. Appendix 7 outlines the experimental results of the model configuration process, and the associated estimation performance for each of the estimated wave conditions.

Similarly for the 1-DOF and 2-DOF models, the performance of the 3-DOF model configuration for H_s and T_1 were optimal when using the same architecture, as displayed in Table 3.13. The model architecture for the 3-DOF μ_h NN is given separately in Table 3.14. Each of the models performed best with three hidden layers in their network trunks, with the same number of nodes. The only difference between the network configurations is that the H_s and T_1 models used a batch size of 16, while the μ_h model used a batch size of 32.

Table 3.13: 3-DOF $\mathrm{H_s}$ and $\mathrm{T_1}$ NN model configuration.

Element	Details
Branch structure	1 hidden layer (16 nodes)
Trunk structure	3 hidden layers ([32,32,16] nodes)
Epochs	100
Batch size	16
Loss rate	0.001

Table 3.14: 3-DOF μ_h NN model configuration.

Element	Details
Branch structure	1 hidden layer (16 nodes)
Trunk structure	3 hidden layers ([32,32,16] nodes)
Epochs	100
Batch size	32
Loss rate	0.001

The results from the k-fold cross-validation of the 3-DOF models are given in Tables 3.17 to 3.16. As expected, the inclusion of all of the data led to best performance of the models.

The 3-DOF H_s model also did not see a large increase in performance when compared with the heave and pitch model. The average RMSE reduced to 5.38mm, with a standard deviation of 0.116mm, and an average R^2 value of 0.990. The similar performance could indicate that similar information was extracted from pitch and roll to estimate H_s.

Table 3.15: 3-DOF $\mathrm{H_s}$ NN model performance.

Metric	Value
RMSE Avg.	5.38mm
RMSE Std.	0.116mm
R-squared	0.990

The T_1 estimation performance was, however, improved more when using all of the vessel response data, with an average RMSE of 0.171s and standard deviation of 0.006s, and an average R^2 of 0.995. The closest performance was achieved with the model trained on the heave and pitch data. This indicates that the information extracted from each DOF was significantly different from each other when responding to varying T_1.

Table 3.16: 3-DOF T_1 NN model performance.

Metric	Value
RMSE Avg.	0.171s
RMSE Std.	0.006s
R-squared	0.995

The μ_h estimator experienced some improvement in estimation performance when heave was included, compared to only pitch and roll, but with a lower average RMSE and slightly greater R^2 value. However, this came at the cost of a greater RMSE standard deviation, meaning that there was a greater variation in model performances between k-fold validations.

Table 3.17: 3-DOF μ_h NN model performance.

Metric	Value
RMSE Avg.	7.21°
RMSE Std.	0.411°
R-squared	0.979752

The residual plots for the 3-DOF wave estimation NNs are presented in Figures 3.28 to 3.29. Each of the models show similar patterns to those of the best performing 2-DOF models.

Figure 3.28 demonstrates a general trend towards larger error for greater H_s estimations. The 3-DOF T_1 residuals in Figure 3.16 shows that the model tends to favour overestimating T_1 values rather than underestimating.

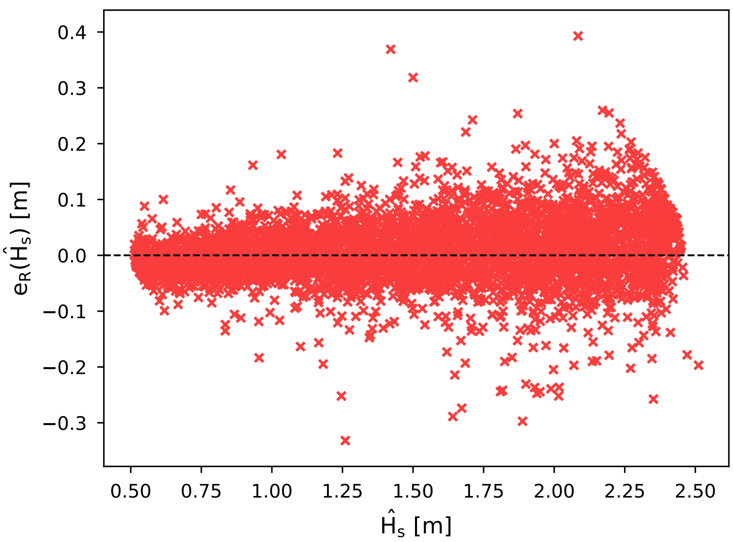

Figure 3.28: Residual plot for 3-DOF H_s estimations.

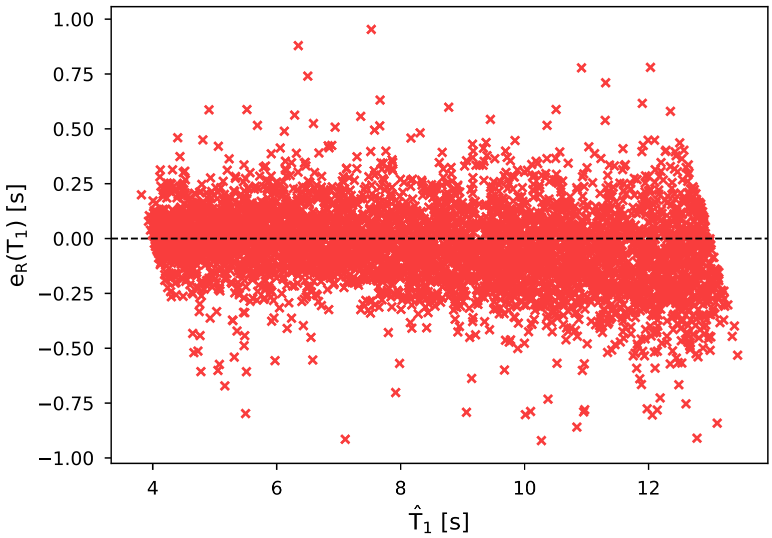

Figure 3.29: Residual plot for 3-DOF $\mathrm{T_1}$ estimations.

In Figure 3.30 it can be seen that the estimations have the greatest accumulation of errors about beam sea. The model is, therefore, tending to incorrectly estimate the vessel as being in beam seas more frequently than in head or following seas. As seen with the μ_h component model estimations, the outlying greatest errors are seen closer to head and following sea, with the model occasionally mistaking the vessel as being in opposing wave headings.

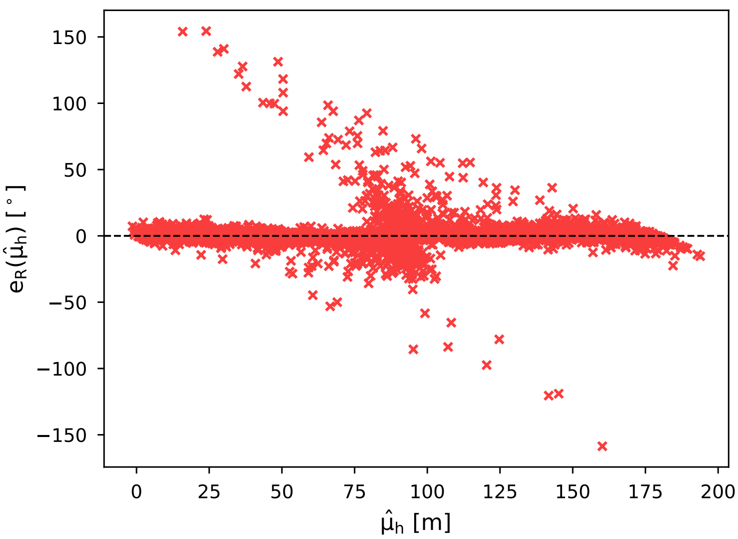

Figure 3.30: Residual plot for 3-DOF μ_h estimations.

The effect of wave heading on the 3-DOF models was also tested by simulating vessel motion across the full range of headings, with the absolute residuals plotted in Figure 3.31. The NN for T_1 estimation had the most evenly spread results across wave headings, mostly accumulated within $1s$ of the real value. The errors produced by the network for H_s estimation were quite evenly spread, with an accuracy within $0.3m$ of the real values, except for wave headings close to 0° which reached errors of around $0.5m$. The errors associated with μ_h were the most unevenly spread, with the worst estimation performance at wave heading angles of approximately 90° and around 30°, each having small spikes specifically at those angles.

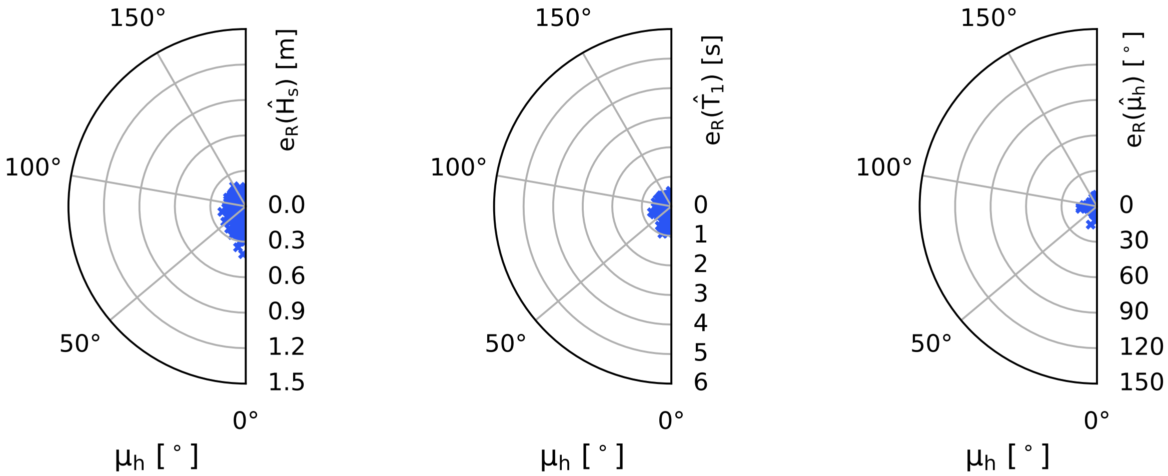

Figure 3.31: Residual polar plots for 3-DOF NNs with fixed U_π of $2.5m/s$ and varying μ_h.

The results presented for the fixed vessel speed and varying wave headings, for each of the wave properties estimated, may not be representative of estimation performance at other fixed speeds. Therefore, any conclusions made must be considered as only providing evidence for the conditions presented.

As a point of comparison with one another, the RMSE for μ_h as a ratio of the range of μ_h values tested was 4.01%. The ratio of RMSE to range for H_s was 2.69%, and the ratio for T_1 was 1.90%. Therefore, the relative estimation performance was best for T_1 and worst for μ_h. However, a comparison with other studies provides a better point of analysis of their performance.

Results Comparison

Due to the vastly different manner in which SSE studies using SAWB have presented their findings, as well as the many different simulation and experimental setups used, it is difficult to directly compare the SSE results with other methodologies. However, two studies have been selected as comparative to this one with both using relatively small vessels and parametric sea states, as well as providing similar result metrics. Results from [17] and [24] are compared with the SSE NNs designed for this project in Table 3.18.

The SSE study by Arneson, Brodtkorb, and Sørensen [17] provides one of the only other ML-based estimation approaches which trained their model using response information in the frequency domain (rather than raw time-series data), and have discussed their results using similar metrics. The authors simulated their vessel motion using a parametric wave spectrum (JONSWAP), then took a Fourier transformation of the time-series responses and integrated over the frequency range to get the total power of the spectra, then used that as input to their models. The authors used quadratic discriminant analysis to estimate wave headings, then partial least squares regression to estimate the significant wave height and peak wave period. There were four major differences in Arneson, Brodtkorb, and Sørensen's approach, however. They were able to utilise the cross-spectra to differentiate between port and starboard wave headings (as well as head and following seas), they simulated sea state conditions all the way through to Sea State Code 8 (very high), they used analysed a 6-DOF system, and they classified wave headings into 10° intervals. Therefore, their model has been compared using only their results for comparable sea states, while the 360° wave heading angle has been compared to the relative heading angle presented in this study.

After calculating the RMSE values, the results were found to be approximately 4° for wave heading, $0.6m$ for significant wave height, and $1.5s$ for peak wave period. The approach of [17] outperforms the one presented in this study in terms of wave heading estimation, both in magnitude, and in being able to differentiate between port and starboard headings (by using the cross spectra). However, the results for significant wave height and wave period estimation are of an order of magnitude more precise when using the presented methodology than in [17]. It should be noted that only five trials were used to calculate the RMSE values in [17] (being the 5 out of a possible 17 cases presented in the study with comparable sea states). Arneson, Brodtkorb, and Sørensen, however, showed their method worked on a much greater range of sea states. Therefore, for a better comparison, the designed methodology should be tested either using experimental data or simulated using a different set of transfer functions which could handle the non-linear relationships resulting from greater sea states. Further, the difference in vessel and idealised wave spectrum would likely have an effect on the results.

In order to compare the obtained results with a model-based approach, the study by Hinostroza and Soares [24] was selected as it also aimed to estimate the wave conditions of a parametric wave spectrum (JONSWAP), as well as presenting their results using comparable metrics. The authors utilised genetic algorithms to calibrate inverse transfer functions which mapped the heave, pitch, and roll responses to the wave spectrum. Further, only wave headings within 180° were simulated (equivalent to the relative wave heading used in this study). Hinostroza and Soares tested their method using two different ship types via simulation, as well as via experiments. The simulation results from the smaller vessel, are compared with the developed methodology, while 2 out of 8 cases were omitted as they simulated Sea State Code 5 (rough seas). Their results showed an RMSE of $0.23m$ for significant wave height, $1.1s$ for peak wave period, and 9° for wave heading. It can be seen that the devised NN models for SSE were more precise than the inverse transfer function approach of Hinostroza and Soares for each wave condition estimated.

Table 3.18: SSE NN performance comparison.

Study	Sig. Wave Height RMSE	Wave Period RMSE	Wave Heading RMSE
Long et al. 2021	$5.38mm$	$0.171s$	7.21°
Arneson et al. [17]	$60mm$	$1.5s$	4°
Hinostroza and Soares [24]	$23mm$	$1.1s$	9°

The NN models for SSE designed as a part of this thesis are promising when compared with similar approaches. With further development, there is potential for the described methodology to be expanded upon for performance augmentation and possible future implementation in real physical systems.

3.4 Conclusion

A new methodology for estimating sea states has been developed, which harnesses the benefits of ML as an alternative to fine tuning physics-based models.

Artificial NN models trained using heave, pitch, and roll vessel response PSD features have been shown to be able to estimate significant wave heights, mean wave periods, and relative wave headings effectively for an idealised sea state (within the given constraints).

The information which could be extracted from each response DOF has been analysed, while the SSE models trained using all three DOFs were shown to perform the best. The influence of wave heading on estimation performance was also investigated, showing different results for each wave condition.

However, the closed-form expressions used to simulate the vessel responses assumed that the vessel DOFs were decoupled, which prevented the distinction between port and starboard wave headings, putting a limitation on the wave heading models estimation capabilities (a limitation of the simulation method, not the SSE model itself).

In the future, a number of extensions could be made to this study. Firstly, more complex vessel dynamic and transfer function models should be used which do not assume decoupling between DOFs so that the cross-spectra can be analysed to differentiate between port and starboard wave headings. This model should also be able to capture non-linear behaviours associated with rougher sea states so that the performance of the NNs can be tested for a greater range of conditions. The model could also be used to simulate additional DOFs to use as input to the NNs. Secondly, the networks could be tested using model-scale testing data under controlled conditions. Thirdly, the PSD analysis procedure could be further optimised and experimented with. Finally, the NN type and architectures could be further optimised to accommodate a greater variety of sea conditions, improving the robustness of the model and estimations.

However, the NN models were shown to perform well, indicating that the hypothesis that spectral vessel response could be used as input for a NN to estimate wave conditions was valid.

Chapter 4: Guidance Strategies for Sea State Estimation using Multiple Ships as Wave Buoys

4.1 Introduction

The use of a single vessel acting as a wave buoy for SSE has some limitations. Firstly, any biases related to the state of the single vessel, when acting as a wave buoy, may be difficult to discern. For example, there may be more errors associated with a particular wave encounter frequency than another. Secondly, any faults in the single vessel's motion sensor, communications, or onboard computer systems may result in incorrectly obtained motion response signals, producing incorrect SSEs. Thirdly, the motion response information available from a single vessel may not be adequate enough to describe complex sea states.

Errors in an estimation task are defined as the difference between the estimated value and the true value [72]. Estimation errors can be caused by any number of factors; however, they are often categorised as either being *random error* or *bias error*. Random error is simply the random variation of estimations about some mean value when given the same input. Bias error is not caused by random chance, but rather biases in the system.

By employing multiple vessels acting as wave buoys for SSE, some of the limitations of a single vessel can be overcome. By averaging the SSEs from multiple vessels, it is hypothesised that both random and bias errors associated with individual vessel SSE errors will be reduced. The multi-vessel SSE study by Nielsen, Brodtkorb, and Sørensen [25] supports this hypothesis. Moreover, it is hypothesised that, by guiding the vessels in such a way that increases variance in the vessel states (in terms of vessel speed and wave heading), SSE performance will be further improved. While a single vessel could achieve this by varying its trajectory, more time would generally be required achieve equivalent accuracies.

This hypothesis is based on the fact that the variation of vessel speed and vessel heading change the FRFs, producing different responses to the same wave input. Therefore, if different filter types converge on the same answer, then it is considered more likely that the results are accurate, reducing filter-specific biases. Further, as seen in Figures 3.8 to 3.13, wave heading has a greater influence on vessel response type to difference wave frequency components than vessel speed for the WG.

As such, three guidance strategies have been investigated to test the effects of vessel trajectory on SSE accuracy. The first strategy involves three vessels moving with a common heading when taking a sea state. The second strategy forces the vessels on trajectories with maximum possible variation in wave heading angle. The third strategy is to send the vessels on randomised headings.

The reduction in SSE errors when using multiple vessels, compared to a single vessel, are explored, as well as the relative difference between the three different multi-vessel SSE guidance approaches. The significance of the differences between SSE performance of the different guidance strategies is also assessed.

The NNs in Chapter 3 were designed to estimate the relative wave heading. As discussed in the previous chapter, the relative wave heading is defined as the angle between the body-fixed coordinates of the vessel, and the direction of wave propagation. As such, estimation of the wave heading angle is appropriate for a single vessel; however, cannot be equally applied to multi-vessel SSE as vessels moving with different vessel headings relative to north-east-down (NED) coordinates, as they will experience different relative wave headings. Therefore, a global *wave direction* has been introduced, characterised as the direction of wave propagation with respect to the global NED coordinates.

While most current ML SAWB SSE techniques have performed SSE using vessel response data for a fixed length of time [17], [18], [20], [21], it is hypothesised that the SSE process can be optimised by taking estimations in real time. The methodology used to design the NNs for SSE in Chapter 3 was hypothesised to be able to estimate wave conditions in real time as the shape of the response spectra is assumed to converge over time due to the statistical interaction with the wave components. The progressive performance of the designed NNs for estimations over time is, thus, evaluated in this chapter. However, it is assumed that the estimations require a minimum of 10 minutes of vessel response data has been taken [14], so no estimations have been taken until 10 minutes of response data was recorded. After the 10 minute time period, it was decided to update the SSEs every 30 seconds.

In order to assess the performance of the SSEs, two estimation uncertainty metrics have been adopted. The first is a time-based metric which evaluates uncertainty related to random errors [72]. The second metric calculates uncertainty by comparing the estimated spectra from each vessel, which can be used to address bias errors in the system [25]. These two metrics have been investigated for both offline and online SSEs.

As there are multiple vessels involved in the SSE task now, given N_π vessels, each vessel is labelled as π_n, where $n = 1, 2, ..., N_\pi$, while the group of vessels are together labelled as Π. The individual estimates from each vessel are, thus, taken as $\hat{H}_{s_n}$, and the combined estimates are defined as $\hat{H}_{s_\Pi}$ (using H_s as the example). Aggregated estimates are simply taken as the average of the individual vessel estimations.

4.2 Guidance Strategies and Wave Direction

It was hypothesised that the variation of vessel trajectories, primarily by varying the wave encounter headings, would improve multi-vessel SSE performance when compared with single vessel SSE or multiple vessels moving with a common heading. Forcing the variation in μ_h between vessels ensures that the vessel FRFs for heave, pitch, and roll are varied. As such, the filtering characteristics of each vessel are changed. This variation means that estimation errors associated with biases for certain headings can be reduced as the estimations from the vessels with varied headings are aggregated.

In order to investigate this hypothesis, three different guidance strategies have been implemented, each comprised of three vessels. The first strategy guides the vessels along trajectories with a common heading. The second strategy forces the vessels on trajectories which are 90° apart, allowing for the greatest possible variance in wave headings (with the estimation range being 0° to 180°). The third strategy is for the vessels to move with random headings. In each instance, the vessel speeds are randomly generated within the operating range ($0m/s$ to $5m/s$).

In Chapter 3, the μ_h angle estimates only required two different coordinate systems, the body-fixed (BF) coordinate system and the wave-relative (WR) coordinate system. However, once multiple vessels with varying μ_h angles were introduced, the transformation of the individual estimated μ_h angles into a global wave direction (μ_w) had to be considered.

The wave direction, as opposed to the wave heading, has been defined in this study as the displacement of the WR coordinate system from the north-east-down (NED) coordinate system fixed to the Earth. As can be seen in Figure 4.1, the origin of the WR frame and the NED frame have been considered as the same point, with the x- and y-axes displaced from one another by μ_w.

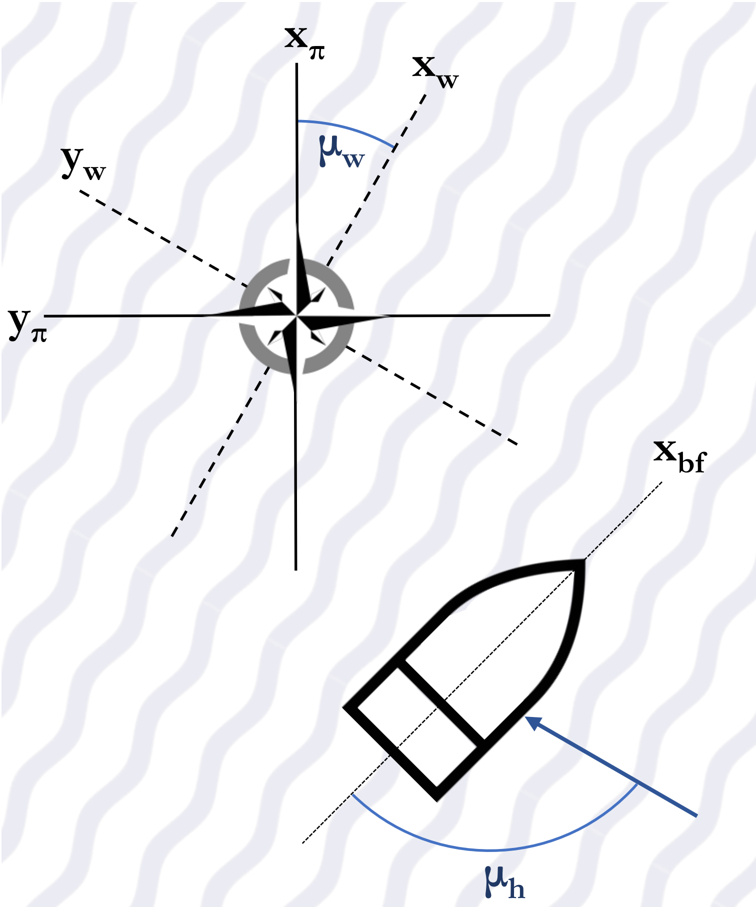

Figure 4.1: The three coordinate systems used for this study: wave-relative, north-east-down, and body-fixed.

As it is assumed that the vessels know their GPS coordinates, allowing them to resolve their location and orientation in the NED reference frame (x_π, y_π, z_π, μ_π). The angle the vessel moves along with respect to the NED coordinates has been termed the vessel heading (μ_π), otherwise known as navigational heading.

With positive x_π considered as north, μ_w is considered as positively increasing in a clockwise direction from x_π to x_w, where the WR coordinate system is considered as only rotationally displaced from the NED coordinate system in the xy-plane, but sharing the same origin. The clockwise rotation from the NED axes to the BF axes is also considered as positive μ_π. For example, in Figure 4.1, $\mu_w = 30°$, while $\mu_\pi = 315°$, and $\mu_h = 105°$.

Although the vessels have equivalent motion when encountering waves at equal wave headings on their port and starboard sides due to ship lateral symmetry, it is assumed that the distinction between port and starboard can be resolved. This could be achieved by analysing the vessel response cross-spectra (though not possible using the closed-form expressions for the vessel transfer functions given in [56] as the DOFs are considered decoupled), or via the inclusion of extra onboard sensors. The distinction between port and starboard is required to effectively resolve the wave direction.

Figure 4.2 shows the varied guidance approach previously described, which was implemented as follows. Firstly, π_1 was given a random vessel heading between $\mu_{\pi_1} = 0°$ and $\mu_{\pi_2} = 90°$. Then another vessel (π_2) traveled with a vessel heading of $\mu_{\pi_2} = \mu_{\pi_1} + 180°$, while a third vessel ($\pi_3$) was sent along a trajectory perpendicular to the other two, at $\mu_{\pi_3} = \mu_{\pi_1} + 90°$.

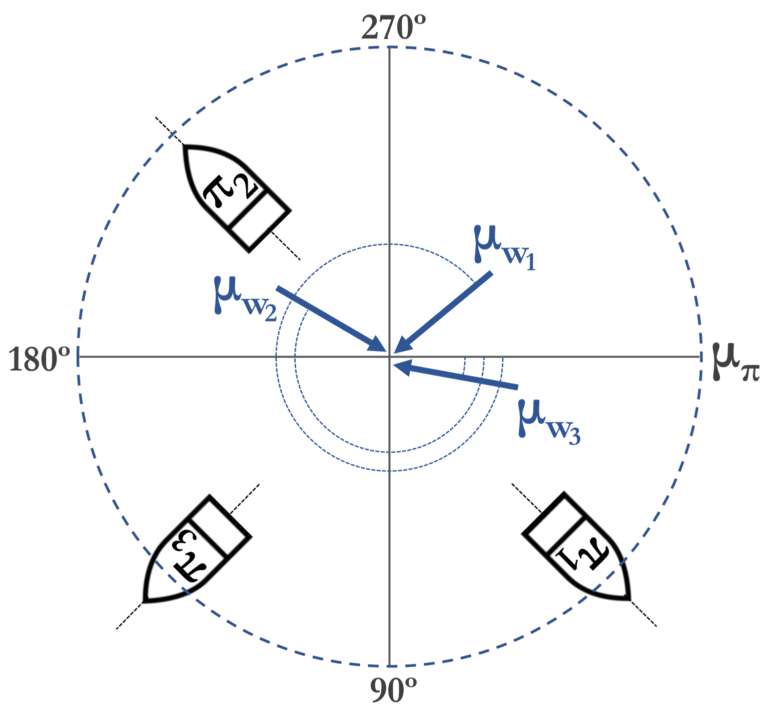

Figure 4.2: Wave direction determination strategy schematic. Three vessels are moving with vessel heading angles 90° apart, and three different wave directions shown relative to the vessels.

Table 4.1 describes the situation portrayed in Figure 4.2 as an example of the difference between vessel headings (μ_π), wave headings (μ_h), and wave direction (μ_w) for three different μ_w angles.

Table 4.1: Vessel heading versus wave direction versus wave heading example.

Wave direction	Vessel heading	Wave heading
$\mu_{w_1} = 320°$	$\mu_{\pi_1} = 45°$ $\mu_{\pi_2} = 225°$ $\mu_{\pi_3} = 135°$	$\mu_{h_1} = 95°$ $\mu_{h_2} = 85°$ $\mu_{h_3} = 5°$
$\mu_{w_2} = 210°$	$\mu_{\pi_1} = 45°$ $\mu_{\pi_2} = 225°$ $\mu_{\pi_3} = 135°$	$\mu_{h_1} = 15°$ $\mu_{h_2} = 165°$ $\mu_{h_3} = 105°$
$\mu_{w_3} = 10°$	$\mu_{\pi_1} = 45°$ $\mu_{\pi_2} = 225°$ $\mu_{\pi_3} = 135°$	$\mu_{h_1} = 145°$ $\mu_{h_2} = 35°$ $\mu_{h_3} = 55°$

In the case of the common heading multi-vessel SSE guidance strategy, the individual μ_h estimations have been aggregated together. However, the varied heading and random heading approaches required the μ_h angles to be transformed to μ_w. The aggregation of $\hat{\mu}_w$ introduced the requirement of combining estimations which vary across the 0° and 360° thresholds. As such, an algorithm was designed to account for the cases where one vessel's μ_w estimation was within the range of $(270°, 360°]$, while the other two were in the range of $[0°, 90°)$, and vise-versa (shown in Algorithm 1). Residual errors for the estimations were handled in a similar way.

Algorithm 1 Wave direction estimation aggregation algorithm.

if $(270° < \mu_{w_1} \leq 360°)$ & $(270° < \mu_{w_2} <= 360°)$ & $(0 \leq \mu_{w_3} < 90°)$ **then**
  $\mu_{w_\Pi} = ([\mu_{w_1} - 360°] + [\mu_{w_2} - 360°] + \mu_{w_3})/3$
else if $(270° < \mu_{w_1} \leq 360°)$ & $(0 \leq \mu_{w_2} < 90°)$ & $(0 \leq \mu_{w_3} < 90°)$ **then**
  $\mu_{w_\Pi} = ([\mu_{w_1} - 360°] + \mu_{w_2} + \mu_{w_3})/3$
else if $(270° < \mu_{w_1} \leq 360°)$ & $(0 \leq \mu_{w_2} < 90°)$ & $(270° < \mu_{w_3} \leq 360°)$ **then**
  $\mu_{w_\Pi} = ([\mu_{w_1} - 360°] + \mu_{w_2} + [\mu_{w_3} - 360°])/3$
else if $(0 \leq \mu_{w_1} < 90°)$ & $(0 \leq \mu_{w_2} < 90°)$ & $(270° < \mu_{w_3} \leq 360°)$ **then**
  $\mu_{w_\Pi} = (\mu_{w_1} + \mu_{w_2} + [\mu_{w_3} - 360°])/3$
else if $(0 \leq \mu_{w_1} < 90°)$ & $(270° < \mu_{w_2} \leq 360°)$ & $(270° < \mu_{w_3} \leq 360°)$ **then**
  $\mu_{w_\Pi} = (\mu_{w_1} + [\mu_{w_2} - 360°] + [\mu_{w_3} - 360°])/3$
else if $(0 \leq \mu_{w_1} < 90°)$ & $(270° < \mu_{w_2} \leq 360°)$ & $(0 \leq \mu_{w_3} < 90°)$ **then**
  $\mu_{w_\Pi} = (\mu_{w_1} + [\mu_{w_2} - 360°] + \mu_{w_3})/3$
else
  $\mu_{w_\Pi} = (\mu_{w_1} + \mu_{w_2} + \mu_{w_3})/3$
end if
if $\mu_{w_\Pi} < 0°$ **then**
  $\mu_{w_\Pi} = 360° + \mu_{w_\Pi}$
end if

In order to handle cases when the μ_h estimations exceeded the range of possible μ_h angles ([0°, 180°]), the μ_h estimate was returned to the bounds. For example, if $\hat{\mu}_h = -3°$, then it will be assumed that $\hat{\mu}_h = 0°$, or if $\hat{\mu}_h = 185°$, then it will be assumed that $\hat{\mu}_h = 180°$.

4.3 Uncertainty Metrics

When considering the SSE methodology described in Chapter 3, one major source of random error is caused by the stochastic nature of sea states [47]. Even during simulation, although a parameterised MPM spectrum was used to generate the wave system, the different components are randomly shuffled to ensure the sea state is random. While this is a property of the simulation setup, the same is equally true of a real sea state, the different wave components will randomly sum to produce the wave elevation. Therefore, the motion of the vessels to the same wave spectrum parameters will randomly vary. As such, inputs to the NNs for SSE will vary for the same vessel trajectory and parametric wave conditions, resulting in random estimation errors output by the NNs.

A major cause of bias errors in the SSE task is from the fact that vessels move along different trajectories. This means that each vessel taking an SSE will encounter waves differently as a function of time and space, resulting in a different motion response to the same global sea state. While the waves encountered are randomly generated, the trajectory of the vessel is not.

Therefore, two uncertainty metrics are presented in this section as tools to analyse the uncertainty related to each type of error. The first is time-based as a function of the random error, and the second is based on spectral estimates and has been used to assess bias errors.

4.3.1 Trial Run Length

In regular seakeeping experiments, the length of time of a seakeeping trial can be associated with the level of uncertainty in the results. Pierce [72] derived a set of equations which can be used to define certainty in a trial as a function of the length of time, along with two parameters defining a wave spectrum PSD: peak spectral frequency (ω_p) and half-power bandwidth ($\Delta\omega_{1/2}$) (seen in Figure 4.3).

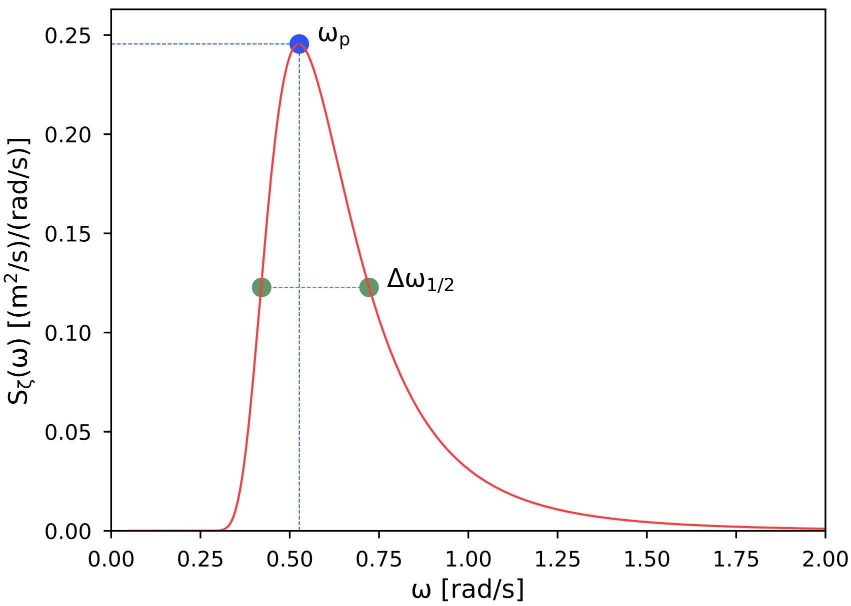

Figure 4.3: Sea state (MPM) spectrum showing peak spectral frequency and half-power bandwidth.

By assuming that the scatter of the SSEs is Gaussian about the mean (even though the process is not), the random error about the estimate standard deviation of the scatter (ξ_T) can calculated using Equation 4.1, where T_{sse} is the time of the trial. After inspecting the residual error distributions shown in Figures 3.28 to 3.30, it has been assumed that the error distributions will be Gaussian when multiple vessels are estimating the same wave conditions.

$$\xi_T = \sqrt{\frac{3\pi}{5\sqrt{2}\,\Delta\omega_{1/2}\,T_{sse}}} \tag{4.1}$$

ξ_T represents a ratio of error, so can be considered the estimated percentage error, or amount of uncertainty in a sea trial. For example, if $\xi_T = 0.05$, then there is estimated to be 95% certainty in the SSE results.

However, Equation 4.1 must be modified for a vessel with forward speed in order to convert the wave spectrum to the encounter wave spectrum. The half-power bandwidth can be transformed into the half-power encounter bandwidth ($\Delta\omega_{1/2_e}$) via Equation 4.2, and is now a function of the peak spectral frequency. α_e is an encounter conversion parameter, defined in Equation 4.3.

$$\Delta\omega_{1/2_e} = \Delta\omega_{1/2}|1 + 2\alpha_e\omega_p| \tag{4.2}$$

$$\alpha_e = -\frac{U_\pi cos(\mu_h)}{g} \tag{4.3}$$

Therefore, uncertainty in the SSEs for a vessel with forward speed can be estimated using Equation 4.4.

$$\xi_T = \sqrt{\frac{3\pi}{5\sqrt{2}\ \Delta\omega_{1/2_e}\ T_{sse}}} \tag{4.4}$$

Pierce intended Equations 4.1 to 4.4 to be used to estimate the time required to confidently assume a ship's motion response characteristics to a given sea state. However, this study hypothesises that they are applicable in the reverse case, that they can be used to define the time required to confidently estimate the causal sea state using vessel response data.

However, in order to estimate the length of a sea trial, the expected spectrum of the waves must be determined as ξ_T is a function of the half-power bandwidth. Therefore, when taking an SSE, if no prior knowledge of the wave conditions are known, ξ_T must be estimated in real time. It is proposed that the minimum requirement of 10 minutes (as stated by Nielsen [14]) can be used as a baseline, where an updating SSE can be used to pre-estimate the wave spectrum for the purpose of calculating the level of certainty in the SSEs. As the spectrum must be converted from incidental to encounter, this same logic applies to an estimation for wave heading. The convergence of uncertainty from the estimated spectra and known spectra over time has been explored in Section 4.5.3.

While Pierce describes a process for trial run estimation using vessel response spectra PSDs, as opposed to deriving a trial run estimate from wave spectra, the process is based on bias errors associated with his specific methodology for obtaining the response PSDs. As this study is not concerned with the error associated with the time-series response transformation methodology, this approach has been deemed inappropriate for the purpose of estimating a trial run length. Bias errors considered more relevant to this study are addressed in Section 4.3.2.

It should be noted that Pierce states that the approximations used for the errors and run length equations are conservative, meaning that there is slightly larger error and slightly longer run times than may be necessary. The appropriateness of the equations, however, have been assessed in Sections 4.5.2 and 4.5.3 of this chapter.

To investigate the effects of varying wave spectrum parameters on the trial run length, 500 randomly sampled significant wave heights and mean wave periods were used to generate MPM spectra, then the peak spectral frequency and half-power bandwidths were taken and used to calculate T_{sse} for a random error uncertainty of 5%. The results are shown in Figure 4.4.

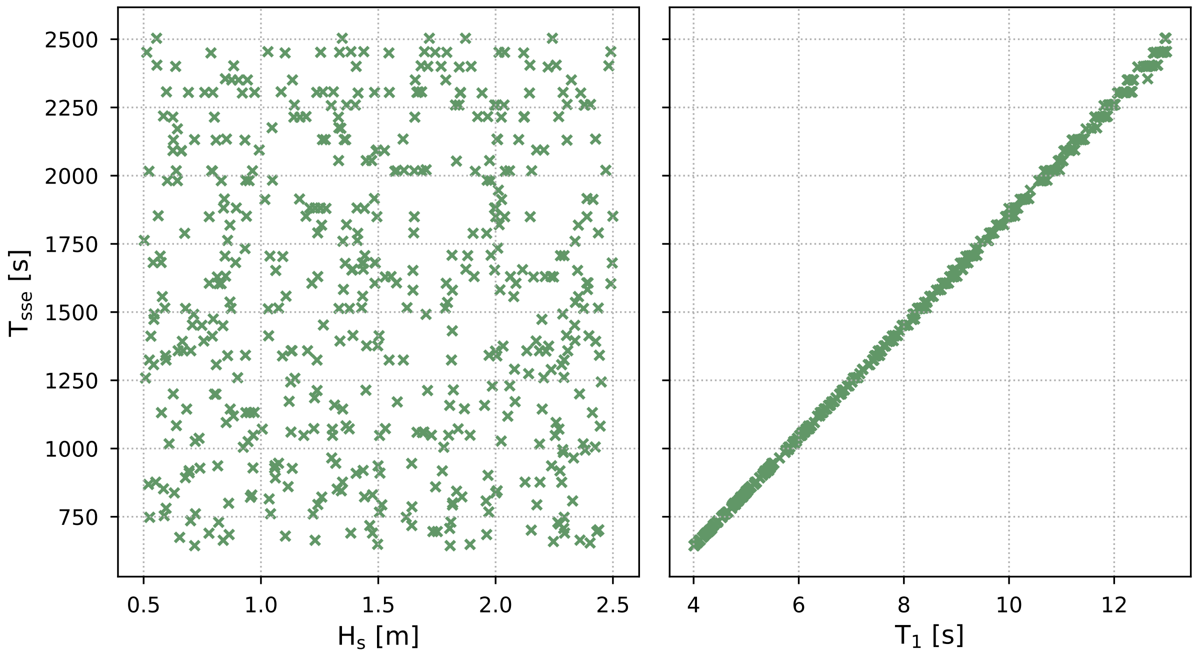

Figure 4.4: SSE 95% certainty trial run length for varying wave spectra at $U_\pi = 2m/s$ and $\mu_h = 130°$.

Immediately, it can be seen that H_s has little effect on the time required for an SSE trial, while T_1 has a clear linear positive relationship with trial time. Shorter wave periods require less time for an SSE, with a minimum time of $645s$ (or 10.7 minutes), while longer wave periods require more time for an SSE, with a maximum of $2503s$ (or 41.7 minutes). These results make sense statistically in terms of the improvement of an estimate when the waves are encountered more frequently (lower T_1 requires less time and vice-versa). Further, the range of SSE trial run times match very well with the minimum time for an SSE of 10 minutes given by Nielsen [14], and the maximum time of 40 minutes set for the simulated time-series used to train the SSE NNs.

The wave spectrum parameters were then fixed, while 500 randomly sampled wave headings and vessel speeds were generated to test their effects on the time required for an SSE trial for a random error uncertainty of 5%. The results have been plotted in 3D (Figure 4.5) to be able to capture the combined effects of U_π and μ_h.

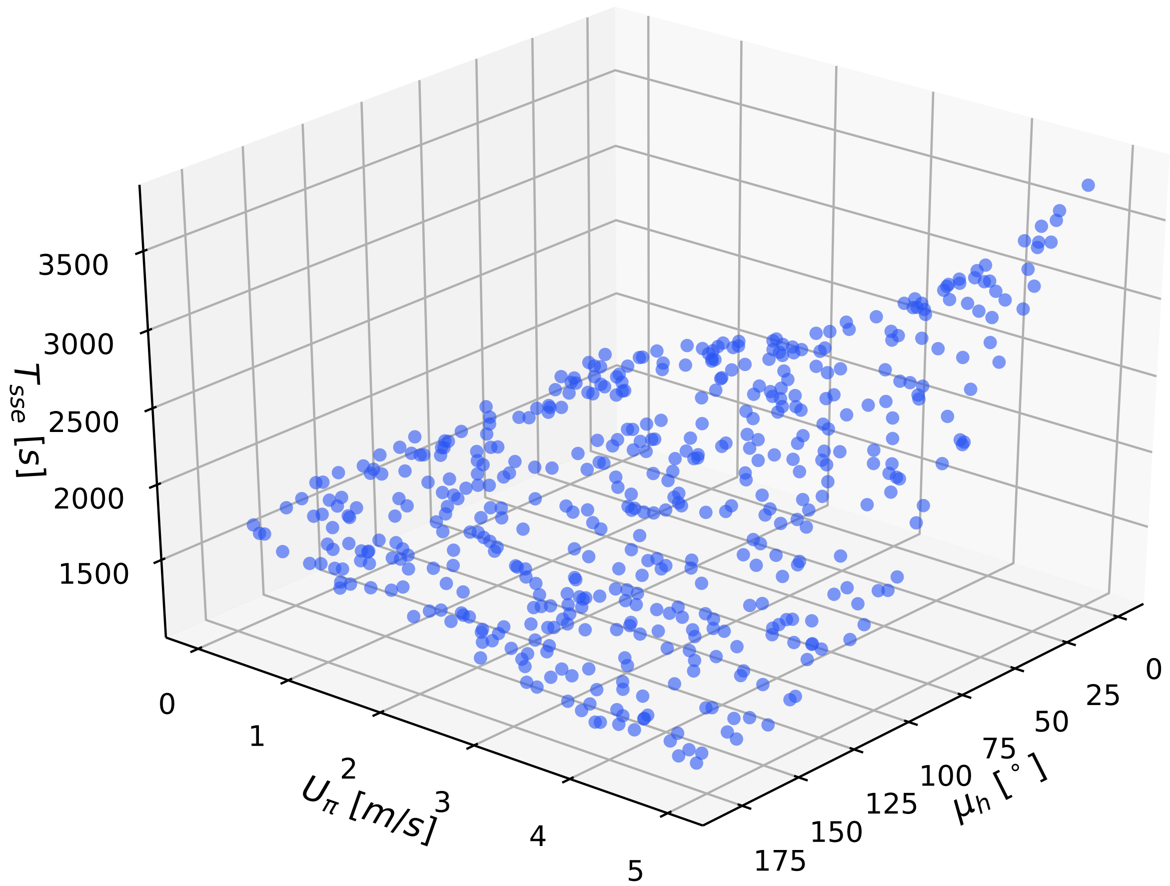

Figure 4.5: SSE 95% certainty trial run length for varying heading and speed in a MPM wave system for $H_s = 1.2m$ and $T_1 = 8.5s$ (3D projection).

The 3D plot reveals a few interesting trends. Firstly, there is an increase in time required for statistical certainty in estimations as the wave heading reduces from head seas to following seas. This is logical as the number of wave components encountered by a vessel will reduce as the wave heading angle is reduced from head sea to following sea, resulting in less information to make an SSE. Secondly, T_{sse} shortens as the vessel speed increases until the wave heading decreases below $\sim 90°$. Both of these trends tend towards equalising the wave encounter frequency (higher U_π and greater μ_h result in an increased encounter frequency). When μ_h reduced below 90°, though, there as a general increase in T_{sse} for greater U_π values. This would result in a maximum trial time exceeding 58 minutes for cases of following seas at high speed, though, so should be avoided when time-sensitivity is a factor (and if otherwise discernible).

4.3.2 Spectral Variation

While the time-based uncertainty metric is applicable to both single vessels and multiple vessels, Nielsen, Brodtkorb, and Sørensen [25] designed an uncertainty metric specifically for multi-vessel SSE.

The variation between each individual vessel spectral SSE ($\hat{S}_{\zeta}$), in this case, an MPM spectrum generated as a function of the estimated significant wave height and mean wave period, was hypothesised to be a useful metric of certainty in the aggregated SSEs [25]. In other words, if the vessels produce very similar spectral estimations, then that would indicate that there is a high level of certainty in the results. Equation 4.5 defines the frequency-wise variation in spectral estimates, where N_{π} is the number of vessels involved in the SSE task.

$$|\Delta\hat{S}(\omega)|^2 = \sum_{n=1}^{N_{\pi}-1} \sum_{m=n+1}^{N_{\pi}} [|\hat{S}_{\zeta_n}(\omega) - \hat{S}_{\zeta_m}(\omega)|^2] \tag{4.5}$$

For example, given $N_{\pi} = 3$ vessels, the right hand side of Equation 4.5 would sum as follows: $|\hat{S}_{\zeta_1}(\omega)-\hat{S}_{\zeta_2}(\omega)|^2+|\hat{S}_{\zeta_1}(\omega)-\hat{S}_{\zeta_3}(\omega)|^2+|\hat{S}_{\zeta_2}(\omega)-\hat{S}_{\zeta_3}(\omega)|^2$. The total uncertainty (ξ_S) is then taken as the frequency-wise integration of the spectral discrepancies ($\Delta\hat{S}(\omega)$), divided by the total energy of the average estimated sea state (the 0th order spectral moment), shown in Equation 4.6.

$$\xi_S = \frac{\frac{1}{N_{pi}} \int_0^{\infty} \Delta\hat{S}(\omega)\, d\omega}{\int_0^{\infty} \hat{S}_{\zeta}(\omega)\, d\omega} \tag{4.6}$$

High certainty, therefore, is achieved when $\xi_S \rightarrow 0$ as the estimations of all vessels are in agreement.

Figure 4.6 portrays three estimated spectra from three different vessels, plotted with the individual ordinates compared in Equations 4.5 and 4.6 highlighted as markers (however, the estimations were fabricated to better highlight the ordinates). As an example, the spectral ordinates for a wave frequency of $\sim 0.46 rad/s$ are highlighted in black.

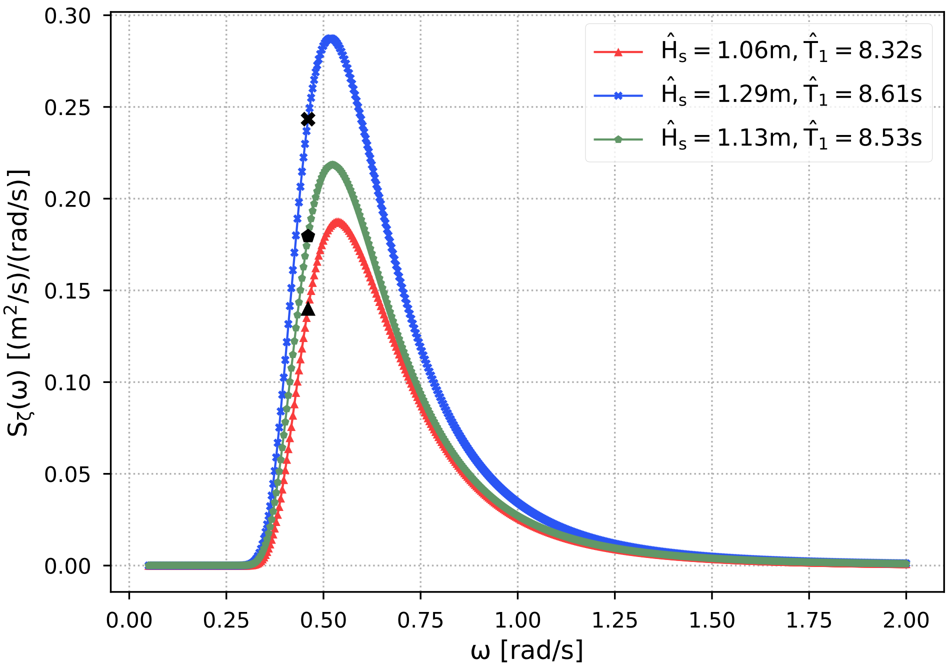

Figure 4.6: SSE spectra for comparison with three estimated spectra showing their ordinates.

4.4 Statistical Analysis

In order to test the proposed hypotheses and analyse the performance of the multi-vessel SSE guidance approaches, three equations must be defined to analyse the results statistically.

Firstly, the ratio of the multi-vessel SSE RMSE to the average single vessel SSE RMSE was used as an indication of the percentage of improvement of estimation precision when using multiple vessels, calculated in Equation 4.7.

$$\%_{RMSE} = 100\,\frac{RMSE_{\Pi}}{(RMSE_1 + RMSE_2 + RMSE_3)/3} \tag{4.7}$$

Secondly, to quantify an amount of variation between vessel headings within a single trial, Equation 4.8 calculates the standard deviation (s_μ) of the heading angles, where μ_{h_n} is wave heading angle of the individual vessels, and $\bar{\mu}_h$ is the average wave heading angle [73].

$$s_\mu = \sqrt{\frac{\sum(\mu_{h_n} - \bar{\mu}_h)^2}{N_\pi - 1}} \tag{4.8}$$

Finally, a t-test was used to measure the statistical significance of the SSE results. A t-test gives an indication of how significant the difference between two sets of data is by comparing their averages. In other words, whether or not the difference between two sets of data is likely to be due to random chance, or if their difference is related to distinct characteristics of the populations themselves [74].

In the case of SSE, the t-tests have been applied to the residual errors of the different multi-vessel guidance approaches in order to determine whether or not the approaches resulted in significantly different SSE performance. The two sample t-test was used to compare the results of the aggregated multi-vessel SSEs for the different guidance approaches.

Equation 4.9 gives the formula for a two-sample t-test [74]. $\bar{x}_1$ and $\bar{x}_2$ are generic variables representing the means of the residual errors for one of the estimated wave conditions for the different combinations of two guidance approaches. The pooled standard deviation (s_x^2) combines the standard deviations of both residual groups ($s_x^2 = \sqrt{(s_{x_1}^2 + s_{x_2}^2)/2}$), and the number of residuals in each group is represented as n_x.

$$t_{sig} = \frac{\bar{x}_1 - \bar{x}_2}{\sqrt{s_x^2(\frac{1}{n_{x_1}} + \frac{1}{n_{x_2}})}} \tag{4.9}$$

The larger the t-value (t_{sig}), the greater the value of the pooled standard deviation is, indicating that there is a more significant difference between the two groups of residual errors. A p-value (p_{sig}) is also often given alongside the t-value, which represents the probability that the two sets of results occurred by random chance. Lower p-values indicate that there is a low probability of the results differing by random chance.

4.5 Results and Discussion

4.5.1 Multi-vessel Sea State Estimation

In order to evaluate the proposed heading variation hypothesis, a sample case was simulated for the varied heading approach, common heading approach, and random heading approach, where the speed of the three vessels was fixed at $U_{\pi_1} = 2.00m/s$, $U_{\pi_2} = 3.2m/s$, and $U_{\pi_3} = 4.5m/s$, as well as the wave properties being fixed at $H_s = 1.2m$ and $T_1 = 8.5s$. The vessels'response PSD total powers for each DOF were then recorded. The total powers of the response spectra provides a single value for the response which can be used to compared the variation between the response spectra.

For all of the vessels in the common heading example, $\mu_h = 130°$. It should be noted that although the vessels share a common heading, they are initialised at different NED locations, meaning that they encounter different wave components at different times from one another. For the varied heading strategy, $\mu_{h_1} = 102°$, $\mu_{h_2} = 78.3°$, and $\mu_{h_3} = 168°$. Finally, for the random heading strategy, $\mu_{h_1} = 85.4°$, $\mu_{h_2} = 134°$, and $\mu_{h_3} = 63.8°$.

Table 4.2 gives the total power of the spectra for the heave responses, Table 4.3 shows the total powers for the pitch responses, and Table 4.4 presents the total powers for the roll responses.

Table 4.2: Example of PSD total powers for heave responses from three guidance strategies.

Guidance approach	π_1	π_2	π_3
Common heading	2.23	2.23	2.25
Varied heading	2.41	2.44	2.43
Random heading	2.51	2.49	2.41

When looking at the total heave response PSD powers for each of the guidance approaches (Table 4.2), all of them showed little variation between the different vessels. Further, the magnitudes of all of the vessel responses were very similar across each of the guidance approaches. By looking at the FRFs for heave in Figure 3.8, the variation in vessel responses to different μ_h angles represented less than 10% of relationship between wave frequency and ship motion. Therefore, it is likely that varying the μ_h angles of the vessels did not result in a very large variation in motion response.

Table 4.3: Example of PSD total powers for pitch responses from three guidance strategies.

Guidance approach	π_1	π_2	π_3
Common heading	9.63	9.70	10.1
Varied heading	1.03	1.02	25.3
Random heading	0.167	12.1	4.82

The difference between guidance approaches was a lot more apparent for the total power of the pitch PSDs (Tables 4.3) than for heave. The common heading strategy shows similar values for the different vessels, as it did for heave. When considering the differences between individual vessels, the random heading approach had the greatest variation in response PSD total power. The varied heading strategy had two vessels with similar PSD variances, and one with a very different total PSD power.

This reveals a consequence of forcing the vessels to travel at headings which are 90° apart. Although the case of π_1 and π_2 travelling in head and following seas, and π_3 travelling in beams seas, maximises the sea conditions encountered by the vessels, if the vessels are travelling such that π_3 is in head or following seas, both π_1 and π_2 will be in beam seas. However, the difference between the total pitch PSD powers for π_1 / π_2 and π_3 for the varied heading approach was greater than between any of the random heading vessels.

The amount of total power in the pitch PSDs reflects how the FRFs vary with μ_h angles (see Figure 3.9). The vessels which are travelling in beam seas experienced lower total power values than those in head seas, with π_1 for the random heading strategy being the closest to $\mu_h = 90°$ and having the lowest total power, while π_3 for the varied heading strategy was the closest to $\mu_h = 180°$ and had the highest total power.

Table 4.4: Example of PSD total powers for roll responses from three guidance strategies.

Guidance approach	π_1	π_2	π_3
Common heading	19.3	20.1	21.2
Varied heading	31.8	30.3	1.60
Random heading	33.1	19.2	23.4

The roll response PSD total powers showed a similar trend to the pitch response total PSD powers. The total powers for the common heading approach were all quite similar, while the random heading strategy had the greatest difference in total powers between individual vessels. The total power in the vessel roll responses for the varied heading strategy was the inverse of the pitch response total powers, with π_1 and π_2 having large total powers, while π_3 had very little power. Again, this trend can be understood by looking at the roll response FRFs in Figure 3.13.

The variation in total powers of the vessels response spectra can also be understood by considering what is actually happening to a vessel as it moves across a wave series. While the vessel will filter out higher wave frequencies, given the small size of the WG, one could imagine it would displace vertically across a similar range of heights when travelling in most directions, albeit at different rates of displacement. The pitch and roll responses; however, would change significantly depending on the μ_h angles they encounter the waves at, as the vessel rotates differently in relation to the wave slopes.

Though the total PSD powers from Tables 4.2 to 4.4 support the idea of more varied information being present in vessels with greater wave heading variation, more data was needed to test the hypothesis.

Therefore, in order to evaluate the performance of the multi-vessel SSE approaches, 600 simulations were run for each of the guidance approaches: common heading, varied heading, and random heading. As such, 1800 individual vessel estimations were taken for each strategy, which aggregated to 600 multi-vessel estimations. The vessel headings were randomly generated in all cases, and the 3-DOF NN SSE models designed in Chapter 3 were used for all strategies.

The RMSE values for each individual vessel's estimations, as well as aggregated multi-vessel estimations, using the common heading approach, are given in Table 4.5. Further, the ratios of the multi-vessel SSE RMSE to the average single vessel SSE RMSE are presented. The individual RMSEs were calculated for each vessel in the same manner as presented in Chapter 3, while the aggregated estimations were found by taking the aggregated estimates (the average of the three vessels), then the RMSEs were calculated using the residuals from the combined estimations.

Table 4.5: Individual vessel versus aggregated multi-vessel SSE for common heading.

	π_1	π_2	π_3	Π	$\%_{\mathrm{RMSE}}$
RMSE ($\hat{\mathrm{H}}_{\mathrm{s}}$)	$4.85mm$	$6.05mm$	$5.03mm$	$4.56mm$	85.9%
RMSE ($\hat{\mathrm{T}}_1$)	$0.155s$	$0.192s$	$0.158s$	$0.142s$	84.4%
RMSE ($\hat{\mu}_{\mathrm{h}}$)	4.94°	3.22°	2.87°	2.81°	76.4%

When evaluating the improved performance of multi-vessel SSE compared to single vessel SSE, the H_s estimates saw an 14.1% reduction in RMSE when using multiple vessels, the T_1 estimates experienced a 15.6% reduction, and the μ_h estimates reduced the RMSE by 23.6%. As such, using multiple vessels with a common heading improved the estimation of μ_h the most, with H_s and T_1 improving by similar amounts.

The individual vessel and aggregated multi-vessel estimation RMSE values for the varied heading guidance approach are given in Table 4.6, as well as the multi-vessel to average single vessel RMSE ratios.

Table 4.6: Individual vessel versus aggregated multi-vessel SSE for varied heading.

	π_1	π_2	π_3	Π	$\%_{\text{RMSE}}$
RMSE ($\hat{\text{H}}_\text{s}$)	$4.97mm$	$4.94mm$	$5.33mm$	$4.08mm$	80.3%
RMSE ($\hat{\text{T}}_1$)	$0.149s$	$0.167s$	$0.146s$	$0.116s$	75.3%
RMSE ($\hat{\mu}_\text{w}$)	4.98°	3.03°	5.14°	2.75°	62.7%

Looking at the RMSE ratios between the multi-vessel RMSEs and individual vessel RMSEs, the $\hat{H}_s$ RMSE reduced by 19.7% and the $\hat{T}_1$ RMSE reduced by 24.7%. The $\hat{\mu}_w$ RMSE showed the best increase in estimation performance, dropping by 37.3%. When comparing this with the common heading guidance strategy, there was the least drop in $\hat{H}_s$ errors. The estimation performance of multiple vessels estimating T_1 and wave propagation angle reduced the RMSE by almost 10% more when using the varied heading strategy compared with the common heading strategy, while the wave propagation angle was the wave condition which saw the largest improvement in SSE for both strategies.

The RMSE values for the individual vessel SSEs and aggregated multi-vessel SSEs are given in Table 4.7, as well as the percentage ratios of average single vessel RMSE to multi-vessel RMSE.

Table 4.7: Individual vessel versus aggregated multi-vessel SSE for random headings.

	π_1	π_2	π_3	Π	$\%_{\text{RMSE}}$
RMSE ($\hat{\text{H}}_\text{s}$)	$4.50mm$	$4.27mm$	$5.85mm$	$3.92mm$	80.4%
RMSE ($\hat{\text{T}}_1$)	$0.195s$	$0.146s$	$0.186s$	$0.123s$	70.0%
RMSE ($\hat{\mu}_\text{w}$)	4.28°	6.87°	4.56°	3.31°	63.2%

The random heading strategy showed similar SSE error reduction, for multi-vessel SSE compared to single vessel SSE, as the varied heading approach. The $\hat{\mu}_w$ RMSE reduced the most of the three wave conditions estimated, dropping by 36.8%, with the $\hat{T}_1$ RMSE decreasing by 30%, and the $\hat{H}_s$ RMSE went down 19.6%.

All of the multi-vessel SSE approaches were shown to reduce estimation error compared to a single vessel. The varied heading and random heading strategies performed better than the common heading approach when estimating the wave propagation angle and T_1, in terms of SSE error reduction. This supports the hypothesis that the variation of vessel state, by varying the relative wave headings, helps to reduce biases related to a common filter type.

However, the estimation of H_s showed similar performance improvement for all of the multi-vessel guidance approaches over single vessel estimations. Given the small variation in PSD total powers for heave, as can be seen in Table 4.2, and the strong relationship between heave and H_s estimation shown in Chapter 3, the similar estimates are likely due to the similar heave response information extracted across the range of wave headings.

Interestingly, the NNs for each of the three estimated wave properties showed better performance after they were trained on the full set of 48,000 simulation runs (original training and testing data), than when trained on 43,200 runs (the original training data). This indicates that the use of more training data could improve SSE NN estimation accuracy.

To investigate whether the differences in performance of the three guidance approaches could be attributed to chance or whether they are a result of the variation in wave encounter headings, two sample t-tests were performed on the different combinations of guidance approaches, with the derived t-values and p-values given in Table 4.8.

Table 4.8: T-test comparison of multi-vessel guidance approaches for SSE.

	Common & Varied	Common & Random	Varied & Random
t_{sig} ($\hat{\mathrm{H}}_\mathrm{s}$)	0.260	-1.40	-1.77
p_{sig} ($\hat{\mathrm{H}}_\mathrm{s}$)	*0.794*	*0.161*	*0.075*
t_{sig} ($\hat{\mathrm{T}}_1$)	-1.75	-2.55	-0.948
p_{sig} ($\hat{\mathrm{T}}_1$)	*0.080*	0.011	*0.343*
t_{sig} ($\hat{\mu}$)	17.4	18.0	2.15
p_{sig} ($\hat{\mu}$)	$1.08 \cdot 10^{-60}$	$2.62 \cdot 10^{-64}$	0.031

Overall, the differences between the three different multi-vessel SSE guidance strategies was found to be most significant for the estimation of wave propagation angle, with the greatest disparity shown between the common and varied, and common and random guidance strategies, which both had **very** little probability that the discrepancy between results was produced randomly. The highest probability that two approaches had different estimation results caused by chance, was for the estimation of H_s between the common and varied approaches, which had a 79.4% probability. Each of the values shown in italics exceed a 5% threshold that the variation in results was caused by chance. Significant wave height was, therefore, the least significantly affected by the guidance strategy used when using multiple vessel for SSE, while only the difference between common and random heading approaches had less than a 5% probability of being due to chance.

The average μ_h variation between the three vessels in the varied heading and random heading strategies was calculated, where the standard deviation for each trial run was taken and then averaged across the 600 trials. The results showed that the varied guidance approach had an average μ_w standard deviation of 48.7°, while the random guidance approach approach had an average μ_h standard deviation of 92.1°. This shows that the vessels using the random heading strategy, on average, had a greater variation in μ_w angles than the forced varied heading strategy. This increase in μ_h variation of the random heading approach did not have a large effect on SSE performance, where only the T_1 estimations had a greater reduction in residual error when compared with single vessel SSE. Therefore, although variation of vessel headings showed improved SSE performance when compared with vessels with a common trajectory, an increase in heading variation did not result in any significant further reduction in SSE error.

To give an example of how the multi-vessel guidance approaches reduced the spread of SSEs errors, residual plots for the varied heading multi-vessel SSE strategy are shown in Figure 4.7. $\hat{H}_s$ is on the left, $\hat{T}_1$ is in the middle, and $\hat{\mu}_w$ on the right. The residuals from the individual vessels are shown in pink, while the aggregated estimations of the multiple vessels are shown in blue.

For all three wave conditions, it can be seen that the spread of residual errors was reduced for the multi-vessel approach when compared with the spread of the single vessel residual errors. This appears most prominent for the spread of $\hat{T}_1$ errors, which aligns with the RMSE ratio results given in Table 4.6. Most major outliers are removed from the data by aggregating the single vessel SSEs.

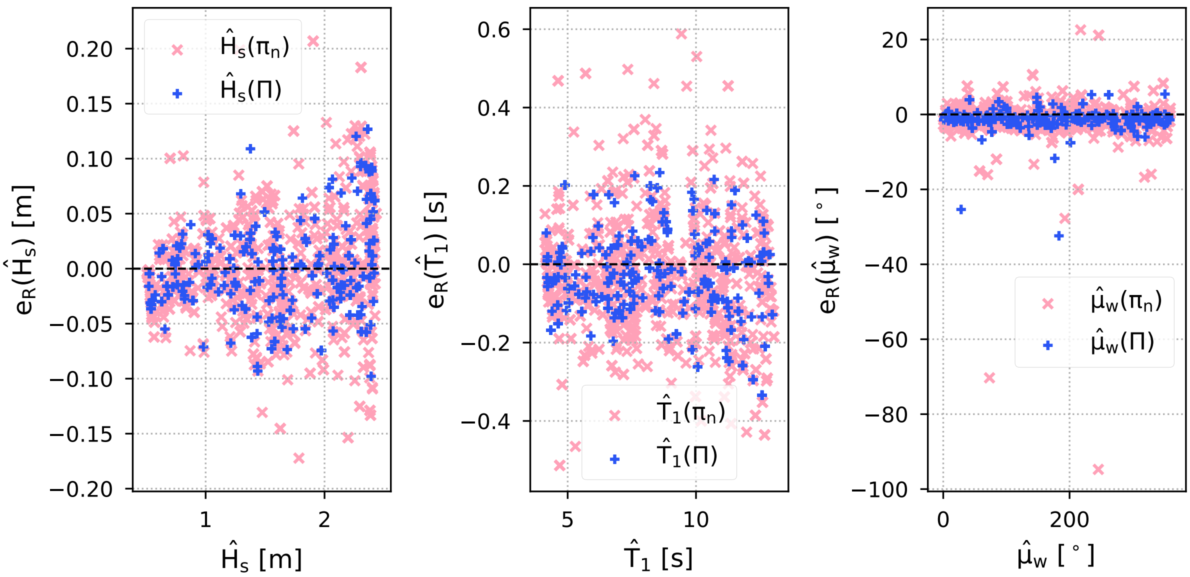

Figure 4.7: Residual plots for multi-vessel SSE using the varied heading approach for $\hat{H}_s$ (left), $\hat{T}_1$ (middle), and $\hat{\mu}_w$ (right).

The general distribution trends for residual errors for $\hat{H}_s$ and $\hat{T}_1$ matched those seen in Figures 3.28 and 3.29. There was an increase in error associated with the estimation of higher $\hat{H}_s$ values, and a quite even spread across $\hat{T}_1$ values. The distribution of errors for $\hat{\mu}_w$ appeared to be more evenly spread than for the μ_h estimations given in Figure 3.30, although there were still more errors congregated around both $\sim 90°$ and $\sim 270°$ (as was seen about $\sim 90°$ in Figure 3.30).

4.5.2 Offline Uncertainty Analysis

Before testing how the uncertainty metrics described in Section 4.3 could be used to assess online SSEs, their applicability to multi-vessel SSE in general has been investigated by applying them to the final wave property estimations.

Although the time-based uncertainty metric described in Section 4.3 was originally designed under the assumption that the sea spectrum was known, the uncertainty results presented here were calculated using the estimated H_s, T_1, and μ_h values from each vessel.

Figure 4.8 displays the residual errors versus time-based vessel-specific uncertainty for the three vessels involved in the varied heading multi-vessel SSE simulations, for each wave condition estimated. The residuals were also grouped into five bins as a function of uncertainty, then the RMSE value for the bin was taken and plotted against the mean of the each bin to give an indication of the error distribution. Table 4.9 then divides the estimation residual errors into five bins as a function of associated time-based uncertainty.

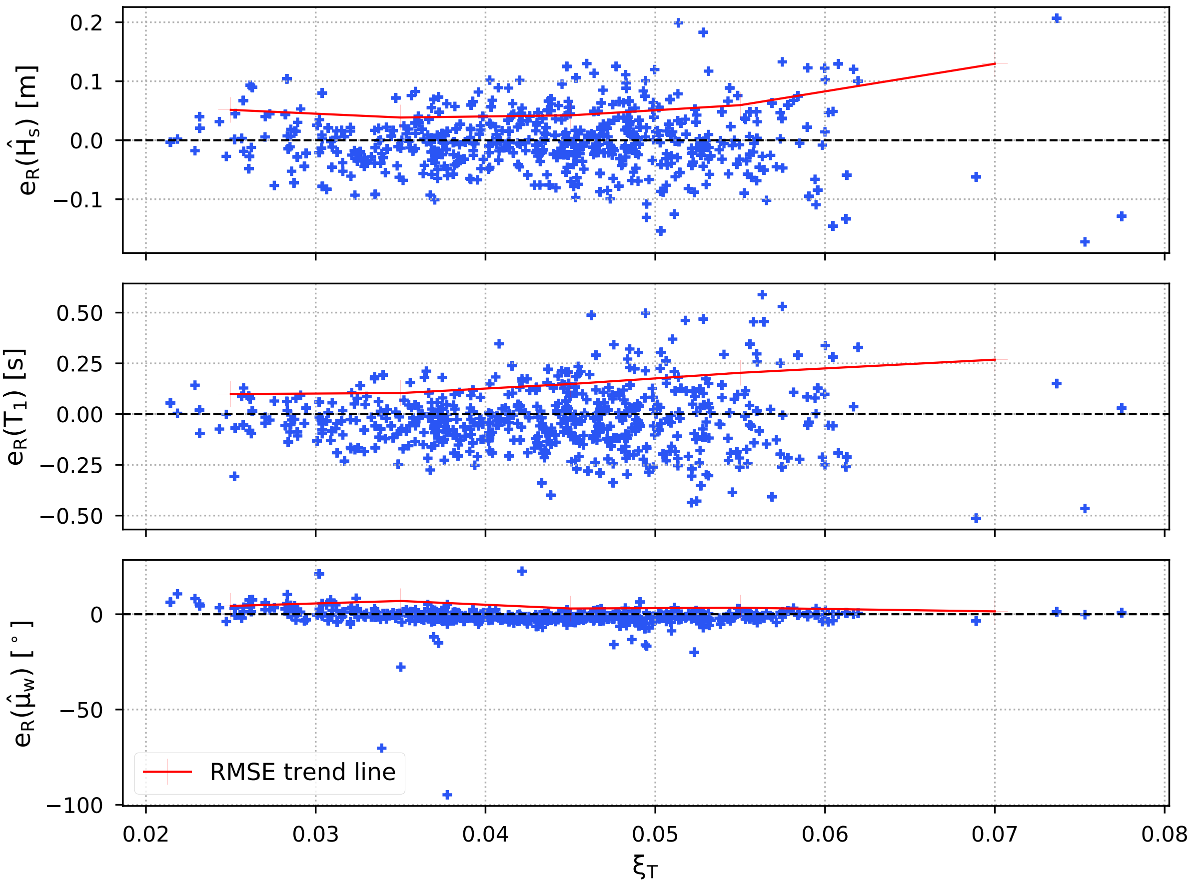

Figure 4.8: Time-based uncertainty versus final SSE residual error for three vessels using the varied guidance strategy.

By looking at the RMSE trend line, it can be seen that there is a clear relationship between the uncertainty in the T_1 estimations and the error associated with the estimations, where the spread of errors reduced as uncertainty reduced. This has been quantified in Table 4.9, where the RMSE values were found to decrease for each grouping as uncertainty decreased. The same pattern was mostly present for $\hat{H}_s$ as well, with the magnitude of errors decreasing as uncertainty decreased, until an uncertainty below approximately 0.035 was reached, where errors increased again. The distribution of errors for $\hat{\mu}_w$ did not follow any obvious trends, with the RMSE values fluctuating across the uncertainty groupings.

Table 4.9: Time-based uncertainty binned data and corresponding RMSE values.

Range (ξ_T)	RMSE ($\hat{H}_s$)	RMSE ($\hat{T}_1$)	RMSE ($\hat{\mu}_w$)
0.02 - 0.03	$5.15mm$	$0.099s$	4.35°
0.03 - 0.04	$3.85mm$	$0.104s$	6.95°
0.04 - 0.05	$4.22mm$	$0.148s$	3.00°
0.05 - 0.06	$5.95mm$	$0.204s$	3.39°
0.06 >	$13.0mm$	$0.268s$	1.49°

The time-based uncertainty metric resulted in the majority of errors being in the uncertainty range of 0.02 and 0.06. Depending on the desired SSE accuracy, the metric gives reasonable results for $\hat{T}_1$ and $\hat{H}_s$; however, was not found to be suitable for $\hat{\mu}_w$. Should a different level of accuracy be desired, the time-based uncertainty metric could be scaled to associate a residual RMSE with an appropriate level of uncertainty. It should be noted that a Shapiro test for normality [75] of the error distribution of the estimations showed that the error distributions had a low probability of being Gaussian ($< 5\%$), which could have impacted the effectiveness of the time-based uncertainty metric.

The distribution of final aggregated multi-vessel SSE errors are plotted against spectral uncertainty in Figure 4.9, then grouped as a function of their associated uncertainty in Table 4.10, with the SSE RMSE values for each group given.

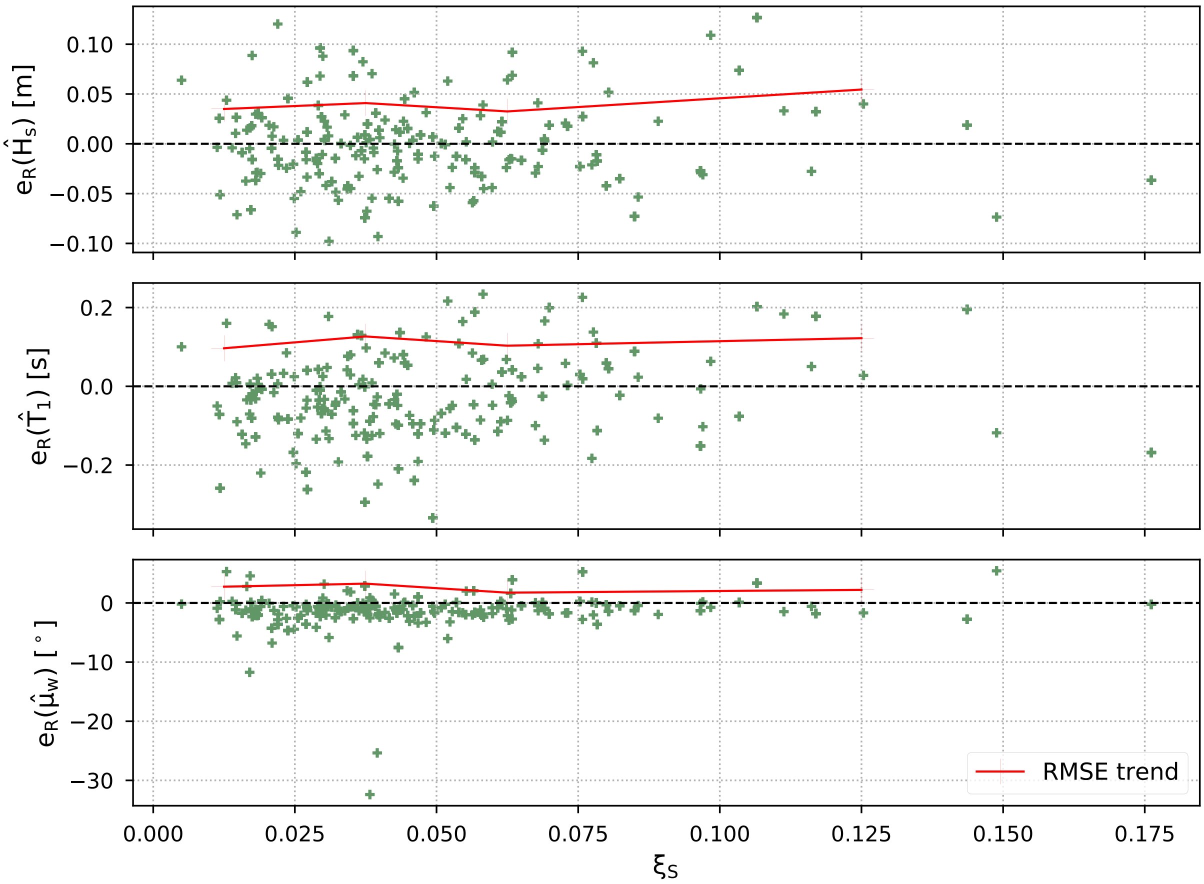

Figure 4.9: Spectral uncertainty versus final multi-vessel SSE aggregated residual error for the varied guidance strategy.

Looking at the errors associated with $\hat{H}_s$, $\hat{T}_1$ and $\hat{\mu}_w$, errors were not found to decrease as uncertainty reduced, rather, there was a fluctuation of error distribution.

Table 4.10: Spectral uncertainty binned data and corresponding RMSE values.

Range (ξ_S)	RMSE ($\hat{H}_s$)	RMSE ($\hat{T}_1$)	RMSE ($\hat{\mu}_w$)
0 < 0.025	3.50mm	0.097s	2.75°
0.025 - 0.050	4.09mm	0.127s	3.27°
0.050 - 0.075	3.25mm	0.103s	1.75°
0.075 >	5.45mm	0.122s	2.21°

Based on the results from this study, the time-based uncertainty metric can be applied to H_s and T_1 offline estimations, while the spectral uncertainty metric failed to accurately describe the offline estimations results. The level of certainty could be scaled to relate to defined RMSE values, depending on the application and desired accuracy of SSE.

The spectral uncertainty metric for multi-vessel SSE using SAWB was originally designed for vessels which estimated the entire range of sea spectrum frequency components [25]. In contrast, the spectra used in Equation 4.6 for this study were reconstructed using the H_s and T_1 wave parameters. Therefore, the discrepancy between each individual estimated spectral ordinate and the actual ordinate could have been higher when using MPM spectra as the parameterisation only reconstructed the same spectral **shape**, rather a component-wise representation, resulting in a higher uncertainty value.

4.5.3 Online Estimation

It was hypothesised that the use of vessel response spectral data for SSE would provide an effective means to estimate sea states in real time, given that spectral data can represent time-series signals statistically. Further, it was hypothesised that the uncertainty metrics defined in Section 4.3 could be applied to the online SSEs to assess their quality in real time.

In order to investigate the performance of the NNs (described in Chapter 3) when updating SSEs online, a sample case was run using the varied heading guidance approach, with the resulting evolution of $\hat{H}_s$ and $\hat{T}_1$ given in Figure 4.10, and $\hat{\mu}_h$ and $\hat{\mu}_w$ shown in Figure 4.11. Estimations were taken every 30 seconds, after an initial 10 minutes of response data was recorded and up until 40 minutes, under stationary conditions (fixed U_π and μ_h). All of the examples are for the same simulation, with the conditions given in Table 4.11.

Table 4.11: Online SSE example conditions (varied headings).

Variable	Value
Sig. wave height (H_s)	1.20m
Mean wave period (T_1)	8.50s
Wave direction (μ_w)	108°
Wave heading for π_1 (μ_{h_1})	113° (starboard)
Wave heading for π_2 (μ_{h_2})	66.2° (port)
Wave heading for π_3 (μ_{h_3})	156° (port)

In Figure 4.10, the H_s estimations of all three vessels evolved similarly, with their initial estimations being of low H_s values, then increasing quite linearly over time. The final estimation for all vessels, as well as aggregated estimation, were all highly aligned with the actual H_s value of 1.2m. However, the accurate estimates were only achieved at the end of online estimation task. This trend was commonly seen in the H_s online estimation process.

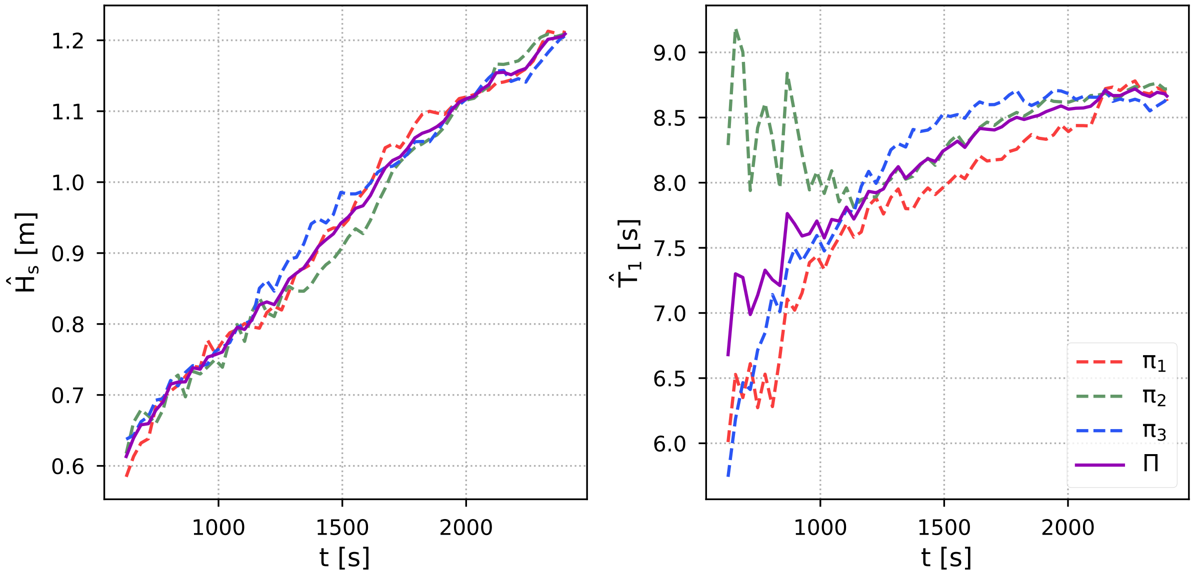

Figure 4.10: Multi-vessel H_s estimation (left) and T_1 estimation (right) versus time for individual vessel and aggregated SSEs (for $H_s = 1.2m$ and $T_1 = 8.5s$).

When estimating T_1, the three vessels initially had a much greater variation in $\hat{T}_1$ than they did for $\hat{H}_s$. The estimations converged to being relatively similar after $\sim 1100s$, slowly increasing in value, resulting in an aggregate overestimation of $\hat{T}_1 = 8.66s$ (for $T_1 = 8.5s$). While the T_1 estimates tended to increase over time, the relationship appeared much less linear, with the estimations flattening towards a solution sooner than for H_s.

Figure 4.11 provide an example of the online estimation process for μ_h and μ_w for the same three vessels, as well as aggregated estimate for μ_w. The transformation of the relative wave heading of each vessel into global wave directions can be discerned.

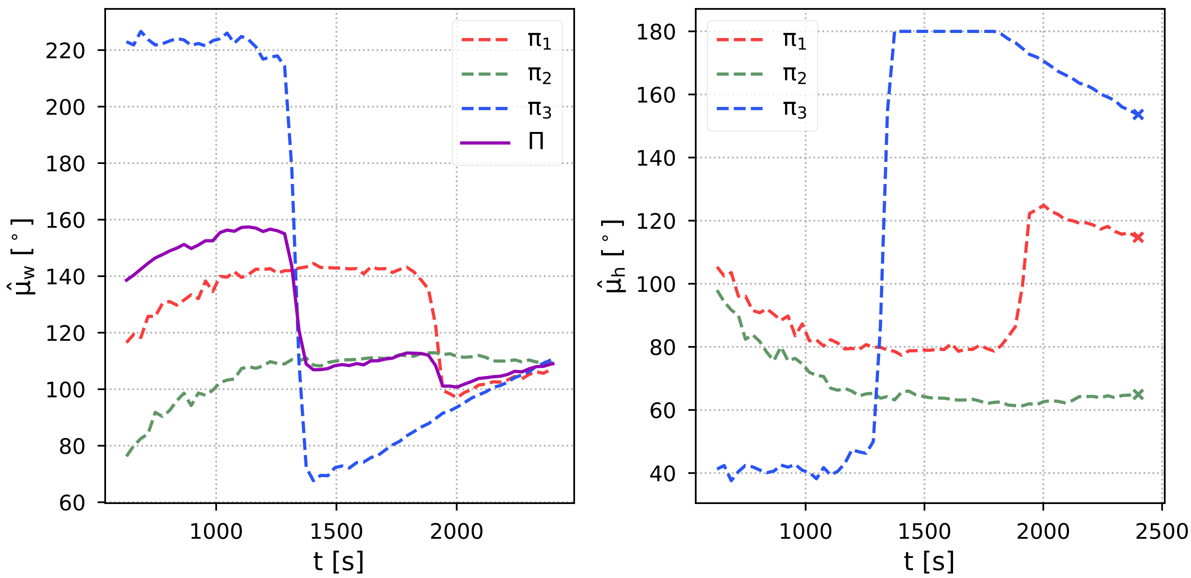

Figure 4.11: Multi-vessel μ_w (left) and μ_h (right) estimation versus time for individual vessel and aggregated SSEs (for $\mu_w = 108°$ and $\mu_h = [113°, 66.2°, 156°]$).

Immediately it can be seen that there is one major change in aggregated μ_w estimate around approximately $1300s$. This was caused by the π_3 vessel changing its μ_h estimate from being in following sea to being in head sea. A second prominent change in $\hat{\mu}_{w_\Pi}$ occurred at around $1900s$, when the H_{s_1} estimate increased from $\sim 80°$ to $\sim 120°$.

One other noticeable feature in Figure 4.11 was that π_3 likely estimated the wave heading as greater than 180° from $\sim 1350s$ to $\sim 1800s$, so the algorithm reduced it to exactly 180° (as desired).

The trends seen in the estimation of wave properties over time, given Figures 4.10 and 4.11, were common across simulation runs. As such, while at times the NNs could converge on an accurate estimations before the 40 minute time period elapsed, overall the SSE models were not effective at estimating wave properties in real time.

The following subsections examine how this affected the uncertainty metrics when predicting the performance of the multi-vessel SSE task, as well as investigate the likely cause for the unreliable online estimation results.

Online Uncertainty Metrics

Figure 4.12 shows how the time-based and spectral uncertainty metrics vary for multi-vessel online estimation over the estimation period of 10 to 40 minutes, for the example simulation conditions given in Table 4.11, where the uncertainty metrics were based on the aggregated estimated wave spectrum (taken every 30 seconds).

As discussed in Section 4.3, the time-based uncertainty metric was originally designed using a known sea spectrum. Therefore, it was decided to test how well the metric performed when using the estimated sea spectrum, instead of the known spectrum.

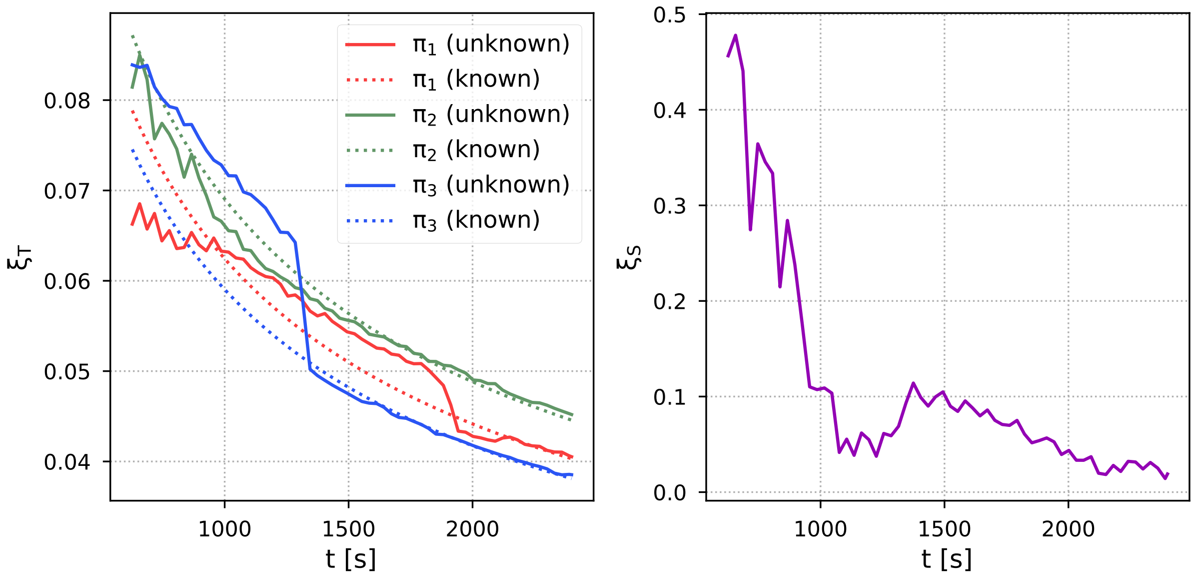

Figure 4.12: Online time-based (left) and spectral (right) uncertainty analyses for multi-vessel SSE versus time.

The spectral uncertainty showed the three different vessels rapidly converging towards the same spectral estimates at around 1100s, before diverging until around 1400s, then slowly converging again. This can be understood by looking at Figure 4.10, where the individual T_1 estimates followed the same pattern (H_s estimations were more similar across the whole time period).

When comparing the time-based uncertainty for the known sea spectra and estimated spectra, it can be seen that the estimated and known uncertainty levels converge as the estimations improve. The rapid changes in μ_h estimates for π_3 and π_1 are easily seen in the variation of their uncertainty. Of course, time-based uncertainty in the results could vary by a few percentage points if there are large errors in the estimations.

While this is only a single example of the progression of uncertainty during a simulated SSE trial, experiments reveal a common trend with the known and estimated time-based uncertainty converging together over time.

Compiling the results from the 600 varied heading multi-vessel SSE simulations, Table 4.12 displays the aggregated RMSE values for each of the estimated wave properties for the case of one, two or three vessels reaching an uncertainty level of 0.05. That is, the RMSE values of residual error assuming the estimation process terminated when the uncertainty level dropped below (or equalled) 0.05. It should be noted that in 18 cases, one vessel never achieved an uncertainty of 0.05, while in 83 cases, two vessels never achieved it, and in 321 cases three vessels never achieved it. These cases were removed from the RMSE calculations.

Table 4.12: RMSE values for each wave condition when one, two, or three vessels have reached a time-based uncertainty of 0.05.

Number of vessels	RMSE ($\hat{H}_s$)	RMSE ($\hat{T}_1$)	RMSE ($\hat{\mu}_w$)
One	$48.7mm$	$0.533s$	89.5°
Two	$37.7mm$	$0.342s$	74.7°
Three	$30.7mm$	$0.215s$	68.0°

There was a clear reduction in error when the estimation process was terminated when more vessels achieved an uncertainty of 0.05 or less. The RMSE reduction ratio for two vessels versus one was better than for three vessels versus two for $\hat{H}_s$ and $\hat{\mu}_w$, but not $\hat{T}_1$.

However, the RMSE for $\hat{H}_s$ was nearly an order of magnitude larger for the termination of 0.05 time-based uncertainty than for the complete 40 minute time period, even for only two vessels. The results for the $\hat{\mu}_w$ estimates were also much worse when terminating the SSE after the uncertainty criteria was achieved, although they were found to reduce when more vessels were required to reach 0.05 uncertainty.

The reduction of the required uncertainty level needed to terminate the SSE process was considered; however, when an uncertainty of 0.035 was tested, only 186, 110, 48 cases achieved this level of certainty (out of the 600), making attaining this level of uncertainty unfeasible.

When inspecting the spectral uncertainty evolution, it can be seen that the spectral estimates became rapidly more similar at the start, as the T_1 estimates converged towards one another (Figure 4.10). This resulted in an uncertainty of ~ 0.05 after around $1100s$, which would identify a high level of certainty in the estimations when $\hat{H}_s$, especially, still had an error of $\sim 40.0mm$. The estimates for both H_s and T_1 diverged again for a period of time, raising the uncertainty, before converging towards a common estimation again, with uncertainty once again dropping below 0.05 after around $2000s$. Estimations were much more accurate at this point in time.

Beyond a single example, the appropriateness of the spectral agreement uncertainty metric, when recreating the spectrum online using its estimated parameters, was assessed by applying it to the 600 varied heading estimation simulations. The aggregated estimations were then recorded when the uncertainty metric reached three levels of uncertainty (0.10, 0.05, and 0.02). The RMSE values were then taken for the estimates, given in Table 4.13. It was found that in 18 cases, the estimations never reached an uncertainty of 0.10, while in 49 cases an uncertainty of 0.05 was never reached, and in 209 cases an uncertainty of 0.02 was never reached.

Table 4.13: RMSE values for each wave condition when one, two, or three vessels have reached a time-based uncertainty of 0.05.

Uncertainty	RMSE ($\hat{H}_s$)	RMSE ($\hat{T}_1$)	RMSE ($\hat{\mu}_w$)
0.10	$52.4mm$	$0.563s$	107°
0.05	$35.7mm$	$0.445s$	81.3°
0.02	$19.5mm$	$0.270s$	42.0°

There was found to be a reduction in estimation error for an increase in estimation certainty for each of the wave conditions. The RMSE values more than halved when comparing an uncertainty of 0.10 to 0.02. For an uncertainty of 0.05, the spectral RMSE values were comparable to those for the time-based RMSEs when using two vessels.This indicates that the spectral uncertainty metric may be better suited to the online estimation of H_s and T_1 than it was at comparing errors for offline SSE.

However, as the estimations of each wave condition tended to only converge towards the actual values after 40 minutes (rarely sooner), the performance of the online wave property estimation was considered to be poor. This was especially true for H_s. As such, it is difficult to assess the applicability of the uncertainty metrics for online estimation.

PSD Variation Over Time

Although PSDs give a statistical representation of the vessel responses to wave excitation for any given time period, a variation in spectral ordinate distribution was found to exist over time. To further study this, an example simulation was run for a single vessel with $U_\pi = 2.38m/s$ and $\mu_h = 51.2°$ in a sea state with $H_s = 1.2m$ and $T_1 = 8.5s$. The spectral ordinates obtained from the time-series for different intervals were then recorded.

Figure 4.13 shows the evolution of the vessel's 30 response PSD components for heave, pitch, and roll for the time periods of 10 minutes, 20 minutes, 30 minutes, and 40 minutes.

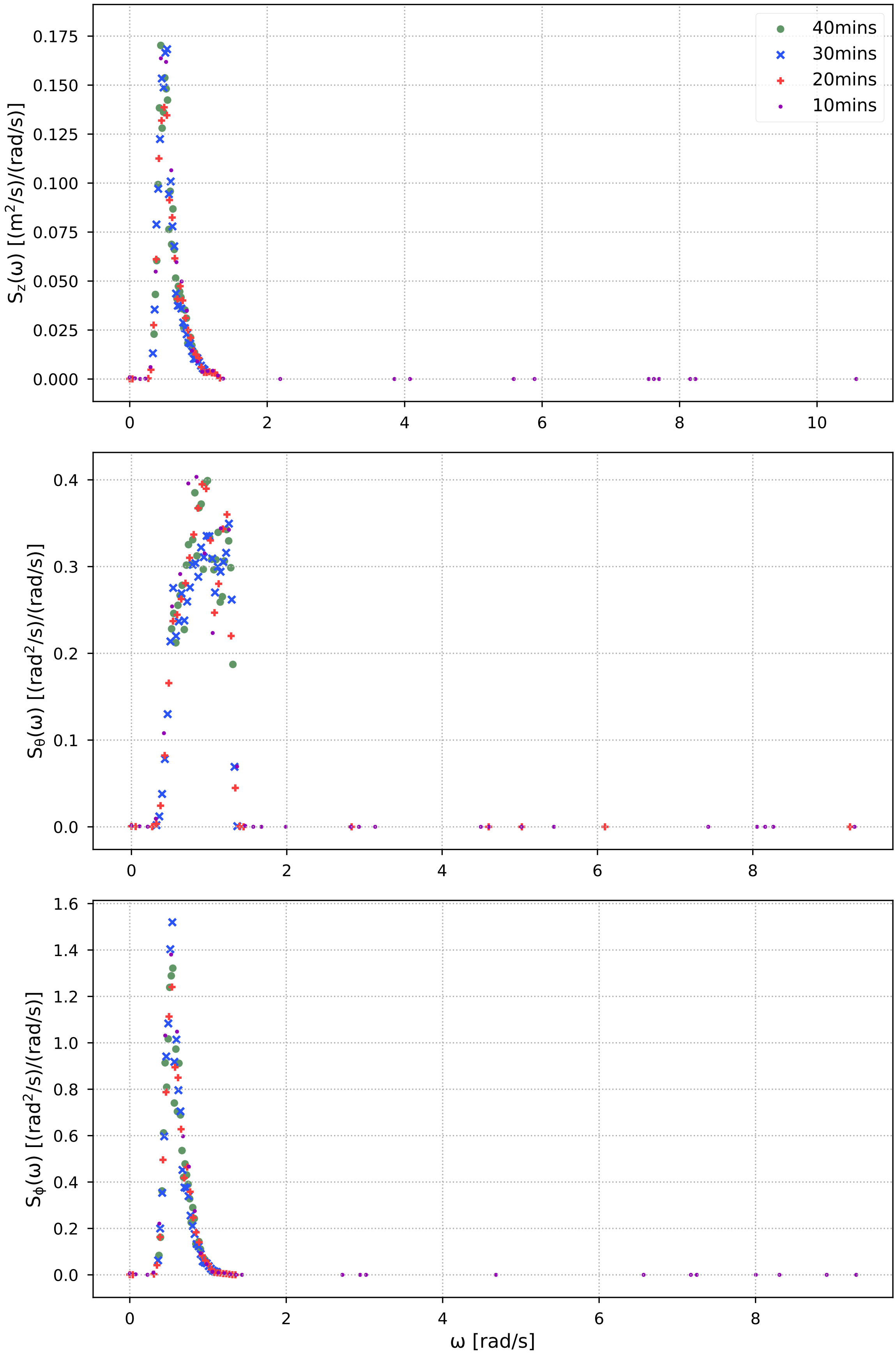

Figure 4.13: Vessel response PSD components versus wave frequency over time.

Inspection of the variation in spectral ordinates for the 30 PSD components with the highest energy over time reveals a common spectral shape for all responses; however, an uneven distribution of the components.

The PSD components for heave, pitch, and roll extracted earlier in the time-series capture more lower energy spectral ordinates than the PSD components extracted later in the time-series. As a result, the spread of frequencies associated with extracted spectral ordinates at 10 minutes is much larger than the spread for any of the proceeding time periods, while the magnitude of the ordinates are also much lower (even approaching zero). This trend is replicated across all of the response PSDs. Further, this pattern evolves over time, with the ordinates captured becoming larger as time continues.

One explanation for this can be understood by considering the workings of Fourier transforms. If a regular sine wave is transformed into the spectral domain, then a single spike at the frequency of that sine wave will exist. If a signal is the summation of two sine waves with different frequencies, then two spikes will exits after being transformed into the frequency domain.

Given that the wave-series simulated for this study were comprised of 500 different wave frequency components, and the fact that the vessels respond differently to every wave frequency (see Figures 3.8 to 3.13), the number of wave components encountered will alter the number of spikes in the vessel response PSDs.

While it was considered that the number of wave components encountered would become trivial over time, and the spectral ordinates captured would be able to converge on similar amounts of energy at times before the 40 minute interval, it appears that the lower density of high energy components after less time resulted in lower energy spectral ordinates being extracted.

One solution to this problem could be to change the Welch parameters when transforming the time-series, such as a different number of windows. Alternatively, the NNs could be trained on response data for different time periods, rather than only on the 40 minute time intervals.

4.6 Conclusion

Although a single vessel acting as a wave buoy has been shown to capture sea state information with relatively high accuracy, biases in the vessel state could lead to unreliable estimation results.

As such, it was hypothesised that the use of multiple vessels acting as wave buoys would be able to reduce errors in the estimations when compared with a single vessel. Further, it was hypothesised that the variation of wave heading would result in improved SSE performance as it would allow more information to be extracted from the vessel motion responses.

Three multi-vessel guidance approaches were introduced in this study, each involving the use of three vessel, designed to compare SSE performance between vessels with a common heading, vessels with a maximum variation in heading, vessels with random headings, and a single vessel.

Significant wave height and mean wave period estimates were aggregated for all three multi-vessel guidance strategies. For the common heading guidance strategy, the individual wave heading results were also aggregated. However, for the varied heading and random heading multi-vessel guidance approaches, the individual wave headings were transformed into a global wave direction so that a common variable estimation could be aggregated.

Each of the three multi-vessel approaches were found to reduce the error in SSEs when compared to using a single vessel as a wave buoy. The aggregated estimates of H_s showed similar performance for all three approaches, while the varied heading and random heading strategies had less errors than the common heading approach for T_1 and wave propagation estimates, with the wave propagation estimates showing the best improvements. As wave heading was cited as being the most difficult wave condition to estimate [18], the use of multiple vessels for wave heading estimation could be prove to be a valuable application.

The differences in SSE errors between the three different multi-vessel guidance approaches was found to be statistically significant for the μ_w discrepancies and between the common and random approaches of T_1, indicating that the different approaches are able to extract different information from the vessel responses when the headings are varied. However, the difference between approaches was found to not be significant for H_s.

Two uncertainty metrics were applied to the SSE results for the varied heading approach, offline, in order to determine their applicability to multi-vessel SSE using SAWB. One metric was time-based and designed to assess random estimation errors, while the other was based on the agreement of sea spectrum estimations, aimed at assessing bias errors. The time-based uncertainty metric showed a correlation between reduction in uncertainty and reduction in estimations errors for H_s and T_1, but not for μ_w. The spectral uncertainty metric, however, was found to not show a relationship between error reduction and uncertainty. A modified version of the combination of the two uncertainty metrics could potentially be effectively applied to future offline SSE results.

The methodology used to design the NNs for SSE were hypothesised to be able to update in real time Further, it was hypothesised that the uncertainty metrics could also be applied in real time, meaning that the estimation process could continue until a level of certainty in the results was achieved. Vessel response PSD information was used as input to the NNs for SSE as response spectra are a statistical representation of the response information. However, the online estimation process was not found to perform effectively due to the lower density of spectral ordinates at earlier times. This also resulted in the uncertainty metrics not being able to assess the estimations accurately in real time. Alternatively, the method could work for a vessel providing an 'online'estimation using a sliding 40 minute window, assuming that the vessel continues to operate in the same region.

The guidance strategies presented in this study were developed to act as a benchmark for future multi-vessel SSE studies using SAWB. There are likely more optimal guidance approaches which better harness multi-agent dynamics to further improve SSEs; however, the information presented in this study could be used to aid in their design.

Chapter 5: Conclusion

The global network of human expansion across the globe has led to the widespread use of maritime platforms for reasons ranging from fishing to trade and defence. As such, the ability to estimate sea states in-situ offers a number of advantages for maritime operations. Primarily, the improvement of passenger, crew and payload safety and comfort, through the reduction of impact loads on the vessel, can be achieved by incorporating sea state information in the vessel's path planning.

It was decided to adopt the ship as a wave buoy technique to address this problem, which involves the establishment of the relationship between a vessel's motion and the wave properties which caused it. As such, the vessel can be deployed on a body of water, its motions recorded, and the causal wave properties calculated. Further, the use of uninhabited surface vessels was suggested as this allows for a cheaper solution, capable of full autonomy.

While a number of SSE techniques have been developed over the years, the use of ML to map the relationship between vessel motions and wave conditions was chosen due its potential for accurate estimation, as well as its removal of the requirement of having in-depth knowledge of a vessel's physical properties to build an SSE model [17]–[21].

It was hypothesised that a NN could be designed such that it took vessel spectral response information as input, then output one of three wave properties: significant wave height, mean wave period, and wave heading. Following a common trend in SAWB SSE, it was decided to use the heave, pitch, and roll responses of vessel motion as the DOFs for analysis [22]–[25].

In order to test this hypothesis, several models had to be developed. Firstly, a sea state was generated by transforming a parametric Modified Pierson-Moskowitz spectrum [2] into a uni-directional wave elevation time-series which propagated in three dimensions. Secondly, the semi-analytical closed-form FRFs derived by Jensen, Mansour, and Olsen [56] were used to model the vessel responses to the wave excitations. It was decided to use a WG as the vessel model as it is a USV, capable of full autonomy, and powered entirely by renewable energy sources. As such, the WGs have increased endurance and range, compared with vessels not powered by renewable energy, meaning that they could be deployed over large distances for long periods of time.

The heave, pitch, and roll time-series responses derived from the sea state and ship dynamics models were then transformed into the frequency domain and processed to be used as input to the SSE NNs. PSDs of the vessel responses were obtained using Welch's method [63], which used overlapping windows to smooth the transformed spectral data and helped to remove noise. The spectral ordinates with the highest energy and their associated frequency components were then extracted from the PSDs, as well as their total power (taken by integrating across the spectra).

As the vessel response information had already been pre-processed, it was decided to use MLPs as the NN type for SAWB-based SSE, which took the spectral ordinates, frequency components, and total PSD powers from each DOF, as well as the forward speed of the vessel as input.

In order to investigate what information was able to be extracted from each vessel DOF, as well as combination of DOFs, different NNs were designed to take single DOF response information as input, and combined two DOF response information as input, to estimate each wave parameter.

48,000 simulations were run for randomised vessel states (forward speed and wave heading) in randomised sea states (varying significant wave height and mean wave period) in order to obtain 40 minute time-series for heave, pitch, and roll responses.

Different NN architectures and configurations were then experimented with for the 1-DOF, 2-DOF, and 3-DOF input models, with k-fold cross validation performed using 90% of the data for each configuration to determine which ones estimated the sea states with the least error (lowest average RMSE) and had the best model fit (highest R^2 value). Therefore, 21 NNs in total were designed, one for each DOF (heave, pitch, and roll), one for each combination of two DOFs, and one using all three DOFs, for each estimated wave condition (significant wave height, mean wave period, and wave heading). It was found that 30 spectral response components provided the best estimates for significant wave height and mean wave period, while 80 spectral response components were needed for optimal wave heading estimation. The final NNs were then tested on the remaining 10% of the simulation data.

A number of interesting observations were made when analysing the performance of the different response components at SSE. There was a strong correlation between the heave response and significant wave height estimation, with estimation errors only slightly reducing when the other DOFs were included in the SSE models. The estimation of the mean wave period, to the contrary, improved for the 2-DOF models, and again for the 3-DOF model, indicating that alternative information could be extracted from each DOF to aid in its estimation. Pitch was found to be the DOF which best estimated wave heading, though a significant improvement was seen when all three DOFs were combined into one network.

The effects of wave heading on estimation performance were then explored, where 4,800 simulations were run with the vessel speeds fixed, then the wave headings were randomly varied. The strongest relationship found between wave heading and estimation performance was for the single DOF models. Pitch struggled to estimate significant wave height and, to a lesser extent mean wave period, when the wave heading angles were in beam seas as its FRFs approached zero as the wave heading approached 90°, meaning the responses were minimal. Similarly, roll performed poorly when estimating significant wave height and mean wave period when the wave heading approached 0° and 180° as its FRF approached zero at those angles.

The final 3-DOF model for significant wave height was capable of estimation with an average RMSE of $5.38mm$ and an R^2 of 0.990. The final 3-DOF model for mean period had an average RMSE of $0.171s$ and an R^2 of 0.995. Finally, the 3-DOF model for wave heading had an RMSE of 7.21° and an R^2 of 0.980. As such, each of the models was found to closely fit the relationship between the vessel response input and sea state output. Overall, by comparing the ratio of the RMSE to the total range of possible wave property values simulated, it was found that mean wave period was estimated with the highest accuracy, while wave heading was estimated with the lowest accuracy.

These results were then compared with two other studies which used similar methodologies in terms of using parametric wave spectra to simulate wave conditions, and selected the same wave properties to estimate: one ML-based SSE approach [17], and one model-based approach [24]. The designed NN strategy was found to estimate the significant wave height and mean wave period with an accuracy an order of magnitude greater than the ML-based approach, as well as reducing errors considerably more than the model-based approach. The ML-based SSE model; however, was able to estimate wave heading with a slightly higher accuracy, while the designed NN performed slightly better than the model-based approach.

Therefore, it was shown that the vessel response spectral data could be used to estimate sea states, with a relatively high degree of accuracy, when using the simulation setup described, supporting the hypothesis that spectral response data could be used to train NNs to estimate sea states. However, while the results for SSE using a single vessel and the designed NNs were promising, they depict a simplified representation of the real interactions between wave systems and maritime platforms.

The use of a single vessel for SSE presents some limitations. If the vessel sensors are faulty, stored and/or transferred data is corrupted, or if the vessel's physical properties have changed in some way since the SSE model was calibrated, then the estimation results may be inaccurate. Further, certain vessel states, in terms of forward speed and heading, may have inherent biases associated with their motion and the SSE model.

The use of multiple vessels acting as wave buoys for SSE helps to overcome the limitations of a single vessel by introducing redundancy and improving reliability of the estimations. By aggregating the estimations of multiple vessels, errors from a biased or faulty vessel may be reduced once combined with more accurate estimations.

There is currently one study that explored the use of multiple vessels for SSE using the SAWB technique [25]. The authors considered vessels already in transit estimating sea states while en route to their destinations, with the results of the aggregated estimations showing improvement when compared to single vessel estimation. However, no study was conducted into the effects of the vessel trajectories on the SSE performance.

It was hypothesised that varying the vessel headings when using multiple vessels as wave buoys for SSE would result in better SSE performance when compared to a single vessel or multiple vessels travelling along common trajectories. This hypothesis is based on the assumption that estimating sea states using only one vessel state (primarily, heading), would result in estimation model biases. However, if estimations from multiple vessels with different states are aggregated, then any errors associated with estimation model biases should be reduced.

In order to test this hypothesis, three multi-vessel SSE guidance strategies were investigated, where each one incorporated three vessels. The first approach involved the three vessels estimating sea states when moving with a common heading. The second approach forced the vessels to move along headings which were 90° apart. The third approach sent the vessels on random headings.

While the wave heading estimates from the three vessels using the common heading approach were able to combine their estimates into an aggregated wave heading estimation, the varied heading and random heading approaches required a new wave property to be introduced, a global wave direction. Wave heading was given as the relative angle between the BF coordinates of the vessel and the WR coordinates. The wave direction was defined as the angle between the NED coordinates and the WR coordinates. Therefore, by assuming that the vessels were capable of determining their NED coordinates, as well as distinguishing between port and starboard wave headings, the relative wave heading estimates from individual vessels moving with different headings could be aggregated into a single wave direction estimate.

600 simulations for each of the three guidance approaches were run in order to study the performance of the three guidance approaches, with respect to an individual vessel and one another, in terms of reduction in error. The common heading approach showed a decrease in error of approximately 15% for significant wave height and mean wave period, when the aggregated estimations were compared with the single vessels SSEs, while the aggregated wave heading estimation errors decreased by $\sim$ 25%. The common heading aggregation, therefore, improved wave heading estimation the most. The varied heading and random heading approaches showed similar improvements in significant wave height and wave direction aggregated estimation, when compared with single vessel estimation, with reductions in error of approximately 20% and 37%, respectively. The varied heading aggregated mean wave period estimation reduced errors by $\sim$ 25%, while the random heading aggregated estimations reduced errors by $\sim$ 30%, compared with single vessel SSE.

The significance of the differences in estimation performance of the different guidance approaches was then tested by applying a two sample t-test to the different pairs of residual errors. The significance of the difference between the wave propagation angle estimations for the varied and random heading strategies, compared to the common heading, was found to be the greatest, with the likelihood of the variation in results obtained being due to chance appearing extremely low. The difference between the common heading and random heading approach was also found to be statistically significant for mean wave period. However, the difference between each of the approaches and significant wave height estimation results were not found to have greater than a 5% probability of being significantly different.

Therefore, as the variation between estimation results of the different guidance approaches were found to be significantly different, and the fact that the varied and random heading approaches showed increased performance when compared with the common heading approach, the hypothesis that varying the vessel trajectories when doing multi-vessel SSE would improve estimation performance was supported. Further, the multi-vessel guidance approaches were generally found to reduce estimation errors by large percentages when compared with single vessel SSE. Overall, using multiple vessels for SSE saw the least improvement for significant wave height, and most improvement for wave propagation angle.

Two uncertainty metrics were selected to assess the quality of the multi-vessel SSEs. The first was a time-based uncertainty metric which was used to reduce random errors in the estimations by ensuring that a statistically significant number of wave components were encountered. The second metric was related to the agreement between estimated wave spectra, and aimed to assess bias errors in the estimations, where a convergence of estimated spectra from different vessels would indicate a high level of certainty in the estimations. When applied to the results from offline SSE, a correlation between time-based uncertainty and estimation error was found for significant wave height and mean wave period estimations, but not wave heading estimation. The spectral uncertainty was not found to be appropriate for the parametric wave spectrum estimation for offline estimation as no relationship between error reduction and uncertainty level was found.

While most other SEE methods using ML estimated the wave properties used vessel response time-series data for a fixed period of time, it was hypothesised that the designed NNs for SSE using vessel response spectral data would be able to estimate wave properties in real time, given that the spectra gave statistical representations of the responses.

Further, it was hypothesised that the time required for an SSE could be optimised by applying the uncertainty metrics to the estimations in real time, such that once a level of certainty in the estimations was reached, the SSE process could be terminated.

However, the NNs for SSE were not able to estimate sea states effectively online, which meant that the applicability of the uncertainty metrics at assessing sea state estimations in real time could not be determined. The reason the NN models were not able to estimate sea states online was found to be due to the difference in the density of spectral ordinates earlier in the time-series as opposed to later. While the response PSD shape was similar across the evolution of time, the spectral analysis captured lower energy spectral ordinates at earlier times than later times, as there were fewer higher energy ordinates. One proposed solution to solve this would be to vary the parameters used to transform the time-series response data to PSDs. Another would be to train the NNs on the spectral information from different points in time, rather than just for

40 minute time-series.

The simulation setup which provided the result for this thesis was relatively simple, which leads to two remarks. The first remark is that although the variation in accuracies for the estimated wave properties is on a scale which may have little relevance in real world applications, if the designed models and strategies are translated to real equipment, the difference in accuracy may be more significant as the estimation errors would likely be greater. The second remark is that, of course, the methodologies used for this simulation setup may not prove to be effective when applied to alternative or more complex situations.

This study aims to be a benchmark for future research on both the use of vessel response spectral data for ML-based SSE using the SAWB technique, as well as to provide a baseline which shows the potential for multi-vessel SSE.

5.1 Further Research

The SSE models trained for this study utilised one type of NN, with a limited range of architecture configurations tested. Any future studies on the use of vessel response spectral data for SSE could explore the effects of different NN types, as well as conduct an optimisation of the network architecture and configuration, in order to further improve SSEs. One example would be to train a network to estimate all three wave properties simultaneously such that any relationships between them could be exploited.

As the networks failed to estimate sea states in real time with a high degree of accuracy, the NNs could be retrained on spectral data transformed from different vessel response time-series intervals. The type of spectral analysis conducted on the vessel response time-series data could also be investigated to see how the information could be manipulated to further improve SSE performance or be used for online SSE. One example could be to use wavelet analysis of the vessel responses, essentially building on Cheng et al.'s study [21] using spectrograms; however, wavelet analysis uses a varying window size depending on the magnitude of the frequency being analysed. This could lead to finer features being extracted from the vessel responses for use as input to ML models for SSE. Further study on the multi-vessel SSE uncertainty metrics presented in this thesis could be undertaken if an effective real time estimation model is used.

While most SSE studies have assumed that stationary conditions must exist for vessels acting as a wave buoys (moving with a constant heading and forward speed), at least one study has shown that SAWB SSE is also possible when the vessel varies its trajectory (Cheng et al. [20] used a zig-zag motion). By removing the stationary requirement for SAWB SSE, a much greater range of guidance approaches are possible, which could result in better SSE performance, and potentially remove the need for multiple vessels as the single vessel would not only have motion data from one stationary condition.

This could be particularly useful when using multiple vessels for SSE, where it is proposed that the vessels could be guided in such a way that their trajectories across a wave system are designed to reduce uncertainty in the online SSEs, further optimising the SSE process.

The use of multiple vessels for SSE could, further, prove to be useful when estimating sea states which vary spatially, where centralised and decentralised coordinated guidance of the vessels could be used to build a map of the varying sea states.

Finally, before the use of any SSE model developed via simulation should be made operational, experimentation using real hardware and wave systems, such as model-scale experiments in a tow tank, or even conducting tests at sea with a WG, should be undertaken to validate the methodology under more realistic conditions.

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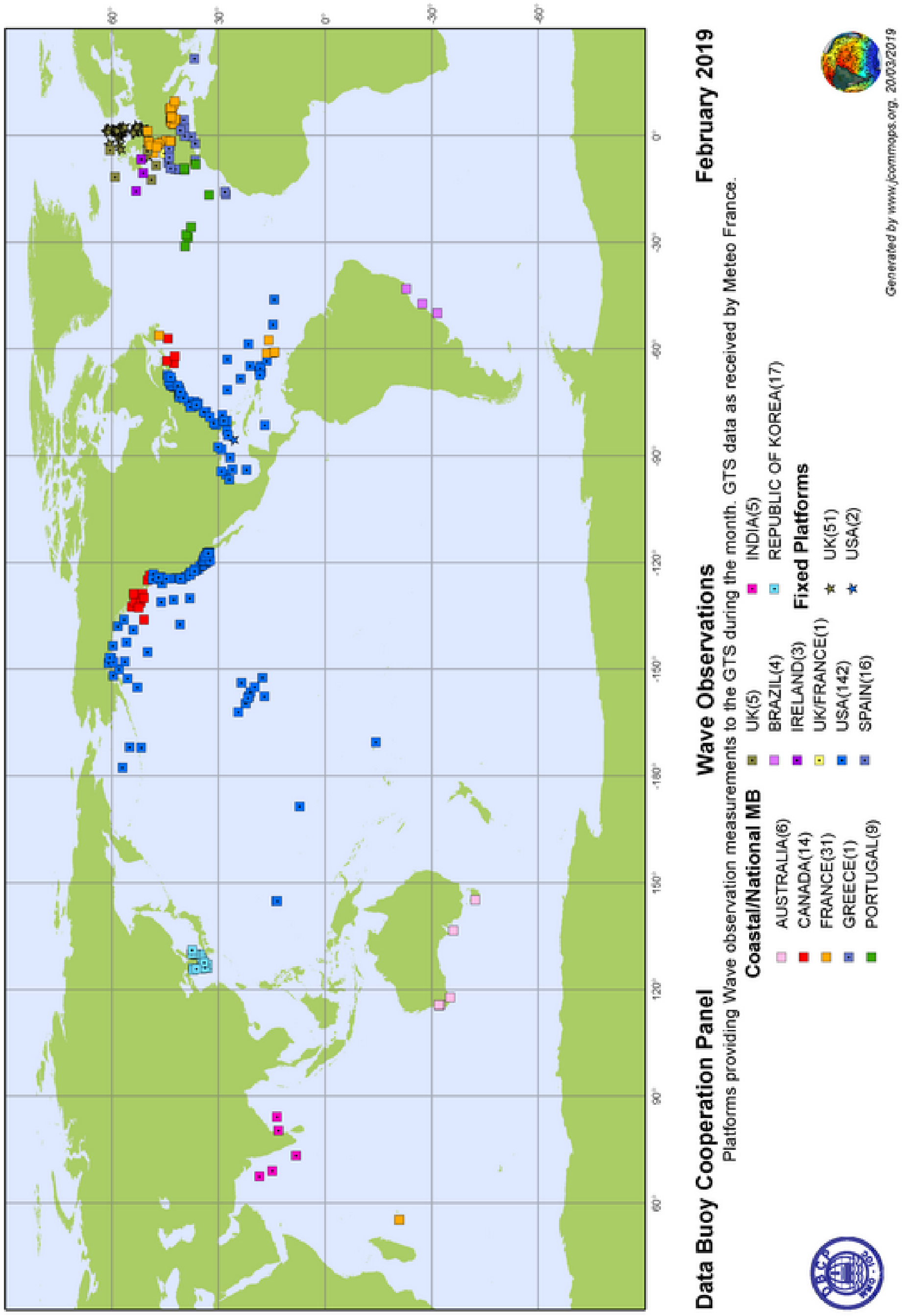
[A] Data Buoy Cooperation Panel: Wave Observations

Global wave buoy locations of wave observation data for February 2019, as collected by the Data Buoy Cooperation Panel (a body created by the World Meteorological Organization and Intergovernmental Oceanographic Commission of UNESCO) [31].



[B] Sea State Estimation Neural Network Construction Function for a 1-DOF Response

Listing 1: Keras NN model construction function example.

```
def sse_nn_model_1dof(opt, loss, n_batch,
                      n_branch_nodes, n_trunk_nodes_1, n_trunk_nodes_2):

    """
    Sea State Estimation Neural Network Model Construction (1-DOF) Function
    ------------------------------------------------------------------------

    This function creates and compiles a NN model using a single DOF
    response from a vessel.


    Arguments:
        opt                   : Optimisation function
        loss                  : Loss function
        n_batch               : Batch number
        n_branch_nodes        : No. of nodes in branch hidden layer
        n_trunk_nodes_1       : No. of nodes in 1st trunk hidden layer
        n_trunk_nodes_2       : No. of nodes in 2nd trunk hidden layer

    Returns:
        model                 : Compiled NN model
    """

    # Input layers
    input_1 = Input(shape=(n_batch,30))
    input_2 = Input(shape=(n_batch,30))
    input_3 = Input(shape=(n_batch,2))

    # Merge PSD input layers
    input_psd_merged = concatenate([input_1,input_2])
```

```
# Hidden PSD layer
branch_1 = Dense(n_branch_nodes, activation='relu')(input_psd_merged)
branch_1 = Model(inputs=[input_1, input_2], outputs=branch_1)

# Merge with integral and speed
input_merged = concatenate([branch_1.output, input_3])

# Output layer
trunk = Dense(n_trunk_nodes_1, activation='relu')(input_merged)
output_layer = Dense(1, activation='linear')(trunk)

# Build model
model = Model(inputs=[branch_1.input, input_3], outputs=output_layer)

#Compile model
model.compile(loss=loss, optimizer=opt, metrics=[rmse, r_squared])

return(model)
```

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[C] NN Training (1-DOF): Heave for Sig. Wave Height and Mean Wave Period

Table 1: Heave SSE NN training for H_s**.**

Branch Structure	Trunk Structure	Epochs	Batch Size	Loss Rate	RMSE Avg.	RMSE Std.	Mean R-Squared
1 hidden layer (16 nodes)	2 hidden layers ([16,8] nodes)	50	32	0.001	0.058717*m*	0.000873*m*	0.988486
1 hidden layer (16 nodes)	2 hidden layers ([16,8] nodes)	100	32	0.001	0.058337*m*	0.000797*m*	0.988796
1 hidden layer (16 nodes)	1 hidden layer (8 nodes)	50	32	0.001	0.059215*m*	0.001919*m*	0.987839
1 hidden layer (16 nodes)	1 hidden layer (8 nodes)	100	32	0.001	0.059056*m*	0.000810*m*	0.988535
1 hidden layer (16 nodes)	1 hidden layer (8 nodes)	30	32	0.001	0.063042*m*	0.002507*m*	0.987142
1 hidden layer (8 nodes)	1 hidden layer (8 nodes)	50	32	0.001	0.064479*m*	0.001521*m*	0.987176
1 hidden layer (16 nodes)	1 hidden layer (4 nodes)	50	32	0.001	0.062486*m*	0.182444*m*	0.783331
1 hidden layer (16 nodes)	1 hidden layer (8 nodes)	50	32	0.01	0.070212*m*	0.004370*m*	0.984689
1 hidden layer (16 nodes)	1 hidden layer (8 nodes)	50	64	0.001	0.060125*m*	0.001977*m*	0.987931
1 hidden layer (16 nodes)	1 hidden layer (8 nodes)	50	16	0.001	0.060176*m*	0.002102*m*	0.987170

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Table 2: Heave SSE NN training for T_1.

Branch Structure	Trunk Structure	Epochs	Batch Size	Loss Rate	RMSE Avg.	RMSE Std.	Mean R-Squared
1 hidden layer (16 nodes)	2 hidden layers ([16,8] nodes)	50	32	0.001	$0.392048s$	$0.033992s$	0.965136
1 hidden layer (16 nodes)	2 hidden layers ([16,8] nodes)	100	32	0.001	$0.367656s$	$0.038651s$	0.969816
1 hidden layer (16 nodes)	1 hidden layer (8 nodes)	50	32	0.001	$0.471461s$	$0.018114s$	0.943554
1 hidden layer (16 nodes)	1 hidden layer (8 nodes)	100	32	0.001	$0.522909s$	$0.041749s$	0.961915
1 hidden layer (16 nodes)	1 hidden layer (8 nodes)	30	32	0.001	$0.609240s$	$0.020501s$	0.939169
1 hidden layer (8 nodes)	1 hidden layer (8 nodes)	50	32	0.001	$0.724540s$	$0.031701s$	0.924248
1 hidden layer (16 nodes)	1 hidden layer (4 nodes)	50	32	0.001	$0.794047s$	$0.027322s$	0.947983
1 hidden layer (16 nodes)	1 hidden layer (8 nodes)	50	32	0.01	$0.717388s$	$0.05257s$	0.926021
1 hidden layer (16 nodes)	1 hidden layer (8 nodes)	50	64	0.001	$0.569358s$	$0.038163s$	0.948388
1 hidden layer (16 nodes)	1 hidden layer (8 nodes)	50	16	0.001	$0.525976s$	$0.028289s$	0.949575

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[D] NN Training (2-DOF): Heave and Roll for Sig. Wave Height and Mean Wave Period

Table 3: Heave and roll SSE NN training for H_s.

Branch Structure	Trunk Structure	Epochs	Batch Size	Loss Rate	RMSE Avg.	RMSE Std.	Mean R-Squared
1 hidden layer (16 nodes)	2 hidden layers ([8,4] nodes)	50	32	0.001	0.060296*m*	0.000755*m*	0.987134
1 hidden layer (16 nodes)	2 hidden layers ([8,8] nodes)	50	32	0.001	0.061959*m*	0.001433*m*	0.987540
1 hidden layer (16 nodes)	2 hidden layers ([16,4] nodes)	50	32	0.001	0.062898*m*	0.00276*m*	0.987404
1 hidden layer (16 nodes)	2 hidden layers ([16,8] nodes)	50	32	0.001	0.063158*m*	0.001405*m*	0.987807
1 hidden layer (16 nodes)	3 hidden layers ([16,8,8] nodes)	50	32	0.001	0.060490*m*	0.000695*m*	0.988463
1 hidden layer (16 nodes)	3 hidden layers ([32,8,8] nodes)	100	32	0.001	0.060440*m*	0.001353*m*	0.987872
1 hidden layer (16 nodes)	3 hidden layers ([16,16,8] nodes)	100	32	0.001	0.057264*m*	0.003059*m*	0.988073
1 hidden layer (16 nodes)	3 hidden layers ([16,8,8] nodes)	100	32	0.001	0.058271*m*	0.183312*m*	0.989230
1 hidden layer (16 nodes)	3 hidden layers ([16,8,8] nodes)	100	64	0.001	0.062125*m*	0.000986*m*	0.988308
1 hidden layer (16 nodes)	3 hidden layers ([16,8,8] nodes)	100	16	0.001	0.057298*m*	0.001365*m*	0.987300
1 hidden layer (16 nodes)	3 hidden layers ([32,8,8] nodes)	100	16	0.001	0.058203*m*	0.001893*m*	0.988342

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Table 4: Heave and roll SSE NN training for T_1.

Branch Structure	Trunk Structure	Epochs	Batch Size	Loss Rate	RMSE Avg.	RMSE Std.	Mean R-Squared
1 hidden layer (16 nodes)	2 hidden layers ([8,4] nodes)	50	32	0.001	0.398500s	0.780224s	0.768064
1 hidden layer (16 nodes)	2 hidden layers ([8,8] nodes)	50	32	0.001	0.389523s	0.012373s	0.970638
1 hidden layer (16 nodes)	2 hidden layers ([16,4] nodes)	50	32	0.001	0.444790s	0.022635s	0.972043
1 hidden layer (16 nodes)	2 hidden layers ([16,8] nodes)	50	32	0.001	0.370983s	0.048912s	0.972695
1 hidden layer (16 nodes)	3 hidden layers ([16,8,8] nodes)	50	32	0.001	0.351504s	0.042241s	0.975387
1 hidden layer (16 nodes)	3 hidden layers ([32,8,8] nodes)	100	32	0.001	0.260598s	0.020827s	0.977520
1 hidden layer (16 nodes)	3 hidden layers ([16,16,8] nodes)	100	32	0.001	0.342289s	0.02728s	0.981448
1 hidden layer (16 nodes)	3 hidden layers ([16,8,8] nodes)	100	32	0.001	0.338673s	0.811427s	0.981885
1 hidden layer (16 nodes)	3 hidden layers ([16,8,8] nodes)	100	64	0.001	0.322174s	0.016518s	0.973792
1 hidden layer (16 nodes)	3 hidden layers ([16,8,8] nodes)	100	16	0.001	0.263133s	0.020973s	0.977277
1 hidden layer (16 nodes)	3 hidden layers ([32,8,8] nodes)	100	16	0.001	0.298001s	0.026680s	0.984971

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[E] NN Training (1-DOF): Pitch for Vessel Heading

Table 5: Pitch SSE NN training for mu_h**.**

Branch Structure	Trunk Structure	Epochs	Batch Size	Loss Rate	RMSE Avg.	RMSE Std.	Mean R-Squared
1 hidden layer (16 nodes)	1 hidden layers (8 nodes)	50	32	0.001	13.221852°	0.666642°	0.932041
1 hidden layer (16 nodes)	2 hidden layers ([16,8] nodes)	50	32	0.001	11.906571°	0.500538°	0.944817
1 hidden layer (16 nodes)	2 hidden layers ([16,8] nodes)	100	32	0.001	11.086158°	0.541064°	0.952232
1 hidden layer (16 nodes)	2 hidden layers ([16,16] nodes)	100	32	0.001	10.520216°	0.557920°	0.957106
1 hidden layer (16 nodes)	2 hidden layers ([16,16] nodes)	100	16	0.001	10.620208°	0.704697°	0.954504
1 hidden layer (16 nodes)	2 hidden layers ([32,8] nodes)	100	32	0.001	10.91997°	0.548440°	0.953744
1 hidden layer (16 nodes)	2 hidden layers ([32,16] nodes)	100	32	0.001	10.636354°	0.68606°	0.956052
1 hidden layer (16 nodes)	2 hidden layers ([32,16] nodes)	100	16	0.001	10.500245°	0.195178°	0.955727

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[F] NN Training (2-DOF): Pitch and Roll for Wave Heading

Table 6: Pitch and roll SSE NN training for mu_h**.**

Branch Structure	Trunk Structure	Epochs	Batch Size	Loss Rate	RMSE Avg.	RMSE Std.	Mean R-Squared
1 hidden layer (16 nodes)	3 hidden layers ([32,16,8] nodes)	100	16	0.001	8.696148°	0.540514°	0.969746
1 hidden layer (16 nodes)	3 hidden layers ([32,32,8] nodes)	100	16	0.001	8.347173°	0.298398°	0.972733
1 hidden layer (16 nodes)	3 hidden layers ([32,16,16] nodes)	100	16	0.001	8.942045°	0.588409°	0.968571
1 hidden layer (16 nodes)	3 hidden layers ([16,16,16] nodes)	100	16	0.001	8.520662°	0.506033°	0.971146
1 hidden layer (16 nodes)	3 hidden layers ([32,16,16] nodes)	100	32	0.001	8.217757°	0.284845°	0.974137
1 hidden layer (16 nodes)	3 hidden layers ([32,32,16] nodes)	100	32	0.001	8.399155°	0.221148°	0.972775
1 hidden layer (16 nodes)	3 hidden layers ([32,32,16] nodes)	100	16	0.001	8.384418°	0.362725°	0.972559

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[G] NN Training (3-DOF): Heave, Pitch, and Roll for Sig. Wave Height, Mean Wave Period, and Wave Heading

Table 7: SSE NN training for H_s.

Branch Structure	Trunk Structure	Epochs	Batch Size	Loss Rate	RMSE Avg.	RMSE Std.	Mean R-Squared
1 hidden layer (16 nodes)	3 hidden layers ([16,8,8] nodes)	100	32	0.001	0.058429*m*	0.184118*m*	0.989445
1 hidden layer (16 nodes)	3 hidden layers ([16,16,8] nodes)	100	32	0.001	0.053467*m*	0.000708*m*	0.990953
1 hidden layer (16 nodes)	3 hidden layers ([32,16,8] nodes)	100	32	0.001	0.053661*m*	0.00225*m*	0.991033
1 hidden layer (16 nodes)	3 hidden layers ([32,16,16] nodes)	100	32	0.001	0.056622*m*	0.001473*m*	0.989913
1 hidden layer (16 nodes)	3 hidden layers ([32,16,4] nodes)	100	32	0.001	0.055576*m*	0.225336*m*	0.990241
1 hidden layer (16 nodes)	3 hidden layers ([32,16,16] nodes)	100	32	0.001	0.053491*m*	0.000962*m*	0.990999
1 hidden layer (16 nodes)	4 hidden layers ([64,32,16,16] nodes)	100	32	0.001	0.054337*m*	0.001072*m*	0.990704
1 hidden layer (16 nodes)	3 hidden layers ([64,32,16] nodes)	100	32	0.001	0.056546*m*	0.001052*m*	0.989214
1 hidden layer (16 nodes)	3 hidden layers ([32,32,16] nodes)	100	32	0.001	0.054345*m*	0.001389*m*	0.990629
1 hidden layer (16 nodes)	3 hidden layers ([32,32,16] nodes)	100	16	0.001	0.053763*m*	0.001156*m*	0.990181

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Table 8: SSE NN training for T_1.

Branch Structure	Trunk Structure	Epochs	Batch Size	Loss Rate	RMSE Avg.	RMSE Std.	Mean R-Squared
1 hidden layer (16 nodes)	3 hidden layers ([16,8,8] nodes)	100	32	0.001	$0.198793s$	$0.011684s$	0.993787
1 hidden layer (16 nodes)	3 hidden layers ([16,16,8] nodes)	100	32	0.001	$0.193745s$	$0.019266s$	0.994027
1 hidden layer (16 nodes)	3 hidden layers ([32,16,8] nodes)	100	32	0.001	$0.190506s$	$0.005479s$	0.994268
1 hidden layer (16 nodes)	3 hidden layers ([32,16,16] nodes)	100	32	0.001	$0.178151s$	$0.015676s$	0.994971
1 hidden layer (16 nodes)	3 hidden layers ([32,16,4] nodes)	100	32	0.001	$0.191851s$	$0.836819s$	0.994173
1 hidden layer (16 nodes)	3 hidden layers ([32,16,16] nodes)	100	32	0.001	$0.187010s$	$0.009823s$	0.994474
1 hidden layer (16 nodes)	4 hidden layers ([64,32,16,16] nodes)	100	32	0.001	$0.186882s$	$0.006201s$	0.994725
1 hidden layer (16 nodes)	3 hidden layers ([64,32,16] nodes)	100	32	0.001	$0.170139s$	$0.004981s$	0.995140
1 hidden layer (16 nodes)	3 hidden layers ([32,32,16] nodes)	100	32	0.001	$0.180047s$	$0.005969s$	0.994846
1 hidden layer (16 nodes)	3 hidden layers ([32,32,16] nodes)	100	16	0.001	$0.170668s$	$0.005969s$	0.995105

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Table 9: SSE NN training for μ_h.

Branch Structure	Trunk Structure	Epochs	Batch Size	Loss Rate	RMSE Avg.	RMSE Std.	Mean R-Squared
1 hidden layer (16 nodes)	3 hidden layers ([32,16,16] nodes)	100	16	0.001	7.514573°	0.524905°	0.977865
1 hidden layer (16 nodes)	3 hidden layers ([32,32,16] nodes)	100	16	0.001	7.459236°	0.388174°	0.978064
1 hidden layer (16 nodes)	4 hidden layers ([32,16,16,8] nodes)	100	16	0.001	7.477144°	1.999657°	0.977879
1 hidden layer (16 nodes)	4 hidden layers ([64,32,16,16] nodes)	100	16	0.001	7.213019°	0.602612°	0.979579
1 hidden layer (16 nodes)	4 hidden layers ([64,32,32,16] nodes)	100	16	0.001	8.117252°	0.602612°	0.973116
1 hidden layer (16 nodes)	4 hidden layers ([32,32,16] nodes)	100	32	0.001	7.214160°	0.411126°	0.979752